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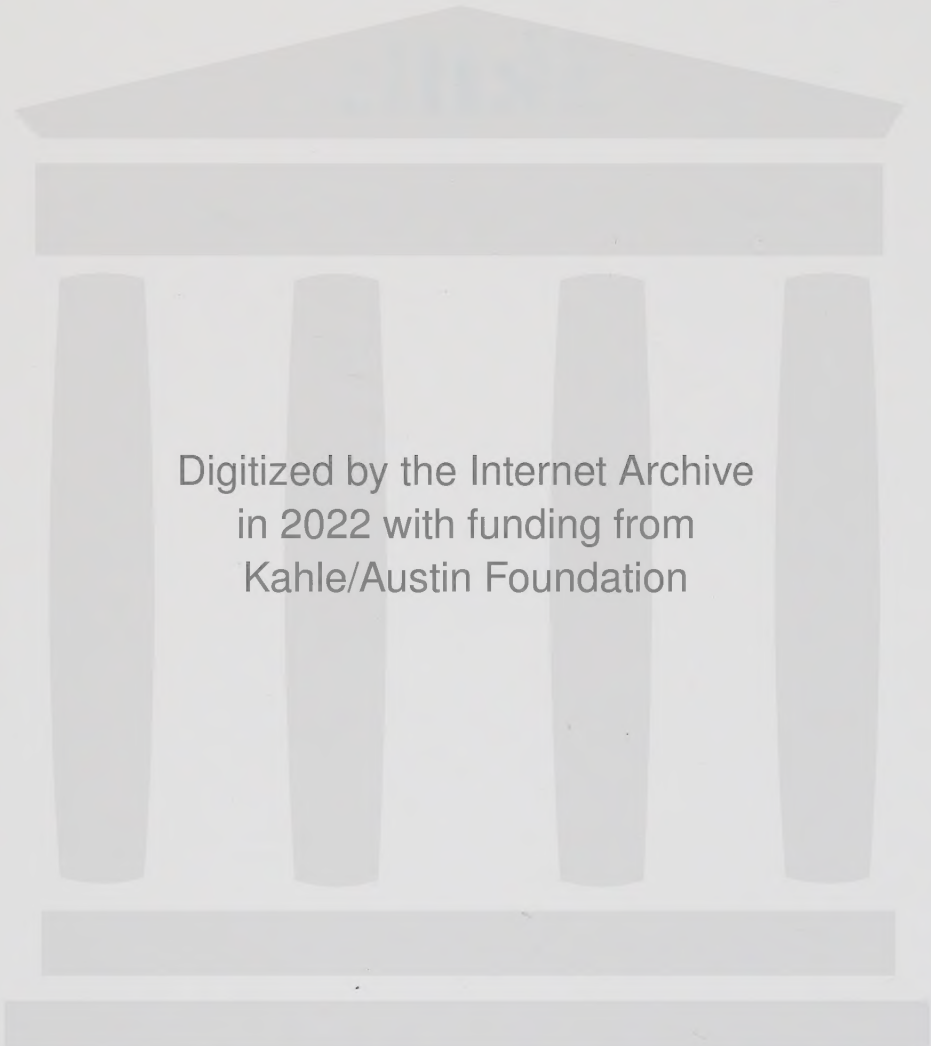
3rd edition

CRITICAL THINKING SKILLS

effective analysis, argument and reflection

STELLA COTTRELL

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Effective analysis, argument and reflection

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Critical Thinking Skills

Effective analysis, argument and reflection

Third edition

Stella Cottrell

macmillan international
HIGHER EDUCATION

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Self-evaluation sheets, planners and activity sheets may be photocopied by individual students for their personal use only.

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Introduction

Building our 'critical muscle'

Nobody is an absolute beginner when it comes to critical thinking. Our most everyday activities require us to make use of some of the basic skills involved in critical thinking, such as:

- working out whether we believe what we see or hear;
- taking steps to find out whether something is likely to be true;
- arguing our own case if someone doesn't believe us.

However, just because we can think critically, this doesn't mean we always do, or that we do it well. This is to be expected, as we don't need to employ the same level of critical thinking for everything we do.

For everyday activities, we take a certain amount on trust, and this saves us from having to recheck every detail. We have to decide how much information is really required and what level of doubt is acceptable for each new circumstance. The levels and types of knowledge we need vary depending on the task, such as whether we are simply switching on a light, inventing a new form of electrical circuit or treating someone for electrocution. Similarly, critical thinking involves:

- identifying correctly when we need to gain more information;
- selecting effectively the right type and level of information for the purpose;
- the appropriate level of critical analysis to use in the specific circumstances.

This book helps you to build on your existing skills, to develop your 'critical muscle'.



not demonstrating much criticality....

Do critical thinking skills matter?

The value of critical thinking skills

Critical thinking abilities are essential – at every age, for everyone, in all contexts, in every aspect of life. Applying critical thinking skills even in everyday situations reaps all kinds of benefits.



It is rewarding to refine your thinking skills – and the benefits can be unexpected. For example, you may find you are:

- absorbing more when watching videos or reading;
- better at telling when other people are being inconsistent or jumping to conclusions;
- quicker at spotting alternative interpretations to issues, stories and case histories;
- faster and clearer in forming an opinion on issues.

Success in most professions requires good critical thinking skills. Academic study also requires increasingly sophisticated levels of critical analysis at every level of study. Whether for work or for study, you may be expected to apply critical thinking to:

- what you hear, see, and do;
- the material you read;
- how you interpret new situations and events;
- what you write, say or present to other people;
- your own learning and professional practice.

About this book

Aims of this book

This book aims to help readers develop an understanding of what is meant by critical thinking and to develop their own reasoning skills. These skills are essential to those progressing to higher levels of academic study, whether at advanced or degree level. However, the underlying concepts are useful to anyone who wishes to:

- understand the concepts used in critical thinking;
- develop clearer thinking;
- interpret and produce arguments more effectively;
- be more observant of what they see and hear.

This book focuses mainly on aspects of critical thinking that can be applied to work and study, and which help individuals to think about how they think. It is not intended to be an advanced study of abstract reasoning or logic. For these, the reader is referred to works such as A. Garnham and J. Oakhill (1994), *Thinking and Reasoning*, and A. Fisher (2004), *The Logic of Real Arguments*. Rather, the purpose of this book is to focus on the basics of clear thinking.

For those new to critical thinking

The book will assist you in practical ways such as helping you to:

- recognise and understand the technical terms in critical thinking so you know what other people are referring to when they mention these, and so you can apply them yourself as relevant;
- build confidence in your own ability to apply critical thinking techniques;
- examine closely the opinions, views and arguments presented by other people;
- challenge other people's views from an informed perspective when this is appropriate.

For students

Students will find the book particularly useful in developing the ability to:

- recognise the arguments of specialist authors;
- locate arguments in key texts with greater speed;
- engage in dialogue with the arguments used by both experts and their peers;
- produce better critical, analytical writing of their own for marked assignments;
- recognise the difference between critical analysis and other kinds of writing, such as description.

Using activities in this book

Critical thinking is an activity. It isn't sufficient to read about it: it has to be practised. The book offers activities to apply the concepts it introduces and to practise new skills. It may be that, after completing one or two of the activities that accompany a new concept, you find that aspect very easy. If so, move on to the next aspect. However, many people find some or all aspects of critical thinking to be difficult at first. If this is true of you, be reassured that this way of thinking becomes easier with practice.

The answers pages do not simply provide a correct answer: they also explain the reasons behind the answers so as to develop further the concept that has been practised. Reading through these should help you to clarify your understanding about that aspect of critical thinking.

A wide range of topics is used as examples and as practice material. You do not need any background knowledge of the subjects covered in these. It is possible to do all the activities no matter what your subject discipline or area of interest. The activities require you only to apply critical thinking to the material provided.

Companion website materials

A companion website has been created to assist your use of the material in this book. This includes templates of some of the questionnaires, and longer texts which support activities. Look for the icon to

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see which material is available online. You might also like to browse the site to see what is available for you there at www.palgravehighered.com/cts3e.

Passages used in the book

All of the passages in the book have been specially designed to illustrate the key points of each chapter and to provide appropriate practice material. They draw on a range of different academic disciplines but are written in such a way that you do not need to be an expert in the subject to understand the material.

None of the passages in this book is reproduced from any other text. However, some draw on the writing of others for background information. Where this is the case, details of the original source are given at the end of the chapter to enable you to follow up subjects that interest you.

Purpose of short and longer passages

For the main body of the book, especially the early chapters, the passages are short in order to introduce new learning points and to give you the opportunity for quick practice in applying these. Usually, at least three examples of each learning point are provided, as that is regarded as the minimum number of practice tries required in order to anchor new learning into memory.

In articles, books and other source material, the learning points will not always be as apparent as in the short passages provided here in the early chapters. It is important to develop the ability to apply critical awareness when reading longer texts. For these, you will need to balance different perspectives and weigh up material from a range of sources, synthesising the material to form your own judgements.

Longer practice passages are provided in Chapters 4 and 11, and in the practice section towards the end of the book. These enable you to work on several aspects of critical thinking simultaneously, and are also available on the companion website, at www.palgravehighered.com/cts3e.

Terminology: Author and audience

The different aspects of critical thinking covered in this book can be applied to material in varied media, whether written, audio or televisual. However, in order to simplify the text, the terms 'author' and 'audience' are used throughout, irrespective of the type of media.

Author

This refers to the person who creates the message, whether this is written, spoken or delivered through another medium. It doesn't necessarily mean the 'author' of a book.

Audience

This refers to whoever receives the message, whether through conversation, books, television, video downloads and/or podcasts, or other medium. The audience, in this respect, may be a viewer, a reader, a listener, or an observer.

Glossary

A glossary of technical terms used in critical thinking is provided on page xiv.

Contents of the chapters

The book is organised to help you build your skills in critical thinking, starting from a basic understanding of what critical thinking is, through to applying techniques and strategies when reading and producing your own critical writing.

Chapter 1 introduces critical thinking, looking at the range of underlying skills and attitudes associated with critical thinking, and why it is beneficial to develop critical thinking skills. It emphasises the importance of self-awareness as an aspect of making accurate judgements and bringing suitable objectivity to critical reasoning.

Many people find critical thinking to be a challenging activity when they first begin. The chapter looks at the barriers that might prevent you from developing critical thinking skills and ways of overcoming these. You are invited to evaluate your current skills in order to focus on those aspects of the book that are the most useful for you.

Chapter 2 looks at important sub-skills of critical thinking skills such as focusing your attention, identifying similarities and differences, sequencing, categorising, and close reading. These are skills that underlie more advanced critical thinking as well as personal management skills, so improving these can benefit many aspects of academic work and personal and working life. The chapter provides an opportunity

for you to evaluate your skills and then to practise those aspects which need further development.

Chapter 3 'What's their point?', introduces argument as a central aspect of critical reading. It identifies the main features and components of arguments within critical thinking, and provides practice in identifying these different elements. This is useful in helping you to find the most significant passages in your specialist texts, and to do so more quickly.

Chapter 4 builds on the previous chapter, looking at the differences between critical arguments and other types of writing that may appear to be arguments, such as disagreements. It also looks at how, when reading, to distinguish critical argument from summaries, explanations and descriptions. As arguments can become lost within other details, this chapter gives practice in identifying more easily the material relevant to the main argument. Such skills are also useful for improving reading speed and accuracy and in helping you to identify whether your own writing has a sufficiently critical focus.

Chapter 5 focuses on the quality of reasoning. It gives you practice in evaluating how well authors present their arguments in terms of structure, logical order, internal consistency, the way in which reasons are used to support each other, and the use of interim conclusions. Understanding the structure of an argument is beneficial both in making reading faster and more effective, and in structuring your own arguments.

Chapters 6 and 7 develop skills in analysing the details of an argument. These skills help you to read texts and interpret arguments at a deeper rather than a superficial level. This is especially important for evaluating academic arguments or, for example, checking that you understand the implications of contracts in the workplace or the nuances of political arguments used at election time. As you develop these skills, you will be better able to engage in debating the issues raised by experts or by specialist authors, checking whether they are consistent in what they are saying and whether their arguments contain flaws that are not immediately obvious.

Chapter 6 focuses on 'reading between the lines', identifying aspects of the author's position and argument that are not directly stated. These include underlying assumptions and 'implicit arguments'. The chapter also looks at what is meant by the 'premises' on which arguments are predicated and at identifying 'false premises'. Finally, it examines what is meant by denoted and connoted meanings, and the importance of identifying hidden connotations within an argument.

Chapter 7 provides a different perspective on evaluating an argument, this time focusing on flaws within the reasoning. It looks at confusions that are made between cause and effect, and introduces the concept of 'meeting necessary and sufficient conditions'. It also introduces the most common types of flawed argument, such as false analogies, unfair use of emotive language, tautology, and misrepresentation.

Chapter 8 focuses on finding and evaluating sources of evidence to support an argument. It examines the difference between primary and secondary sources, looks at how to conduct a literature search, and provides criteria for evaluating and selecting different kinds of evidence. Concepts such as authenticity, validity, currency and reliability are introduced. The chapter also looks at a range of methods used to ensure the evidence is robust, such as checking for representative sample sizes and levels of probability, and triangulating evidence.

Chapter 9 looks at specific ways of applying critical thinking to reading and note-making, such as orientating to the task of critical reading, making accurate interpretations, and categorising and selecting material in order to make the process of reading and note-making more effective. It examines the relationship of theory to argument, and looks at ways of categorising theories in order to ease comparison between different arguments. The chapter also emphasises the importance of noting the sources of evidence, as an essential aspect of critical note-making.

The next two chapters focus on the application of critical thinking to the act of writing.

Chapter 10 looks at characteristics of critical writing, and especially the importance of maintaining a focus on your potential readers, setting the scene for them. It gives details about how to use language to structure and signpost arguments so that the reader is clear about which stage of the argument is being presented and the direction of your argument. Critical writing uses tentative language to express conclusions and this is also examined. The chapter looks specifically at how students can apply what they have learnt about critical thinking to each stage of writing their own essays.

Chapter 11 provides an opportunity to evaluate two critical essays. The emphasis in this chapter is not on identifying and evaluating arguments, but rather on evaluating longer texts as pieces of critical writing. The two essays differ in how effective they are at applying the conventions required for critical, analytical writing. Checklists and commentaries are provided to help you approach the task and to evaluate your responses. A further checklist is provided as an optional tool for you to use, or adapt, to evaluate your own critical writing. Additional practice activities are provided on the companion website at www.palgravehighered.com/cts3e.

Chapter 12 looks at critical thinking from a different perspective. Critical reflection is used, increasingly, within professional practice and for student assignments. The chapter provides practical means of addressing this challenging form of critical activity, taking you through the steps of planning your reflection, relating personal experience to theory and practice in a critical way, and presenting these skilfully for assessment.

Chapter 13 is about applying critical thinking skills when looking for work and applying for jobs. It looks at the different ways that critical thinking is relevant to your career path, from the way you consider your options, through to making a critical evaluation of your own job applications so that these give you the best chance of success. The chapter looks at employers' need for employees who can apply critical abilities, and at where job applicants go wrong in failing to demonstrate such skills.

Critical reflection

As with all academic work and professional good practice, you will benefit from reflecting upon the points raised in each chapter and, in particular, your own current ways of approaching these. Some chapters provide prompts to assist such reflection.

You are likely to gain more from using the book if, as you work through a section, you pause to consider from time to time how that aspect of critical awareness would benefit your own study, writing or professional work.

It is well worth taking such time to pause and consider the implications of the key points in order to help you see the significance and relevance of the materials and critical strategies to your own work or study.



Using critical reflection

At various points in the text, you are encouraged to pause and reflect on a particular point. You may find it is beneficial to jot down your thoughts as this can help you to formulate and clarify your thinking. It is useful to have a light notebook for your reflective activity, or to set up a folder or a notes page for this purpose on your device.

Glossary

When we discuss arguments, a number of specific terms are sometimes employed. Some that are useful to know in the initial stages of learning about critical thinking are listed below.

Argument Using reasons to support a point of view, so that known or unknown audiences may be persuaded to agree. An argument may include disagreement, but is more than simply disagreement if it is based on reasons.

Argument – the overall argument The overall argument presents the author's position. It is composed of contributing arguments, or reasons. The term 'line of reasoning' is used to refer to a set of reasons, or contributing arguments, structured to support the overall argument.

Arguments – contributing arguments Individual reasons are referred to as arguments or 'contributing arguments'.

Assertions Statements which are made without providing any supporting evidence or justification. These may turn out to be true or untrue.

Conclusion Reasoning should lead towards an end point, which is the conclusion. The conclusion should normally relate closely to the author's main position. In critical thinking, a conclusion is usually a deduction drawn from the reasons, or evidence. The final section of an essay is also referred to as the conclusion. This is examined in detail on page 166.

Conclusion – intermediate conclusions The author may draw interim conclusions during the course of an argument, before arriving at final conclusions. Each interim conclusion is based on only some of the evidence or a particular set of reasons. These intermediate conclusions may be used to provide evidence, or to serve as reasons, in the next stage of the argument.

Consistency – internal consistency An argument is *internally consistent* when all parts of the line of reasoning contribute to the conclusion. Nothing then contradicts or undermines the main message. An argument may be internally consistent but still be inconsistent in other respects, such as not being consistent with the evidence or with the opinions of experts in the field.

Consistency – logical consistency An argument is logically consistent when the reasons are provided in a logical manner – that is, in the best order, with each linked to previous or following arguments so as to build up a case. A logically consistent argument will be internally consistent. In a logically consistent argument, the reasons support the conclusion.

Discursive Discursive writing develops and elaborates an argument, moving successively from one point to the next in a given direction, towards conclusions. It does this in a thoughtful way that engages critically with the evidence base and the theories and arguments of others, drawing out implications and significance.

Line of reasoning The line of reasoning is established through the order in which reasons and evidence are presented. This order should make it clear to the reader how the argument is to be interpreted and what the structure of the argument is. The line of reasoning should lead forwards with a clear direction, with one piece of reasoning leading in an obvious way to the next, rather than hopping from one point to another in a random way, or leading the audience round in circles.

Logical order Good arguments present reasons and evidence in a structured way, so that information builds on what has already been said. See 'line of reasoning'.

Position A point of view, supported by reasoning.

Predicate The foundation of the argument; the aims of the argument; an underlying point of

view; the assumption that underlies the argument. For example: *the argument was predicated on a Marxist interpretation of wealth; the programme was predicated on the assumption that the prisoner was innocent.*

Premises Propositions believed to be true and used as the bases for the argument; the basic building blocks for the argument – that is, the reasons for believing that the conclusion is true. Premises that are not well founded are referred to as *false premises.*

Propositions Statements believed to be true and presented as arguments or reasons for consideration by the audience. A proposition may turn out to be true or false.

Reasons Contributing arguments put forward to support the overall argument or line of reasoning.

Reasons – independent reasons The author may use several reasons to support the conclusion, each of which may be valid in its own right but may have nothing to do with the other reasons given.

Reasons – joint reasons Reasons provided to support an argument when they are connected in some way and mutually reinforce each other.

Salience 'Salient' simply means 'relevant to the argument'.

Substantive point The central point that is being made, or the core of the argument. This expression is used to focus attention on the main point, especially if an argument has been diverted towards more minor issues and when the key message is becoming obscured.

Tautology Unnecessary repetition, when the author makes the same point but in different words. For example, in poor arguments, a tautology may be used to make it appear as if there are two reasons to support a conclusion, when the first reason has merely been reproduced in a different way.

Example of key terms used together

- *Proposition 1*: One of the expedition team is suspected of having pneumonia.
- *Proposition 2*: A serious storm has been predicted in the area.
- *Proposition 3*: The mountainside can be dangerous during some storms.
- *Proposition 4*: Some members of the team are not familiar with the area or with mountaineering.
- *Conclusion*: It isn't a good moment to launch an expedition into the mountains.

Premises

It is not a good time for the expedition to go into the mountains as a storm is expected and some of the team may not have the health or experience to cope with this.

False premises

The argument against launching the expedition sounds convincing. However, it could be based on false premises: a storm may not be due, the dangers might be exaggerated, or the team may be more experienced than described, or the team member may have only a minor cold. In that case, the argument against launching the expedition would be based on false premises.

Predicate

The argument against the expedition is predicated on an assumption that the safety of the team should take priority over the requirements of the expedition.

Salience

The question of safety is salient to the debate about whether to launch the expedition. Other things may not be salient to that argument. For example, the facts that a team member was good at sports at school 20 years ago, or had hiccups yesterday, are probably not salient to the discussion.

Acknowledgements

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Chapter 1
What is critical thinking?

Learning outcomes

This chapter gives you opportunities to:

- understand what critical thinking is
- recognise some of the benefits associated with critical thinking skills
- recognise the personal qualities associated with critical thinking
- recognise barriers to the development of good critical thinking skills
- assess your current understanding of critical thinking and identify your priorities for improvement

Introduction

This chapter provides a general orientation to critical thinking. It examines what is meant by 'critical thinking', the skills associated with it, and the barriers that can hinder effective development of critical approaches. Many people can find it difficult to order their thoughts in a logical, consistent and reasoned way. This book starts from the premise that skills in reasoning can be developed through a better understanding of what critical thinking entails, and by practice.

Critical thinking is a cognitive activity, associated with using the mind. Learning to think in critically analytical and evaluative ways means using mental processes such as attention, categorisation, selection and judgement. However, many people who have the potential to develop more effective critical thinking can be prevented from doing so for a variety of reasons apart from a lack of ability. In particular, personal and emotional, or 'affective', reasons can create barriers.

You are invited to consider, in this chapter, how far such barriers could be affecting your own thinking abilities and how you will manage these.



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Critical thinking as a process

Critical thinking is a complex process of deliberation which involves a wide range of skills and attitudes. It includes:

- *identifying other people's positions*, arguments and conclusions;
- *evaluating the evidence* for alternative points of view;
- *weighing up opposing arguments* and evidence fairly;
- *being able to read between the lines*, seeing behind surfaces, and identifying false or unfair assumptions;
- *recognising techniques* used to make certain positions more appealing than others, such as false logic and persuasive devices;
- *reflecting on issues* in a structured way, bringing logic and insight to bear;
- *drawing conclusions* about whether arguments are valid and justifiable, based on good evidence and sensible assumptions;
- *synthesising information*: drawing together your judgements of the evidence, synthesising these to form your own new position;
- *presenting a point of view* in a structured, clear, well-reasoned way that convinces others.

Scepticism and trust

Ennis (1987) identified a range of dispositions and abilities associated with critical thinking. These focused on:

- the ability to reflect sceptically;
- the ability to think in a reasoned way.

Scepticism in critical thinking means bringing an element of polite doubt. In this context, scepticism doesn't mean you must go through life never believing anything you hear and see. That would not be helpful. It does mean holding open the possibility that what you know at a given time may be only part of the picture.

Critical thinking gives you the tools to use scepticism and doubt constructively so that you can analyse what is before you. It helps you to make better and more informed decisions about whether something is likely to be true, effective or productive. Ultimately, in order to function in the world, we have to accept the probability that at least some things are as they seem. This requires trust. If we can analyse clearly the basis of what we take as true, we are more able to discern when it is reasonable to be trusting and where it is useful to be sceptical.

Method rather than personality trait

Some people seem to be more naturally sceptical whilst others find it easier to be trusting. These differences may be because of past experiences or personality traits. However, critical thinking is not about natural traits or personality; it is about a certain set of methods aimed at exploring evidence in a particular way. Sceptical people can require structured approaches that help them to trust in the probability of an outcome, just as those who are more trusting require methods to help them use doubt constructively.

Critical thinking and argument

The focus of critical thinking is often referred to as the 'argument'. Chapter 3 identifies the features of an argument in critical thinking. The argument can be thought of as the message that is being conveyed, whether through speech, writing, performance or other media. Critical thinking helps you to identify the obvious and the hidden messages more accurately, and to understand the process by which an argument is constructed.

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Knowing our own reasons

Critical thinking is associated with *reasoning* or with our capacity for *rational* thought. The word 'rational' means 'using reasons' to solve problems. Reasoning starts with ourselves. It includes:

- having reasons for what we believe and do, and being aware of what these are;
- critically evaluating our own beliefs and actions;
- being able to present to others the reasons for our beliefs and actions.

This may sound easy, as we all assume we know what we believe and why. However, sometimes, when we are challenged on why we believe that something is true, it becomes obvious to us that we haven't really thought through whether what we have seen or heard is the whole story or is just one point of view. There are also likely to be occasions when we find we are not sure what we consider to be the right course of action or a correct interpretation.

It is important to examine the basis of our own beliefs and reasoning, as these will be the main vantage points from which we begin any critical analysis.

Challenging our own assumptions

Our brains like to assume that they are right. Research has shown that we are wired to make quick assumptions – to take the easiest route to jump to the most likely conclusion rather than to slow down and examine our reasoning (Kahneman, 2011). This means we can easily miss essential information and omit relevant considerations.

Focusing on our reasons, and examining the foundations of these systematically, can help us uncover our assumptions. When we are more aware of these, we can test them out, too, in a systematic way.

Critical analysis of other people's reasoning

Critical reasoning usually involves considering other people's reasoning. This requires the skill of grasping an overall argument, but also skills in analysing and evaluating it in detail.

Critical analysis of other people's reasons can involve:

- identifying their reasons and conclusions;
- analysing how they select, combine and order reasons to construct a line of reasoning;
- evaluating whether their reasons support the conclusions they draw;
- evaluating whether their reasons are well founded, based on good evidence;
- identifying flaws in their reasoning.

Constructing and presenting reasons

Reasoning involves analysing evidence and drawing conclusions from it. The evidence may then be presented to support the conclusion. For example, we may consider that it is a cold day. Someone who disagrees may ask why we believe this. We may use evidence such as a thermometer reading and observation of weather conditions. Our reasons may be that the temperature is low and there is ice on the ground.

We use basic examples of reasoning such as this every day. For professional and academic work, we are usually required to present our reasoning using formal structures such as essays, or reports with recommendations. This requires additional skills such as knowing how to:

- select and structure reasons to support a conclusion;
- present an argument in a consistent way;
- use logical order;
- use language effectively to present the line of reasoning.

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Why do critical thinking skills matter?

Sharpening our minds

As we have seen, we often assume that we have the full story, the right answer, or the best solution when that is not the case. It is easy to slip into simply repeating something we have heard, or describing what we have read, without much thought. We may consider that we are using critical skills when we are merely stating what we believe to be self-obviously true.

Such thinking leads to mistakes, weak understandings, unconscious bias, unfairness, and errors of judgement. Most of these won't be significant, but some could have serious consequences. Critical awareness sharpens our minds so that we are better able to identify where we need to slow down and apply more systematic critiques of our thinking processes and actions.

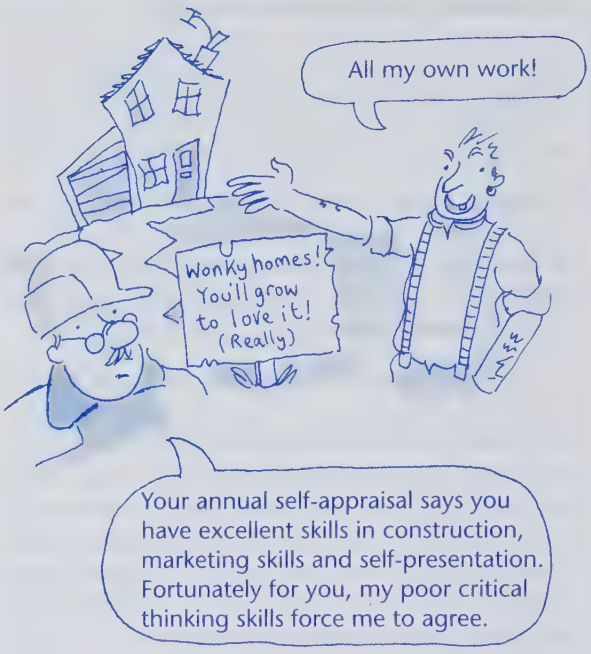
For academic and professional life

Advances in knowledge and professional practice are made through recognising where improvements can be made to what has gone before. This involves being able to break down existing understandings and practice into their component parts, such as what are assumed to be facts, or good evidence, or sound methods, or the assumptions made about how different pieces of information are connected.

Academic study and research-based enquiry require us to slow down our processing of information. The methodologies used for conducting research, and the feedback received from peers, help in identifying flaws in the way we arrive at conclusions. That has an impact on the speed, accuracy, efficiency, and fairness of our thoughts and actions.

Realistic self-appraisal

Good critical thinking skills, if applied well, can help us to make much more realistic and accurate appraisals of our own abilities, interests and thinking processes. This is useful in helping us to make decisions about where to focus our energies when looking for work, pursuing further training, or making life choices.



Benefits of good critical abilities

1. Ability to spot your own and other people's assumptions.
2. Ability to spot inconsistencies and potential errors that merit further investigation.
3. Ability to make fair, sound decisions.
4. Less likelihood of being misled or cheated.
5. Ability to notice what is relevant and significant – so saving time and effort.
6. Ability to bring greater accuracy and precision to different parts of a task.
7. Clearer thinking and communication.
8. Better problem-solving skills, such as in identifying where improvements could be made and evaluating potential solutions.
9. Ability to take a systematic approach, to ensure essentials are not overlooked.
10. Greater speed and accuracy in analysing complex information.
11. Confidence in taking on more complex problems and challenges.
12. Possibility of seeing the world through different eyes – with sharper awareness.

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Underlying skills and attitudes

Critical thinking rarely takes place in a vacuum. Higher-level critical thinking skills usually require some or all of the skills and attitudes listed below.

Underlying thinking skills

Critical thinking assumes abilities in a range of skills such as categorising, selection and differentiation, comparing and contrasting. These skills are examined in Chapter 2.

Knowledge and research

Good critical thinkers can often detect a poor argument without a good knowledge of the subject. However, critical thinking usually benefits from background research. Finding out more about a subject helps you to make a more informed judgement about whether relevant facts, alternative explanations and options have been covered sufficiently.

Emotional self-management

Critical thinking sounds like a dispassionate process but it can engage emotions and even passionate responses. This should not surprise us when we consider that reasoning requires us to decide between opposing points of view. In particular, we may not like evidence that contradicts our own opinions or beliefs. If the evidence points in a direction that is unexpected and challenging, that can rouse unexpected feelings of anger, frustration or anxiety.

The academic world traditionally likes to consider itself as logical and immune to emotions, so if feelings do emerge, this can be especially difficult. Being able to manage your emotions under such circumstances is a useful skill. If you can remain calm, and present your reasons logically, you will be better able to argue your point of view in a convincing way.

Perseverance, accuracy and precision

Critical thinking involves accuracy and precision and this can require dedication to finding the right answer. It includes:

- *Attention to detail*: taking the time to note small clues that throw greater light on the overall issue.
- *Identifying trends and patterns*: this may be through careful mapping of information, analysis of data, or identifying repetition and similarity.
- *Repetition*: going back over the same ground several times to check that nothing has been missed.
- *Taking different perspectives*: looking at the same information from several points of view.
- *Objectivity*: putting your own likes, beliefs and interests to one side with the aim of gaining the most accurate outcome or a deeper understanding.
- *Considering implications and distant consequences*: what appears to be a good idea in the short term, for example, might have long-term effects that are less desirable.



Reflection: Emotional self-management

For me, the emotions that are most difficult to manage when others disagree with me are:

I deal with these by:

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Self-awareness for accurate judgement

Good critical thinking involves making accurate judgements. We noted above that our thinking might not be accurate if we are not fully aware of the influences that affect it. These can include such things as our own assumptions, preconceptions, bias, dislikes, beliefs, things we take for granted as normal and acceptable, and all those things about our selves and our world that we have never questioned.

People who are outstanding at critical thinking tend to be particularly self-aware. They reflect upon and evaluate their personal motivations, interests, prejudices, expertise and gaps in their knowledge. They question their own point of view and check the evidence used to support it.



Reflection: Influences on my thinking



For me, the influences on my own thinking that I need to be most aware of so they don't prejudice my thinking are:

I will deal with these by:

Becoming more self-aware takes courage. It can be unsettling to find out things about ourselves we didn't know, as most of us like to think we know ourselves very well. It is also challenging to question our belief systems. We think of these as part of our identity and it can be unsettling if we feel our identity is called into question.

Furthermore, the result of your critical thinking might place you in a minority amongst your friends, family or colleagues. Nobody else might interpret the evidence in the same way as you. It takes courage to argue an alternative point of view, especially when it is possible that you might be wrong.



Reflection: Challenging opinions



For me, the things I find most difficult about challenging the opinions of other people are:

I deal with these by:

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Personal strategies for critical thinking

Below, three lecturers describe how they view critical thinking.

Example 1

- I may make a quick first reading to get the overall picture and check my initial response. I see whether it rings true or contradicts what I believe to be true.
- I compare what I read with what I already know about the topic and with my experience.
- I summarise as I go along, and hold the overall argument in my head to make sense of what comes next.
- I look for the author's position or point of view, asking 'What are they trying to "sell me"?'
- As I read, I check each section and ask myself if I know what it means. If not, I check again – sometimes it is clearer when I read the second time. If it is still unclear, I remind myself to come back to it later as the rest of the passage may make it clearer.
- I then read more carefully, seeing what reasons the writers present and checking whether I am persuaded by these.
- If I am persuaded, I consider why. Is it because they make use of experts in the field? Is there research evidence that looks thorough and convincing?
- If I am not persuaded, then why not? I check whether this is a 'gut level' thing or whether I have good reasons for not being convinced. If I have relied on a gut response, I check for hard evidence such as whether I have read other material that contradicts it.
- I then create my own position, and check that my own point of view is convincing. Could I support it if I was challenged?

Here the lecturer is describing an overall strategy for reading and analysing the text in a critically analytical way. The final point refers to 'creating' a personal position by synthesising the available material – and then submitting this to critical analysis too.

Example 2 indicates that, as well as the words on the page or material being critiqued, there are wider contextual and other considerations to be taken into account.

Example 2

I put my energy into looking for the heart of the issue: what is really being said, and why? The answers may not be on the page; they may be in the wider history of a debate, a cultural clash, or conflicting bids for project money. It is surprising how often the wider context, popular debates, even a desire to be seen to be saying what is currently in fashion, have a bearing on what a given passage is really saying.

The third lecturer wouldn't disagree with what has gone before, but adds another dimension. Analysis encourages a focus on the detail, and on considering many different angles. This can generate a large body of evidence or a long list of points for consideration. An important aspect of your critical analysis is to sift through this wealth of information, and make good judgements about what is the most significant.

Example 3

The trick is being able to see the wood for the trees; identifying what is relevant amongst a mass of less relevant information. It isn't enough just to understand; you have to be constantly evaluating whether something is accurate, whether it gets to the heart of the issue, whether it is the most important aspect on which to focus, whether it is the best example to use – and whether what you are saying about it is a fair representation of it.

All three examples illustrate different aspects of the critical thinking process:

- an analytical strategy for the material;
- understanding of the wider context;
- an evaluative and selective approach;
- being self-critical about your own understanding, interpretation and evaluation.

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Development of understanding

Students are expected to develop critical thinking skills so that they can dig deeper below the surface of the subject they are studying and engage in critical dialogue with its main theories and arguments. This is usually through engaging in critical debate in seminars, presentations or writing produced for assessment or publication.

One of the best ways of arriving at a point where we really understand something is by doing, or replicating, the underlying research for ourselves. However, as undergraduates, and indeed in everyday life, there simply isn't time to research everything we encounter. The depth of understanding that comes through direct experience, practice and experimentation has to be replaced, at times, by critical analysis of the work of other people.

Students need to develop the ability to evaluate critically the work of others. Whilst some find this easy, others tend to accept or apply the results of other people's research too readily, without analysing it sufficiently to check that the evidence and the reasoning really support the main points being made. Bodner (1988), for example, describes chemistry students as being unable to 'apply their knowledge outside the narrow domain in which it was learnt. They "know" without understanding.' Bodner suggests that, instead of focusing primarily on standard chemical calculations in books, students should be looking for answers to questions such as 'How do we know ...?' and 'Why do we believe ...?'

Bodner's description is likely to be just as true of students in other subjects. It is not unusual for students, and for people generally, to rely unquestioningly on research that is based on a small sample of the population, or that is based on faulty reasoning, or that is now out of date. Evidence from small or isolated projects is often treated as if it were irrefutable proof of a general principle, and is sometimes quoted year after year as if it were an absolute truth. Chapter 8 looks further at critically examining and evaluating evidence.



Reflection: 'Knowing without understanding?'



Do you recognise anything of yourself in Bodner's description of students? What effect would the approach he suggests have on your learning and understanding?

Both positives and negatives

In academic contexts, 'criticism' refers to an analysis of positive features as well as negative ones. It is important to identify strengths and satisfactory aspects rather than just weaknesses, to evaluate what works as well as what does not. Good critical analysis accounts for *why* something is good or poor, why it works or fails. It is not enough merely to list good and bad points.

Comprehensive: Nothing is excluded

For most academic programmes, students are expected to take a well-reasoned, evidence-based, critical approach to what they hear, see, read, and learn. That is the case even when considering the work of respected academics.

Normally, any theory, perspective, data, area of research or approach to a discipline could be subjected to critical analysis. Some colleges, such as religious foundations, may consider certain subjects to be out of bounds, but this is not typical.

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The idea or the action, not the person

A distinction is usually drawn between the idea, work, text, theory or behaviour, on the one hand and, on the other, the person associated with these. This is also true when making critical analyses of other students' work, if this is a requirement of your course. Even so, it is worth remembering that people identify closely with their work and may take criticism of it personally. Tact and a constructive approach are needed. Giving difficult messages in a way other people can accept is an important aspect of critical evaluation.



Irma wasn't famed for her tact.

Non-dualistic

In our day-to-day lives, we can slip into thinking everything is right or wrong, black or white. In the academic world, answers may occur at a point on a continuum of possibilities. One of the purposes of higher-level thinking is to address questions which are more complicated and sophisticated, and which do not lend themselves to straightforward responses. You may have noticed yourself that the more you know about a subject, the more difficult it becomes to give simple answers.

Dealing with ambiguity and doubt

With the internet at our fingertips, we are used to obtaining answers within minutes of formulating a question. However, in the academic world, questions are raised in new areas and answers may not be found for years, or even lifetimes. This can feel uncomfortable if you are used to ready answers.

This does not mean, though, that vague answers are acceptable. If you look at articles in academic journals, you will see that they are very closely argued, often focusing on a minute aspect of the subject in great detail, and with precision. Students, too, are expected to develop skills in using evidence, even if drawn from other people's research, to support a detailed line of reasoning.

It is worth remembering that in academic work, including professional research for business and industry, researchers often need to pursue lines of enquiry knowing that:

- no clear answers may emerge;
- it may take decades to gain an answer;
- they may contribute only a very small part to a much larger picture.

Critical thinking as a student means:

- finding out where the best evidence lies for the subject you are discussing;
- evaluating the strength of the evidence to support different arguments;
- coming to an interim conclusion about where the available evidence appears to lead;
- constructing a line of reasoning to guide your audience through the evidence and lead them towards your conclusion;
- selecting the best examples;
- providing evidence to illustrate your argument.

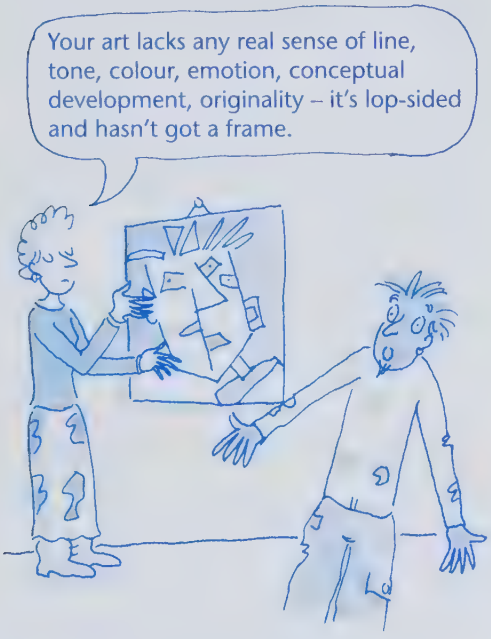
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Barriers to critical thinking (1)

Critical thinking does not come easily to everyone. Barriers vary from person to person, but can usually be overcome. This section looks at some key barriers to critical thinking and encourages you to consider whether these might be having an impact on you.

Misunderstanding what is meant by criticism

Some people assume that 'criticism' means making negative comments. As a result, they refer only to negative aspects when making an analysis. This is a misunderstanding of the term. As we saw above, critical evaluation means identifying positive as well as negative aspects, what works as well as what does not.



Others feel that it is not good to engage in criticism because it is an intrinsically negative activity. Some worry that they will be regarded as an unpleasant sort of person if they are good at criticism. As a result, they avoid making any comments they feel are negative and make only positive comments. They may not provide feedback on what can be improved. This is often an unhelpful approach, as constructive criticism can clarify a situation and help people to excel.



Over-estimating our own reasoning abilities

Most of us like to think of ourselves as rational beings. We tend to believe our own belief systems are the best (otherwise we wouldn't hold those beliefs) and that we have good reasons for what we do and think.

Although this is true of most of us for some of the time, it isn't an accurate picture of how humans behave. Most of the time our thinking runs on automatic. This makes us more efficient in our everyday lives: we don't have to doubt the safety of a tooth-brush every time we brush our teeth.

However, it is easy to fall into poor thinking habits. People who get their own way, or simply get by, with poor reasoning, may believe their reasoning must be good as nobody has said it isn't. Those who are good at winning arguments can mistake this for good reasoning ability. Winning an argument does not necessarily mean that you have the best case. It may simply mean that your opponents didn't recognise a poor argument, or chose to yield the point for their own reasons, such as to avoid conflict. Imprecise, inaccurate and illogical thinking does not help to develop the mental abilities required for higher-level academic and professional work.

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Barriers to critical thinking (2)

Lack of methods, strategies or practice

Although willing to be more critical, some people don't know which steps to take next in order to improve their critical thinking skills. Others are unaware that strategies used for study at school and in everyday situations are not sufficiently rigorous for higher-level academic thinking and professional work. With practice, most people can develop their skills in critical thinking.

Reluctance to critique experts

There can be a natural anxiety about critically analysing texts or other works by people that you respect. It can seem strange for students who know little about their subject, to be asked to critique works by those who are clearly more experienced. Some students can find it alien, rude or nonsensical to offer criticism of practitioners they know to be more expert than themselves.

If this is true of you, it may help to bear in mind that this is part of the way teaching works in most universities. Critical analysis is a typical and expected activity. Researchers and lecturers expect students to question and challenge even published material. It can take time to adapt to this way of thinking.

If you are confident about critical thinking, bear in mind that there are others who find this difficult. In many parts of the world, students are expected to demonstrate respect for known experts by behaviours such as learning text off by heart, repeating the exact words used by an expert, copying images precisely, or imitating movements as closely as possible. Students of martial arts such as tai chi or karate may be familiar with this approach to teaching and learning.

Affective reasons

We saw above that emotional self-management can play an important part in critical thinking. To be able to critique means being able to acknowledge that there is more than one way of looking at an issue. In academic contexts, the implications of a theory can challenge deeply held beliefs and long-held assumptions. This can be difficult to accept, irrespective of how intelligent a student might be.



This is especially so if 'common sense' or 'normality' appears to be challenged by other intelligent people or by academic research. It can be hard to hear deeply held religious, political and ideological beliefs challenged in any way at all. Other sensitive issues include views on bringing up children, criminal justice, genetic modification, and sexuality.

When we are distressed by what we are learning, the emotional response may help to focus our thinking, but very often it can inhibit our capacity to think clearly. Emotional content can add power to an argument, but it can also undermine an argument, especially if emotions seem to take the place of the reasoning and evidence that could convince others. Critical thinking does not mean that you must abandon beliefs that are important to you. It may mean giving more consideration to the evidence that supports the arguments based on those beliefs, so that you do justice to your point of view.

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Mistaking information for understanding

Learning is a process that develops understanding and insight. Many lecturers set activities to develop expertise in methods used within the discipline. However, students can misunderstand the purpose of such teaching methods, preferring facts and answers rather than learning the skills that help them to make well-founded judgements for themselves.

Cowell, Keeley, Shemberg and Zinnbauer (1995) write about 'students' natural resistance to learning to think critically', which can mean acquiring new learning behaviours. Cowell et al. outline the problem through the following dialogue:

> Student: *'I want you (the expert) to give me answers to the questions; I want to know the right answer.'*
>
> Teachers: *'I want you to become critical thinkers, which means I want you to challenge experts' answers and pursue your own answers through active questioning. This means lots of hard work.'*

If you feel that critical thinking is hard work at times, then you are right. There are lecturers who would agree with you. However, if it wasn't difficult, you would not be developing your thinking skills into new areas. In effect, you are developing your 'mental muscle' when you improve your critical thinking skills.

Insufficient focus and attention to detail

Critical thinking involves precision and accuracy, and this, in turn, requires good attention to detail. Poor criticism can result from making judgements based on too general an overview of the subject matter. Critical thinking activities require focus on the exact task in hand, rather than becoming distracted by other interesting tangents.

When critically evaluating arguments, it is important to remember that you can find an argument to be good or effective even if you don't agree with it.

Which barriers have an effect upon you?

On the table below, tick (✓) all those barriers that you consider might be affecting your critical thinking abilities.

Barrier	Has an effect?
Misunderstanding what is meant by criticism (page 10)	
Over-estimating my reasoning abilities (page 10)	
Lack of methods and strategies (page 11)	
Lack of practice (page 11)	
Reluctance to criticise those with more expertise (page 11)	
Affective reasons (page 11)	
Mistaking information for understanding (page 12)	
Insufficient focus and attention to detail (page 12)	

Reflection: Managing barriers

Consider what you could do to manage these barriers in the next few months.

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Self-evaluation

For each of the following statements, rate your responses as outlined below.
Note that 'strongly disagree' carries no score.

4 = 'strongly agree' 3 = 'agree' 2 = 'sort of agree' 1 = 'disagree' 0 = 'strongly disagree'

		Rating 4–0
1	I feel comfortable pointing out potential weaknesses in the work of experts.	2
2	I can remain focused on the exact requirements of an activity.	3
3	I know the different meanings of the word 'argument' in critical thinking.	3
4	I can analyse the structure of an argument.	2
5	I can offer criticism without feeling this makes me a bad person.	2
6	I know what is meant by a line of reasoning.	1
7	I am aware of how my current beliefs might prejudice fair consideration of an issue.	3
8	I am patient in identifying the line of reasoning in an argument.	2
9	I am good at recognising the signals used to indicate stages in an argument.	3
10	I find it easy to separate key points from other material.	3
11	I am very patient in going over the facts in order to reach an accurate view.	2
12	I am good at identifying unfair techniques used to persuade readers.	3
13	I am good at reading between the lines.	2
14	I find it easy to evaluate the evidence to support a point of view.	3
15	I usually pay attention to small details.	3
16	I find it easy to weigh up different points of view fairly.	3
17	If I am not sure about something, I will investigate to find out more.	1
18	I can present my own arguments clearly.	2
19	I understand how to structure an argument.	2
20	I can tell descriptive writing from analytical writing.	3
21	I can spot inconsistencies in an argument easily.	2
22	I am good at identifying patterns.	3
23	I am aware of how my own up-bringing might prejudice fair consideration of an issue.	3
24	I know how to evaluate source materials.	3
25	I understand why ambiguous language is often used in research papers.	3
	Score out of 100	62

Interpreting your score

Going through the questionnaire may have raised some questions about what you know or don't know about critical thinking. The lower the score, the more likely you are to need to develop your critical thinking skills. A score over 75 suggests you are very confident about your critical thinking ability. It is worth checking this against objective feedback from your tutors or colleagues, for example. If your score is less than 100, there is still room for improvement! If your score is under 45 and remains so after completing the book, you may find it helpful to speak to an academic counsellor, your tutor or a supervisor to root out the difficulty.

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Priorities: Developing critical thinking abilities

- In column A, identify which aspects of critical thinking you want to know more about. Give a rating between 5 and 0, giving 5 for 'very important' and 0 for 'not important at all'.
- In column B, consider how essential it is that you develop this aspect soon. Give a rating between 5 and 0, where 5 is 'very essential' and 0 is 'not essential at all'.
- Add the scores in columns A and B to gain an idea of where your priorities are likely to lie.
- Column D directs you where to look for more information on that point.

Aspects I want to develop further **I want to:**	**A** **Want to know more?** Rate from 0 to 5 5 = 'very important'	**B** **How essential to develop it now?** Rate from 0 to 5 5 = 'very essential'	**C** **Priority score** Add scores for columns A and B	**D** **See Chapter**
1 understand the benefits of critical thinking	1	1	2	1
2 remain focused on the exact requirements of an activity	3	2	5	2 ✓
3 pay better attention to small details	1	1	2	2
4 know what is meant by a line of reasoning	4	4	8	3 ✓
5 identify the component parts of an argument for critical thinking	1	0	1	3
6 recognise the words used to signal stages in an argument	1	0	1	3, 10
7 distinguish argument from disagreement	0	0	0	4
8 distinguish argument from summaries, descriptions and explanations	0	0	0	4
9 pick out the key points from background information	0	0	0	4
10 be able to analyse the structure of an argument	0	0	0	5, 10
11 evaluate whether arguments are internally consistent	2	3	5	5 ✓
12 understand what is meant by an intermediate conclusion	3	3	6	5 ✓
13 be able to structure an argument	3	2	5	5, 10, 11 ✓

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Aspects I want to develop further I want to:	A Want to know more? Rate from 0 to 5 5 = 'very important'	B How essential to develop it now? Rate from 0 to 5 5 = 'very essential'	C Priority score Add scores for columns A and B	D See Chapter
14 be better at reading between the lines	4	4	8	6 ✓
15 recognise underlying assumptions	1	1	2	6
16 recognise when an argument is based on false premises	1	1	2	6
17 recognise implicit arguments	—	—	—	6
18 understand what is meant by denoted and connoted meanings	3	3	6	6
19 be aware of how cause, effect, correlation and coincidence can be confused				7
20 be able to check for 'necessary and sufficient conditions'				7
21 identify unfair techniques used to persuade readers				6, 7
22 recognise tautology				7
23 recognise flawed reasoning				6, 7
24 be able to evaluate source materials				1, 8
25 understand what is meant by authenticity, validity, and reliability				8
26 evaluate when samples are representative				8
27 understand what is meant by 'triangulation'				8
28 check for levels of probability				8
29 apply critical thinking when making notes				9, 10
30 use language more effectively to structure argument				3, 10, 11
31 present my own arguments clearly in writing/for essays				10, 11
32 be able to undertake good critical reflection for assignments or my job				12

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Priorities for action

- Look back over the priorities table. Identify the three aspects to which you gave the highest scores. If more than three have the highest score, select three to start with.
- Write the three priorities here as actions starting with 'I will …', using words that are meaningful to you – e.g. 'I will find out what tautology means.'

1. I will	
2. I will	
3. I will	

Summary

Critical thinking is a process that relies upon, and develops, a wide range of skills and personal qualities. Like other forms of activity, it improves with practice and with a proper sense of what is required. For some people, this may mean changing behaviours, such as paying attention to detail or taking a more sceptical approach to what they see, hear, and read. Some need to focus on developing critical thinking techniques, and this is the main purpose of the book.

For others, weaknesses in critical thinking abilities may stem from attitudes to criticism, and anxiety about potential consequences. Barriers associated with attitudinal and affective responses to critical approaches were considered in this chapter. Sometimes, it is sufficient to become more aware of these barriers, and to recognise the blocks to effective thinking, for the anxiety to subside.

If you find that these difficulties persist, it is worth speaking to a student counsellor about your concerns. They will be familiar with such responses and may be able to help you to find a solution that fits your personal circumstances.

Developing good critical thinking skills can take patience and application. On the other hand, the rewards lie in improved abilities in making judgements, seeing more easily through flawed reasoning, making choices from a more informed position and improving your ability to influence others.

Having undertaken an initial personal evaluation of your critical thinking skills, you may now wish to follow up the priorities you identified. This is a particularly useful approach if you have already worked on your critical thinking skills. If you are new to critical thinking, you may find it useful to progress directly to Chapter 2 in order to test, and practise, your underlying thinking skills. Alternatively, proceed now to Chapter 3 and work through the chapters in turn, to develop a systematic approach to enhancing your critical thinking abilities.

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Chapter 2

How well do you think?

Develop your thinking skills

Learning outcomes

This chapter offers you opportunities to:

- identify foundation thinking skills which contribute to critical thinking
- assess your recognition of patterns and your attention to detail
- practise focusing attention, to help develop your critical thinking abilities

Introduction

We use basic thinking skills in everyday life, usually with little difficulty. However, many people find it difficult to apply these same skills automatically to new contexts, such as more abstract problem-solving and academic study. This is partly because, although people use these skills in contexts that are familiar to them, they are not always sufficiently aware of the underlying strategies that they are using so as to be able to adapt them to new circumstances. The more used we are to applying skills easily in one context, the more difficult it can be to identify the underlying skills.

Critical thinking skills are based on underlying sets of thinking skills or sub-skills, such as:

- focusing attention so as to recognise the significance of fine details;
- using attention to fine detail in order to recognise patterns, such as similarities and differences, absence and presence, order and sequence;
- using recognition of pattern in order to compare and contrast items and to predict possible outcomes;
- sorting and labelling items into groups, so that they form categories;
- using an understanding of categories to identify the characteristics of new phenomena and make judgements about them.

These skills are not only useful for critical thinking in academic and professional life, but are tested as part of the procedures for selecting job applicants for interviews. Developing these sub-skills can help with your general 'employability'.

The next pages provide several short self-assessment activities for you to assess how good you are already at these skills. In particular, these give you practice in attending to fine detail, and the level of mental discipline that critical analysis involves. Student critical writing is often weakened by a lack of sustained and close attention to detail in one or more stages of the critical process. Improving your ability to pay attention to detail, selectively and at speed, is likely to have benefits for your study and assignments.

If you find the assessment easy, then progress to a chapter that is more useful for you. Otherwise, use the rest of this chapter to practise these skills further.

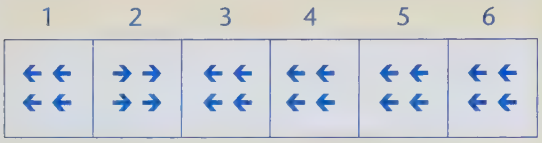
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Assess your thinking skills (1)

Comparison

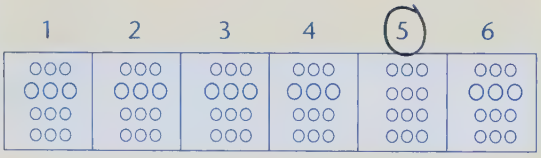
The following activity enables you to check how good you are at identifying similarity and difference, an essential sub-skill for critical thinking. For each set of boxes below, identify which box is the odd one out. An example is given first.

Example

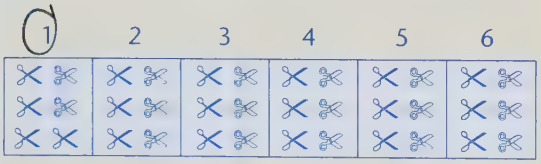


Here, box 2 is the odd one out as the arrows point in a different direction from those in the other boxes. Now try the following.

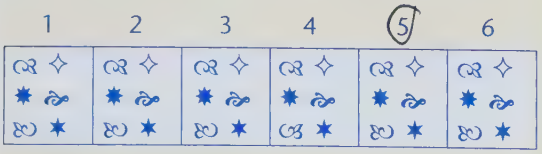
A



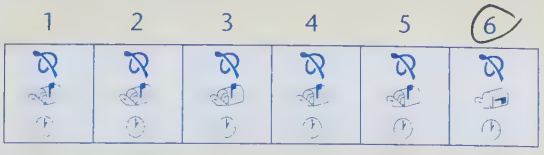
B



C



D



Sequence

This activity assesses your ability to recognise how a sequence is structured. Each set of boxes forms a sequence. Below each sequence is a set of options. Choose one option to replace the question mark and complete the sequence.

Example sequence

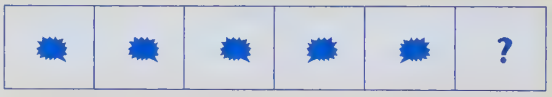


Options for the example sequence

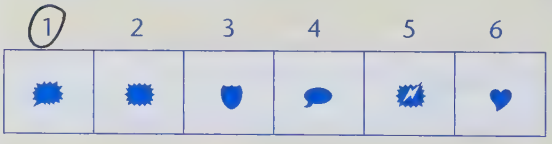


The answer is 5, as the sequence in the example is one of alternating diamonds and stars. Now complete the following sequences.

Sequence A



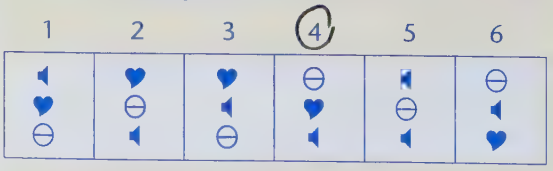
Options for sequence A



Sequence B



Options for sequence B

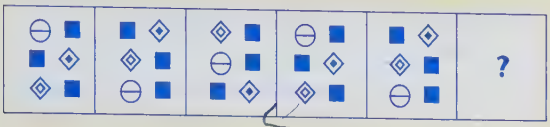


Answers: see p. 275.

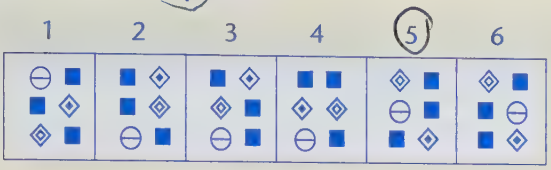
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Assess your thinking skills (2)

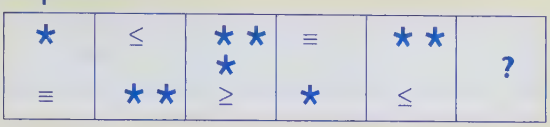
Sequence C



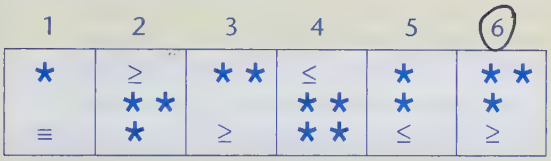
Options for sequence C



Sequence D



Options for sequence D



Categorising

This activity enables you to check your ability to categorise information. For each box, organise the set of items into two groups. Each group should be distinct, with its own particular characteristic. Identify the characteristic of each group. It is possible that there might not be the same number of items in each group.

A mouse typing drive
printer talking monitor
screen scrolling eating

B pyramid vast oasis gigantic
palm-tree desert massive
enormous Nile immense

C topaz agate silver ruby
gold opal platinum

D Empty Gate Shoal Divan burst
chops Kenya hertz micro Pound

Following directions

Answer the following questions as directed.

A Use the multiple-choice responses to answer the question: How many legs has a cow?

1 A cow has three legs.
2 A cow has two legs and two tails.
3 A cow has four legs and a tail.
4 A cow has four legs.

B Use the multiple-choice responses to answer the question: Of which atoms is water comprised?

1 Water is comprised of oxygen and hydrogen.
2 Water can exist as a solid, fluid or gas.
3 Water is comprised of oxygen and hydrogen, and is rarely, if ever, found on any planet except earth.
4 Water forms ice when it is frozen and is then considered to be a solid.

Answers: see pp. 275–6.

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Assess your thinking skills (3)

Close reading

This activity enables you to check your close reading skills. Each passage is followed by a series of questions about the text. For each question, circle:

A If the statement follows logically from the information given in the passage.

B If the statement is untrue or it does not follow logically from the information given in the passage.

C If the passage does not give enough information for you to say whether it is true or follows logically from the information that is given. Consider what other information you would need.

Passage 2.1

The Arctic

The Arctic region at the top of the northern hemisphere forms a broad cold band that encompasses both Greenland and Siberia. Conditions in the Arctic are harsh. Vegetation is sparse and temperatures are low for most of the year. Inhabitants of the region enjoy three months of continual daylight in the summer. However, in winter the sun never appears. For three months of the year there is perpetual night, the only natural light coming from the moon, the stars and the northern lights.

1 The main argument is that summers in the Arctic are very short.
A B C

2 The vegetation of the Arctic is not good to eat.
A B C

3 The sun never appears.
A B C

4 There is daylight for some or all of the day during nine months of the year.
A B C

5 There is no electricity in the Arctic.
A B C

Passage 2.2

George Washington Carver

George Washington Carver Jr. (1865–1943) was an agricultural scientist of repute. His research contributed to the development of over 300 products from peanuts alone. Amongst over 100 industrial applications he helped to develop from agricultural products such as soybeans, were rubber substitutes, paints and textile dyes. His work was celebrated by President Franklin D. Roosevelt with a national monument in 1943. Carver became a symbol to many different groups. As the first black student to attend his college, he was proof of what a former slave could achieve through education. Given Carver's claim that God was his inspiration, religious groups embraced his discoveries as a sign of God's blessing for materialism. Southern businessmen saw in him a living example of New South Philosophy and materialism. Through him, the South was transformed from being a one-crop cotton-based producer to a multi-crop economy with industrial outlets. As with many great men, the stories surrounding Carver's life have taken on mythic aspects, which are now hard to disentangle from the truth.

6 Franklin D. Roosevelt succeeded George Washington Carver as President of the USA.
A B C

7 Roosevelt constructed a monument to Carver after his death in 1943.
A B C

Answers: see p. 276.

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Assess your thinking skills (4)

8 Carver was not really a great man as his story is based on myths.
A B C

9 Religious groups thought God favoured materialism.
A B C

10 Before Carver, no black students had attended college in the USA.
A B C

11 Before Carver, the Southern USA was a multi-crop economy with industrial outlets.
A B C

12 Carver helped to invent over 100 industrial applications from soybeans.
A B C

Recognising similarities

Which of the following two passages, Option 1 or Option 2, is closest in meaning to Passage 2.1, *The Arctic* (p. 20)?

Option 1

As you move further north, the environment deteriorates to one of icy climatic conditions for much of the year, with few plants, and months without any daylight at all.

Option 2

The Arctic region in the northern hemisphere is best avoided as conditions are so harsh, and yet the inhabitants enjoy living there. They like the continual daylight in the summer and the natural light of the moon and the stars.

Which of the following three passages, Option 3, 4 or 5, is closest in meaning to Passage 2.2, *George Washington Carver* (p. 20)?

Option 3

Carver claimed that he owed his success to God's inspiration. This shows that if you put your faith in God, he will help you to be successful as an inventor and as a materialist.

Option 4

Carver was an important symbol for black groups in the South. He was born a slave and was the first black student to go to his college. Southern businessmen and religious groups would not have expected that education would have such an effect upon people they had regarded as slaves. Carver's celebration by a President of the United States was given in recognition that the economy of the Southern states no longer relied on cotton.

Option 5

The economy of the South was diversified by Carver's discoveries in agricultural science, resulting in Carver becoming something of a symbol and mythic figure for Southern groups.

Answers: see p. 276.

Use the scoring sheet on the next page to add up your score and evaluate your performance.

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Scoring sheet

Work out your score



Check the answers on pages 275–6. Use the scoring sheet to add up your score.

Item	Possible score	Your score
Comparison	4	3
A	1	1
B	1	1
C	1	1
D	1	1
Sequence	9	6
A	2	2
B	2	2
C	2	2
D	3	1
Categorising	14	10
A	3	1
B	4	4
C	4	4
D	3	1
Following directions	4	4
A	2	2
B	2	2
Close reading	15	15
1	1	
2	1	
3	1	
4	1	
5	1	
6	1	
7	2	
8	1	
9	1	
10	2	
11	1	
12	2	
Recognising similarities	4	1
Arctic	2	
Carver	2	
Total	50	
Double your total to find your percentage score	100	54%

Evaluate your score

This is only a rough test for you to see for yourself how easily you performed some of the underlying skills associated with critical thinking. People who find critical thinking hard often have difficulty in one or more of these skills. However, this is not always the case. There can be many reasons why people do well at one task and not another, so do not be discouraged if you found this difficult or gained a low score.

86–100 Excellent! This score suggests that you already have very good overall basic thinking skills to apply to critical thinking skills.

60–85 Good! If this is distributed evenly across all the items in the assessment, then you are likely to have a good basis for developing critical thinking skills. Note whether there were any areas where your score was lower than others, or which you found more difficult. You might benefit from more practice in those areas before proceeding or if you find critical thinking difficult.

30–60 Well done! You have obviously already gained a basis for developing your basic and critical thinking skills. Some people find critical thinking easier when it applies to real situations rather than in abstract activities, and that might be true of you. However, you are likely to benefit from practising these skills further, both through the activities in the rest of this chapter and taking more time on tasks which require close attention to detail.

Less than 30 Well done for sticking with the task. It may be that today wasn't your day for this kind of activity. Have a go with the activities in this chapter and see if you develop some strategies for improving these skills. If not, and if you find critical thinking to be difficult, it is worth speaking to a tutor or lecturer to discuss the assessment. Your college or university is likely to have academic skills support staff who can help.

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Attentional processes

Attention isn't the same as concentration. Concentration is associated with maintaining a focus on a task, even when it is difficult to do this. Attentional processes may include such concentration, but do not necessarily do so. Attentional processes that are important to critical thinking include:

- knowing where to look, where to focus the attention;
- being aware of when we are unable to maintain attention, and when to rest, so that we are able to maintain a sharp focus;
- becoming used to conventions, whether for reading, writing, tests or exams, so that we can use these to help us to focus attention efficiently;
- being aware of the conventions or rules for jokes, puzzles, television programmes, videos, different types of text or spoken information, so that we can use that knowledge to direct our attention efficiently;
- being aware of where there may be tricks, false impressions or illusions;
- remembering previous experience so that we can use it to direct our attention.

Automatic thinking and frames of reference

We can train our attention so that we are better able to notice relevant information. Much of the time, our mind operates 'on automatic pilot'. This makes it efficient for a range of activities other than critical thinking. Our brain tends to be effective at finding ways of saving us mental energy. Where it can, it takes short cuts and uses what it already knows to make sense of anything new it encounters. It tends to do this on an approximate basis, so it isn't always accurate.

Our mind uses its previous experiences to provide frameworks, known as 'frames of reference', to help categorise incoming information. When the brain thinks it knows what it is looking at, it will naturally stop trying to categorise the experience. That is how optical illusions or magical tricks work: our brain thinks it knows what it is seeing even though it is being tricked, so it stops looking for further explanations.

Our frames of reference can be more or less sophisticated. At a basic level, we sort situations into safe/not safe, as this helps our survival. We can usually pick out our name from background noise, even when we are not paying attention, as that also helps our survival. We sort other information according to our vocabulary, knowledge and experiences. The more experience we have of consciously thinking through how our experiences are interconnected, and labelling those experiences, the more likely it is that we can organise our thoughts and direct our attention in particular ways when we want to.

Activity 6 Find the 't'

Read the following text. Read it a second time, and identify how many times the letter 't' occurs.

Terrifying torrents and long dark tunnels are used to create the excitement of the thrilling train ride at the park.

Answer: see p. 277.

The activities in the self-assessment on pages 18–21 focus on some attentional processes that underlie critical thinking. To complete these activities, we need to pay attention to detail so as to notice patterns within the 'bigger picture'. If we can recognise patterns, we can compare items, drawing out similarities and differences. If we can recognise sequence, we are in a better position to identify trends, predict the next step and distinguish between cause and effect.

The following pages give more practice in developing attentional skills.

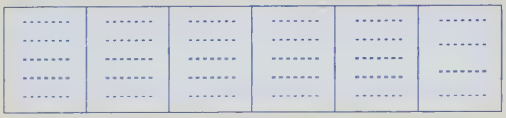
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Focusing attention: Identifying difference

This activity gives you further practice in comparing different patterns in order to identify which item does not belong to the set. For each set of boxes, identify which box is the odd one out.

1

a b c d e f



2

a b c d e f



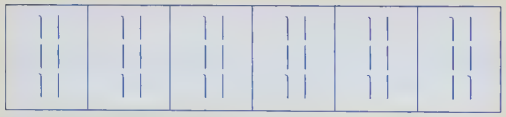
3

a b c d e f



4

a b c d e f



5

a b c d e f



6

a b c d e f



7

a b c d e f



8

a b c d e f



9

a b c d e f



10

a b c d e f



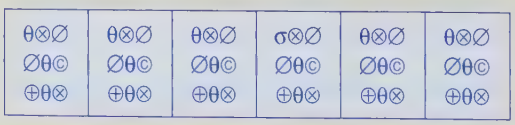
11

a b c d e f



12

a b c d e f



Answers: see p. 277.

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Focusing attention: Recognising sequence (1)

The following activity gives you further practice in recognising sequence. As in the previous assessment exercise, work out what the sequence is. Select an answer from the choice below, to replace the question mark and complete the sequence in each case.

1

a	b	c	d	e	f
#	# # # 	# # # # #	## # # # #	## ## ## # #	?

Options for sequence 1

1a	1b	1c	1d	1e	1f
# # #	## ## ##	#	# #	## ## #	## #

2

a	b	c	d	e	f
ωξ	ωξ	ωξ	ωξ	ωξ	?

Options for sequence 2

2a	2b	2c	2d	2e	2f
ωξ	ωξ ω	ωξ ωξ	ω	ωξ	ω ξ

3

a	b	c	d	e	f
XχX χXχ XχX	χXχ XχX χXχ	XχX χXχ XχX	χXχ XχX χXχ	XχX χXχ XχX	?

Options for sequence 3

3a	3b	3c	3d	3e	3f
XχX χXχ	χXχ χXχ χXχ	χXχ XχX χXχ	χXχ XχX XχX	XχX XχX XχX	XχX χXχ XχX

4

a	b	c	d	e	f
⊃	⊃ ⊄	⊃ ⊄ ∉	⊃⊃ ⊄ ∉	⊃⊃ ⊄⊄ ∉	?

Options for sequence 4

4a	4b	4c	4d	4e	4f
⊃⊃ ⊄ ∉	⊃⊃ ⊄⊄ ∉	⊃ ∉	⊃⊃ ⊄⊄ ∉∉	⊃⊃ ∉	⊃ ⊄

5

a	b	c	d	e	f
← Ψ → ↓ ⇦ ↓	Ψ ↓ ← ↓ → ⇦	↓ ↓ Ψ ⇦ ← →	↓ ⇦ ↓ → Ψ ←	⇦ → ↓ ← ↓ Ψ	?

Options for sequence 5

5a	5b	5c	5d	5e	5f
← ← Ψ ⇦ ↓ ↓	Ψ Ψ → ⇦ ↓ ↓	Ψ ← ⇦ ↓ ↓ ↓	⇦ Ψ → ↓ ↓ ↓	→ ← ⇦ Ψ ↓ ↓	← ← ⇦ ↓ Ψ ↓

6

a	b	c	d	e	f
□○ □○	□○ □○	■○ □○	■○ □○	■● *○	?

Options for sequence 6

6a	6b	6c	6d	6e	6f
■○ □○	■● *○	■○ □□	■○ ■○	■● *●	□○ □○

Answers: see p. 277.

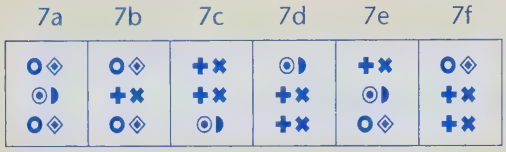
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Focusing attention: Recognising sequence (2)

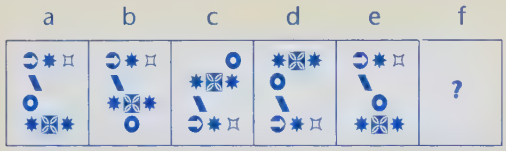
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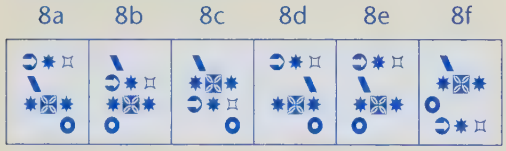
Options for sequence 7



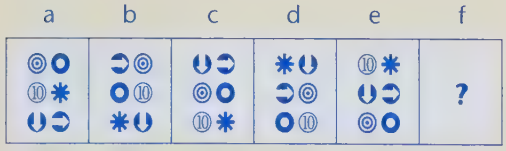
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Options for sequence 8



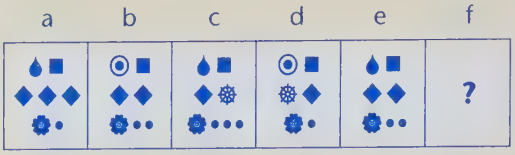
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Options for sequence 9



10



Options for sequence 10



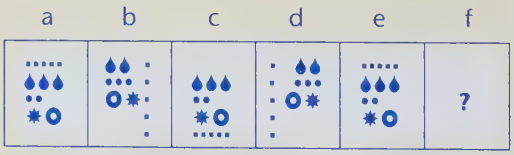
11



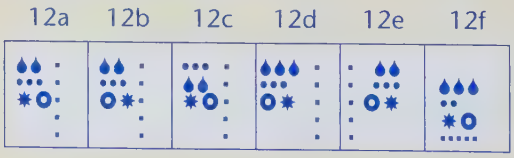
Options for sequence 11



12



Options for sequence 12



Answers: see p. 277.

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Categorising

Categorising skills are important for critical thinking as they enable you to sort information into appropriate groups and recognise which information has relevant connections to other kinds of information. In critical analysis, this helps you to compare the right things, so that you compare 'like with like'. This is necessary for constructing sophisticated arguments, such as in debate or for essays and reports.

Comparisons

Drawing comparisons is essentially about finding similarities and identifying differences. The same two items may be considered to be similar or different depending on the context and the criteria used for comparison, as the following set of questions demonstrates.

Q1 What do these eight items have in common?

zebra cat puppy goldfish
whale kitten seal elephant

Q2 What do these items have in common which makes them different from the other items in the list in Q1?

cat goldfish kitten puppy

Q3 What do these items have in common which makes them different from other items in the list?

kitten puppy

The items in Q1 are all animals. Q2 has focused in on animals that are common domestic pets and Q3, on *young* domestic pets. In each case, the selection focuses in more detail on a narrower range of shared characteristics.

Salient characteristics

'Salient' simply means 'relevant to the argument'. In the above examples, your existing knowledge of animals and pets probably made it easy to recognise the characteristics that the items in each group shared. When you recognise the characteristics that a set of items holds in common then, in effect, you are sorting these into groups, or categorising. A category is simply a group of items with shared characteristics. Any kind of category is possible: tall pointed objects; green vegetables; current prime ministers, etc.

Activity 6 Categorising

Identify the following categories (in other words, what does each group have in common?).

(a) pond lake sea pool
(b) Indian Irish Iranian Bolivian
(c) lair den pen burrow hutch
(d) biology chemistry physics geology
(e) creates stellar engines soothes
(f) decide deliver denounce devour
(g) never seven cleverest severe
(h) memory language problem-solving
(i) appendicitis tonsillitis colonitis
(j) rotor minim deed peep tenet
(k) cheluviation illuviation leaching salination
(l) 21 35 56 84 91
(m) oligarchy exarchy plutarchy democracy
(n) cete herd colony flock drove

Answers: p. 278.

Categorising involves not merely identifying shared salient characteristics, but also having the right background knowledge and vocabulary to label the group once identified. You may have found this an issue when trying to describe some of the groups above. Good background knowledge and vocabulary do make it easier to find, sort and use information at speed, making critical thinking more efficient.

The above items were easier to categorise because you already knew that they formed a category. This meant you only had to find the salient characteristics of ready-formed groups. Pattern-finding skills also make it easier to identify similarities when a group is not already formed.

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Activity: Categorising text

For each of the following sets of three texts, identify which two carry the most similar message.

Passage 2.3

Matter

(a) Different ages have classified matter in different ways. Aristotle's view was that all material substances consisted of air, earth, fire and water. This view held sway for a long time. Today, we describe liquids, solids and gases in terms of their chemical properties.

(b) Systems for classifying matter have varied over time. Although we now analyse matter in terms of chemical properties, Aristotle's division into earth, fire, air and water was used for a long time.

(c) Different ages have classified matter in different ways. Aristotle's view that all material substances consist of air, earth, fire and water was clearly erroneous and was replaced by the correct version we hold today. We describe liquids, solids and gases in terms of their chemical properties.

Passage 2.4

Anointing oil

(a) When Elizabeth II was crowned queen of England, she was anointed with a mixture of essential oils. A mixture of oils including cinnamon, rose, jasmine and civet was invented in the days of Egforth of Mercia, who was the first English king to be anointed in the manner of the Old Testament. This has been used ever since and is prepared by the royal physicians before a coronation.

(b) The anointing of kings and queens has always been an aromatic affair. Coronation oil is prepared by royal physicians, and contains a mix of essential oils. For that of Elizabeth II, these included cinnamon, rose, jasmine, civet, musk and neroli. Such a mix of oils has been used for hundreds of years, perhaps even all the way back to 785, when Egforth of Mercia was the first English king to be anointed in the manner of the Old Testament.

(c) Essential oils have always played an important part in coronation rituals. When Elizabeth II was crowned queen of England, a mixture that included cinnamon, jasmine, benzoin and neroli was prepared by the Surgeon-Apothecary for anointing her in the manner of the Old Testament. It is possible that such ingredients were used even as long ago as 785, when Egforth of Mercia was crowned.

Passage 2.5

The right hemisphere

(a) The brain controls our capacity to identify what is real. When either hemisphere is damaged, it is difficult for people to perform tasks such as differentiating their mother from a cupboard. When the right hemisphere is damaged, some individuals lose their power of imagination. They also may not be able to envisage what problems they have. This is not the case with left hemisphere damage.

(b) The right hemisphere of the brain controls the ability to recognise what is real in the outer world, such as differentiating your mother from your cupboard. When the right hemisphere is damaged, some individuals find it impossible to recognise or imagine the problem they have. This is not the case with left hemisphere damage.

(c) Our capacity to recognise what is real is controlled by the brain's right hemisphere. If this is damaged, the individual may find it impossible to imagine what problems are created by the damage. Individuals with syndromes of the left hemisphere are able to grasp what problems are created by the syndrome.

Answers: see p. 278.

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Close reading (1)

Critical thinking frequently requires you to read in very precise ways, paying attention to fine detail in order to make an accurate interpretation. This may require you to read more slowly, although, with practice, your critical reading skills can also become faster. This chapter provides practice in reading precisely.

In the activity below, you are required to answer a series of multiple-choice questions about a passage. There is a set of questions for each passage. Go through the passages in turn. You may find it helpful to check the answers for one passage before going on to another.

If you do not get all the answers correct, give some thought to where you were going wrong before moving on to the next passage.

Activity

For each statement, circle:

A If this is consistent with what is said in the passage or follows logically from the information given.

B If this is not consistent with what is said in the passage or does not follow logically from the information given.

C If the passage does not give enough information for you to conclude whether a statement is true or follows logically from the information given. Consider what other information you would need.

Passage 2.6

Traditional legends

Traditional legends from the Americas are drawn from very diverse people and regions. They cover what would have been the particular experiences of many groups: coping with natural disasters, migrations, encounters with animals, journeys and events, distilled into myths in which many peoples can see their own story. The legends illustrate cosmic themes, such as the directions of North, South, East and West. Legends are not just quaint stories. They continue a tradition of beliefs and religion that link people culturally and ethically.

1 Legends serve social and cultural purposes.

A B C

2 The themes found in legends are common to many different peoples.

A B C

3 All legends address cosmic themes such as the directions of North, South, East and West.

A B C

4 The passage implies that if you understand these myths, you will have a better sense of direction.

A B C

5 Although different people produced the legends, the legends are all the same.

A B C

6 The passage suggests that people share common experiences and can see their stories reflected in the legends produced by others.

A B C

Answers: see p. 278.

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Close reading (2)

Passage 2.7

Transformation

Disease and developmental disorders can bring about unexpected benefits. Whilst the conditions may bring unhappiness and pain, they can also bring us face to face with what we want the most out of life; many people have seen illness as a transformative event. Whilst certain opportunities close down, others can open up unexpectedly. Where certain neural pathways of the brain are blocked, for example, others may be forced into action, bringing about new ways of doing things, and sometimes even new ways of being.

7 People who travel a lot are more likely to get diseases.

A B C

8 Disease and developmental disorders are essential to neural development.

A B C

9 Most people find illness to be a transformative event.

A B C

10 If a person loses the capacity to perform an activity because of a disease, it is possible that they can learn an alternative method.

A B C

Passage 2.8

Clinical trials

Before a new pharmaceutical drug reaches the market, it must undergo clinical trials. The raw data from clinical trials of drugs are rarely published and what is published may be very misleading. The results of clinical trials that indicate that a drug is beneficial are likely to be published whereas tests that suggest the same drug is not effective are not published, and the public do not get to hear about them. The result is that even academic articles written about new drugs, which are normally based on the same data, can be very inaccurate. Not only that, but even our understanding of what an illness is can become distorted. For example, it was widely believed that depression was caused by people having a serotonin deficiency. Clinical trials suggested taking drugs that raised serotonin levels would result in greatly reducing the risk of suicide. This is contested in *The New Brain Sciences* (Rose, 2004), which argues that such drugs, far from reducing the effect of suicide, may even increase its risk, and that there is very little evidence that depressive illness has anything to do with serotonin levels.

11 The reasons given to explain depressive illness are flawed.

A B C

12 Raw data from clinical trials are less likely to be published than the results of trials that indicate that a drug is beneficial.

A B C

13 Clinical trials are not undertaken frequently enough to establish whether the effects of drugs change as a person grows older.

A B C

14 Academic articles are normally more accurate than the results of trials produced by drug companies.

A B C

15 Decreased serotonin levels lead to an increased risk of suicide.

A B C

Answers: see pp. 278–9.

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Close reading (3)

Passage 2.9

Helping others

If a person carrying a cane collapses, they are likely to receive help very quickly. This suggests that people are altruistic, but not everyone receives such help. A third more people are likely to offer help to a person who appears to be lame than to those who are bleeding or have a facial disfigurement. If a person appears seriously wounded, helpers are more likely to offer indirect assistance by seeking more expert help. People are more likely to help if the cost of helping is low or if intervention is perceived as likely to be effective. Almost all potential helpers take flight if the victim appears drunk. However, in the USA, black people were more likely to help a black drunk and white people to help a white drunk (Piliavin et al., 1981). In other circumstances, there were no obvious racial differences in willingness to assist a victim.

16 If a drunk person is bleeding, they are less likely to receive help than if they are not.

A B C

17 White people are generally more likely to help other white victims than black victims, except when they are drunk.

A B C

18 People with facial disfigurements are less likely to receive help than people who are lame.

A B C

19 People are more likely to help if they feel they can be of use.

A B C

20 It is likely that people think either that the cost of helping a person with a cane is low or that help in such cases is likely to be effective.

A B C

Check your method

If you did not get all the answers correct, give some thought to where you were going wrong before moving on to the next passage. If it is not obvious why you got an answer wrong:

- Read the passage several times, focusing on the lines related to the question you got wrong.
- Focus on the exact wording. Check whether you misread anything.
- Check whether you jumped to conclusions for which there isn't evidence in the passage.
- Check whether you read more into the passage than is actually written on the page or more than can logically be deduced from the information given.
- Did you bring other information to the passage which is not actually written down? The instructions are to use only information which is in the passage – not your general knowledge.

Answers: see p. 279.

Information about the sources

Carwell, H. (1977) *Blacks in Science: Astrophysicist to Zoologist* (Hicksville, NY: Exposition Press).

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Piliavin, J.A., Dovidio, J.F., Gaertner, S.L. and Clark, R.D. (1981) *Emergency Intervention* (New York: Academic Press).

Rose, S. (2004) *The New Brain Sciences: Perils and Prospects* (Milton Keynes: The Open University).

Sachs, O. (1985) *The Man Who Mistook His Wife for a Hat* (London: Picador).

Worwood, V.A. (1999) *The Fragrant Heavens: The Spiritual Dimension of Fragrance and Aromatherapy* (London: Bantam Books).

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Chapter 3
What's their point?
Identifying arguments

Learning outcomes

This chapter offers you opportunities to:

- identify the key components of an argument
- develop strategies for identifying reasons, conclusions and arguments within a message
- practise identifying simple arguments

Introduction

Critical thinking focuses on 'argument'. This chapter looks at what is meant by 'argument' within the context of critical thinking, and how to recognise its key features. If you are able to detect the main argument, you are better able to direct your attention to the salient, or most appropriate, material. This, in turn, enables you to direct your reading to the most relevant material, and to save time by reading more efficiently.

There are many short passages used in this chapter to help you practise critical thinking skills. It is worth noting that the activities may ask you to make judgements about the arguments, but none ask whether you agree with them. You may not agree with the reasons or conclusions given. However, critical thinking requires an evaluation of arguments in terms of the merit of their formal features, such as the quality of the reasoning, and not whether these support our own opinions. Good critical thinking includes recognising good arguments even when we disagree with them, and poor arguments even when they support our own point of view.



Doug never recognised a good argument.

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The author's position

When we read, watch television or listen to people talking, we are presented with other people's arguments. Underlying those arguments are points of view or 'positions' that they aim to convey to us, their audience.



Note how the positions of the authors above relate to the overall arguments opposite.

Key terms: Argument

The word 'argument' is used in two ways in critical thinking:

- **Contributing arguments** Individual reasons are referred to as 'arguments' or 'contributing arguments'.
- **The overall argument** This is composed of contributing arguments, or reasons. The overall argument presents the author's position. The term 'line of reasoning' is used to refer to a set of reasons, or contributing arguments, structured to support the overall argument.

Overall argument	Contributing arguments
(1) Longer prison sentences should be introduced.	Heavy punishments deter criminals. Current penalties for crime are too lenient and don't deter criminals. Since prison sentences were reduced, crime has increased. Victims need to see that perpetrators of crimes are punished.

Overall argument	Contributing arguments
(2) Increasing prison sentences is not the way to stop crime.	Crime was high even when punishments were more weighty. Prison teaches people how to be more skilled as criminals. Criminals who are imprisoned are more likely to take part in increasingly serious crime when released. Most crime is committed by people who are illiterate and lack work-related skills. Education rather than punishment is needed.

Overall argument	Contributing arguments
(3) We should invest in space travel.	Many discoveries have come about through space travel. It is important for us to learn more about the universe we live in. The fuel needed for space travel may not be around for much longer so we should use it while we have the chance.

Overall argument	Contributing arguments
(4) We should stop investing in space travel.	Space travel is expensive and the costs far outweigh the benefits. There are much more urgent projects that need investment more than space travel. Better alternatives for fuel for space travel may be available in the future.

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Activity: Capturing the author's position

Activity

Read through the following passages and identify the author's position:

- Skim quickly over the passage and note your first impressions, aiming to capture the author's position (the main message of the passage).
- Follow up your rapid read by closer reading to check whether you were right. This will give you an idea of how accurately you capture a message when reading at speed.

Passage 3.1

Barristers do not have much direct contact with their clients, but it is possible to find a legal job that suits your preferences for court work. However, if an aspiring barrister wants to spend time in court, they need to select their field carefully, to see if the work patterns associated with it match their preferences. Every field is different. Criminal lawyers may spend most of their time in court. Tax lawyers, on the other hand, may spend only a day a month or less in court. Advocacy work requires less time in court than in the office.

Passage 3.2

The nature and origin of disease was unclear until relatively recently. At the end of the nineteenth century, Koch, a Prussian scientist, introduced a set of procedures now known as *Koch's postulates*. He experimented with bacterial colonies cultivated in the laboratory, made from the blood of dying cattle. When these cultures were injected into healthy live cattle, these also caught the same disease. At the time, these findings were astonishing. Koch had been able to provide proof to support the theory that disease was spread by germs. He contributed one of the most important methodological advances in the history of medicine.

Passage 3.3

The Sahara is a region worthy of serious investigation by travellers interested in the past. Ancient architectures are no doubt hidden beneath the Saharan sands. Somewhere in the eastern Sahara, there may lie the long-lost oasis of Zerzura. In the west, there lies the fabled city of Timbuktu. Many have tried to find traces of the cultures that once straddled the great area covered by the desert.

Passage 3.4

It was initially believed that young children could not understand other people's points of view or undertake tasks such as counting and measuring until they were at least seven years of age. However, it seems the problem does not lie in children's capacity to do these things so much as in their understanding of what is being asked and why. If there is no obvious purpose, or they do not understand the language used, children find tasks difficult. Even young children can perform tasks formerly considered too advanced for them, as long as these are set up in ways that make sense to them. Problems that involve teddies or drinks, for example, may be meaningful to a very young child, whereas tasks with counters and beakers are not.

Answers: see p. 280.

Further activity

Read the introduction and conclusion of three books or articles in your subject.

How well does the introduction present the author's position: is it clear what the author is trying to persuade you to accept?

How well does the conclusion make clear what the author's position is?

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Argument: Persuasion through reasons

Persuasion and reasons

In everyday language, an 'argument' can suggest poor communication, a difficult relationship, hard feelings and, possibly, aggression.

This is not the case with argument as part of critical thinking, where an 'argument' merely means presenting reasons to support your position or point of view. If other people accept those reasons, they are more likely to be persuaded to your point of view.

An argument includes:

- a position or point of view;
- an attempt to persuade others to accept that point of view;
- reasons given to support the point of view.

To identify an argument, it is useful to keep in mind such questions as:

- 'What was the point of producing this text or programme?'
- 'What is the main message I am supposed to take from this?'
- 'What does the author/producer want me to believe, accept or do?'
- 'What reasons have they offered to support their position?'

In most circumstances, authors aim to persuade us to a particular point of view because they believe in what they are saying. However, in some cases, they may have an obvious or a hidden vested interest. It may be that they have a long-standing rivalry with academics from a different school of thought. It may be that they work for a company that wants the audience to buy its products or to subscribe to a particular view on health or pollution or genetics.

Authors may also intentionally, or unintentionally, interpret information through the filter of their own political, religious or ideological perspectives. That doesn't necessarily make their argument invalid, but it is often important to know their theoretical position in order to identify the influences on their line of reasoning.

Ambiguous arguments

Sometimes, for everyday purposes, a statement may be clear and uncontroversial. For example:

- 'It's raining' – when clearly it is raining.
- 'Everyone who ate the fish is ill' – when this is an observation of fact.
- 'I ran a mile in 4 minutes' – when this has been timed and observed.

More often, there are complexities in what we hear, see and read. It may not be obvious what point someone is trying to make, or we may suspect that there are half-truths in what they say. We recognise this in speech when we make comments such as 'What's your point?' or 'What are you trying to say?' We may wonder how someone has arrived at a particular conclusion: what they say just doesn't seem to 'add up'. When this is obvious, we may be able to point it out and resolve the misunderstanding.

However, when we are reading books or watching television, the author isn't available to answer queries about what is meant. The argument may be very complicated and it can take time to clarify the line of reasoning through careful analysis and close reading or observation. The author may also have presented the information in such a way that the lack of evidence, the illogical arguments or false conclusions are not immediately apparent. Critical thinking skills are then particularly important because we cannot always ask directly for explanations and clarifications.

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Identifying the argument (1)

Key terms: Proposition and conclusion

Propositions Statements believed to be true and presented as arguments or reasons for consideration by the audience. A proposition may turn out to be true or false.

Conclusion Reasoning should lead towards an end point, which is the conclusion. The conclusion should normally relate closely to the author's main position.

This section looks at ways of isolating key information in a passage in order to identify the argument. You may find it useful to check the glossary of terms on page xiv before proceeding. Before you read on, see if you can identify the main argument in Passage 3.5.

Passage 3.5

This area has become well known, but for unfortunate reasons. The junction of Green Road and Mill Street has been the site of over a dozen major road traffic accidents in the last five years as drivers take the corner too quickly. A local artist has made a rather grim photographic record of all the main accidents that have taken place. Some tourists have been victims. New speed cameras have now been placed at the corner of the road and this will reduce the number of accidents.

You may have noticed that some statements and pieces of information in Passage 3.5 add to our knowledge of the accident site but do not contribute to the overall argument. This general background and detail is highlighted in italics.

> *This area has become well known, but for unfortunate reasons.* The junction of Green Road and Mill Street has been the site of over a dozen major road traffic accidents in the last five years as drivers take the corner too quickly. *A local artist has made a rather grim photographic record of all the main accidents that have taken place. Some tourists have been victims.* New speed cameras have now been placed at the corner of the road and this will reduce the number of accidents.

If we remove these, the propositions, or main statements of the argument, become clearer:

> The junction of Green Road and Mill Street has been the site of over a dozen major road traffic accidents in the last five years as drivers take the corner too quickly. New speed cameras have now been placed at the corner of the road and this will reduce the number of accidents.

We can then isolate the main points in our own words:

- *Proposition 1:* Many road traffic accidents occur at the junction of Green Road and Mill Street.
- *Proposition 2:* Drivers take the corner too quickly.
- *Proposition 3:* New speed cameras are in place at the junction.
- *Conclusion:* There should now be fewer accidents.
- *Overall argument:* Speed cameras will reduce the number of accidents at the junction.

The overall argument becomes more obvious when the propositions and the conclusion are identified.

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Identifying the argument (2)

Key terms: Premises; predicate

Premises Propositions believed to be true and used as the bases for the argument; the basic building blocks for the argument.

False premise A proposition that later turns out not to be true or correct.

Predicate The foundation of the argument; the aims of the argument; an underlying point of view; the assumption that underlies the argument. For example: *the argument was predicated on a Marxist interpretation of wealth; the programme was predicated on the assumption that the prisoner was innocent.*

For Passage 3.5, we can say further that:

- This is an argument because reasons are given to support a conclusion. The reasons supporting the conclusion are:
  - that drivers drive too quickly around the corner;
  - and that accidents happen as a consequence.
- The argument is based, or 'predicated', on the assumption that drivers will take notice of the speed cameras and reduce their speed on that corner. If you disagree with this, you might consider that the conclusion is predicated on a 'false premise'.

Before you read on, see if you can identify the main argument in Passage 3.6a.

As before, general background and detail is highlighted in italics in Passage 3.6b, so that the key points in the argument stand out more clearly. As you read Passage 3.6b, consider whether the conclusion comes at the beginning, the end or elsewhere within the text.

Passage 3.6a

Pit's End should become a site of major archaeological importance. Formerly, it was believed that the three large granite stones found near the village were deposited after the melting of glaciers at the end of the last ice age. Eleven new stones were unearthed during recent excavations. The area had been covered in farmland. Aerial photography suggested that the area was worthy of excavation. The layout of the 14 stones suggests they were originally part of an unusual oval formation. They are spaced evenly at approximately two metres, which suggests they were laid out by a former settlement, possibly for religious purposes. Geologists confirm they are unlikely to have been laid down by glacial or other natural causes. Tools unearthed there recently are amongst the oldest ever discovered in this country, making them of significant interest. The dig has been funded by the national lottery.

Passage 3.6b

Background information

Pit's End should become a site of major archaeological importance. *Formerly, it was believed that the three large granite stones found near the village were deposited after the melting of glaciers at the end of the last ice age.* Eleven new stones were unearthed *during recent excavations. The area had been covered in farmland. Aerial photography suggested that the area was worthy of excavation.* The layout of the 14 stones suggests they were originally part of an unusual oval formation. They are spaced evenly at approximately two metres, which suggests they were laid out by a former settlement, possibly for religious purposes. Geologists confirm they are unlikely to have been laid down by glacial or other natural causes. Tools unearthed there recently are amongst the oldest ever discovered in this country, making them of significant interest. *The dig has been funded by the national lottery.*

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Identifying the argument (3)

If we remove the general background material from Passage 3.6a, the premises and the conclusion become clearer:

Passage 3.6c

Pit's End should become a site of major archaeological importance. Eleven new stones were unearthed. The layout of the 14 stones suggests they were originally part of an unusual oval formation. They are spaced evenly at approximately two metres, which suggests they were laid out by a former settlement, possibly for religious purposes. Geologists confirm they are unlikely to have been laid down by glacial or other natural causes. Tools unearthed there recently are amongst the oldest ever discovered in this country, making them of significant interest.

We can then isolate the main points, and give them in our own words. Note that the conclusion comes at the beginning of the argument in this instance, and the reasons to support the point of view then follow:

- *Conclusion*: Pit's End should become a site of major archaeological importance.
- *Proposition 1*: 11 new stones have been unearthed that change what was previously known about the area.
- *Proposition 2*: The oval layout of the total 14 stones is unusual.
- *Proposition 3*: The even spacing of the stones suggests they were laid down as part of a human settlement.
- *Proposition 4*: The stones are unlikely to have been laid down by natural causes.
- *Proposition 5*: Tools unearthed there recently are amongst the oldest ever discovered in this country.
- *Argument*: The unusual oval stone formation and the very ancient tools found at Pit's End should make it into a site of great archaeological importance.

While we are looking at this passage we can say further that:

- This is an argument because reasons are given to support a conclusion. The reasons given are that the stones form part of an unusual man-made formation, and that the tools are amongst the oldest in the country. These are likely to create interest.
- The argument is predicated on the assumption that the stone formation dates back to the time of the tools or pre-dates them. However, this may be a false premise as the stones could have been laid down later. That may influence whether the site is interesting archaeologically.



You can see that when presented with a string of information such as in each of the passages above, it isn't always immediately obvious what are propositions, what is the conclusion, and what is simply additional or irrelevant information. If you search out the conclusion first, it is easier to identify the relevant supporting information.

TIP

Search for the conclusions first, and the key messages may become clear quickly.

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Activity: Identifying simple arguments

Activity

Read through the passages. Identify which are arguments, and which are not. If you are unsure, re-order the sentences to see if that helps you identify a conclusion and supporting reasons.

Passage 3.7

I like that picture. The colours create the powerful effect of a sunset, which is pleasant to look at. The figures are interesting and very well drawn. It is a good picture.

Passage 3.8

Biscuits can be bad for your teeth. We often eat mid-morning when the effects of breakfast have worn off. Biscuit companies, like other food manufacturers, require their employees to wear hats to hold back their hair for health and safety reasons.

Passage 3.9

Quantum physics has identified many more dimensions than height, width, depth and time, which most people are familiar with. Such research can take a long time. Discoveries have also been made on other aspects of the time–space continuum.

Passage 3.10

The Pied Piper played a magical pipe and the side of the mountain opened. He encouraged the children from the town to enter into the mountain, which closed behind them so they were lost forever. Their parents never saw them again and he intended this to be the case.
The Pied Piper was angry at the townspeople because they refused to pay him for removing rats from the town. His action wasn't accidental; it was one of revenge.

Passage 3.11

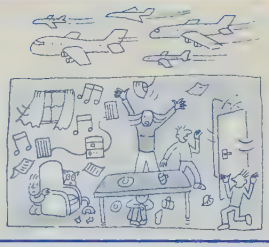
The train is late. There must have been a signal failure.

Passage 3.12

The eclipse was expected over Scotland at 9 a.m. yesterday. Lots of people turned up. The sky was still visible when they arrived but it became cloudy. When you watch an eclipse, you have to protect your eyes and you mustn't look directly at the sun.

Passage 3.13

The windows rattled and the doors banged. The air felt charged. We were all frightened. A strange sound filled the air. It must have been a ghost.



Passage 3.14

Many adults learn to read later in life. Although John and Miranda found it difficult to read as children, as adults they caught up with their peers. They enjoyed attending local literacy classes. Almost a million people have improved their literacy skills through adult classes in recent years.

Passage 3.15

Plants need nitrogen in order to grow. Although there is nitrogen in the air, plants cannot absorb it by taking it in from the air. Instead, they are reliant on bacteria in the soil to absorb nitrogen in a process known as 'nitrogen fixation'. The bacteria turn the nitrogen into nitrates which are easier for plants to absorb through their roots.

Answers: see pp. 280–1.

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Activity: Reasons and conclusions

Activity

Identify the main argument, reasons and conclusions for the following passages. It may help to highlight the reasons and conclusions with a marker pen.

Passage 3.16

A human skeleton was found near the river late last month by a senior couple walking their dog. They believed it was a murder caused by a troublesome local family. The police interviewed the family but ruled out their involvement. The bones are believed to be several hundred years old. Historians confirm that the river Marle passes close to ancient burial grounds and that there are records of other bodies being carried away by the river in the distant past. This was the first for over 150 years. Recent storms have caused the river to rise by half a metre. It is probable that the skeleton was dislodged from its resting place by the river rather than murdered by the local family.

Passage 3.17

There are only 60 species of monocotyledons, more usually known as sea grasses. Nonetheless, sea grasses make important contributions to the coastal ecosystems of every continent except Antarctica. This wasn't discovered until the end of the twentieth century. In shallow waters such as estuaries and bays, sea grasses are indeed the most dominant form of vegetation, supporting a host of marine life. They act as nurseries for fish, including commercial varieties. Moreover, without sea grasses, the bio-diversity of coastal regions would be severely impoverished. The United Nations has sponsored the *World Atlas of Sea Grasses* (Green and Short, 2004) to raise awareness of their importance.

Passage 3.18

According to Csikszentmihalyi, fundamental unhappiness arises in today's world because we are too ruled by the way the world is, rather than the way it ought to be. Although most of us know we gain from being kind, supportive and considerate, we forget this very quickly. We act to satisfy our wants, even though we know there are people who have nothing. We consider them to be far away or less important than ourselves and so we buy another television or pair of trainers rather than give money to strangers. We often ignore basic rules which help to keep the environment in balance. For example, we know carbon-based resources are in short supply and yet we use coal, gas and oil as if they were limitless. When we do this, there are consequences which bring unhappiness. In our everyday lives, we often focus on instant rewards and short-term gains, without thinking of the long-term consequences for human happiness. The challenge facing humans is to find a way of acting more co-operatively with each other and more in harmony with our universe.

Passage 3.19

It is important that pregnant women and those with poor immune systems become aware of the potential risks posed by cats. Many of us keep cats as house pets without realising the dangers they may be harbouring. Cats are host to infectious *toxoplasma gondii*, protozoa that cause disease, toxoplasmosis, in mammals such as humans. The protozoa are crescent-shaped and common in nature, but in the infectious stage rely on cats as host bodies. Adult humans rarely show signs of significant disease if they become infected. However, if pregnant women become infected, the foetus can become infected by parasites and suffer serious congenital damage. In the worst cases, infants may lose their eyesight and acquire motor deficits. In people with poor immune systems or AIDS, toxoplasmosis can cause seizures and death. The symptoms of the disease are not evident in cats so there is no way of knowing if a particular cat is a risk.

Answers: see p. 281.

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Key terms: Conclusions as deductions

In critical thinking, a conclusion is usually a deduction, which draws together the argument, and makes a reasoned assumption about how to interpret the propositions. It may offer an interpretation about what something means, or the best course of action to take.

Activity

Before reading on, check how skilled you are at spotting signals that indicate the conclusion. Look again at Passages 3.5–3.6 and look at Passages 3.16–3.19. For each, identify what signals the conclusion. Then read the commentary.

Commentary

Location of the conclusion

The end of the passage is a good place to start looking for a conclusion, as in Passages 3.5, 3.16 and 3.18. Some authors prefer to state their reasons first. They then sum these up as part of their conclusion before going on to make a deduction about the significance of the reasoning.

However, it is useful to check the opening sentences of a passage too, as in Passages 3.6, 3.17 and 3.19. Some authors choose to state the conclusion near the beginning to establish their position. They then provide the reasons to back it up, showing how they arrived at the deductions they have made.

Conclusions as interpretative summaries

In Passage 3.16, the conclusion is also an interpretive summary. The final sentence pulls together the information from the passage, which has built the case for the logical deduction about what has occurred. A conclusion in critical thinking is more than a mere summary. However, a summary can form part of a conclusion. In Passage 3.16, the conclusion is more than a summary as:

- it contains a selection of salient points which provide an interpretation of events;
- it also makes a judgement about the likelihood of this interpretation.

The author uses these devices to persuade the reader to accept the interpretation offered.

Challenges and recommendations

The conclusions in Passages 3.18 and 3.19 make challenges or recommendations. These are deductions about actions that need to be taken to achieve a desired effect. Such deductions often signal the conclusion.

Signal words

Authors can use words to 'signal', or indicate, that a conclusion is about to be made, such as 'nonetheless' in Passage 3.17. Such words are not always followed by a conclusion, but they signal that it is worth checking. Look for words such as: 'therefore', 'so', 'as a consequence', 'finally', or other phrases that imply 'therefore'.

Words that indicate deduction

As a conclusion is concerned with making deductions, it is worth looking for other words that indicate that the author is making a deduction. These include phrases such as 'this ought to', 'as a result', 'this will', 'this would have', 'this should', 'this must', 'this means that', 'in effect'. Remember to look for words that express this in the negative too, such as 'this ought never to', 'this should never'.

Examples

Passage 3.5: *this will reduce the number of accidents*

Passage 3.6: *Pit's End should become a site of major archaeological importance.*

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Summary of features

Not all messages contain an argument. When we are reading or listening to a message critically, we can save time if we check for the key features of an argument. Later in the book, we will be looking at implicit arguments, where the argument may be hidden. Here, we are concerned with explicit arguments, where the argument is expressed in a relatively open way. So far, we have identified six items to look for in identifying an argument. These are summarised in the table.

Is it an argument?	
1 Position	Authors have a position, or point of view that they attempt to persuade their audience to accept.
2 Reasons/ propositions	Reasons are provided to support the conclusion. Reasons are also referred to as 'contributing arguments' and 'propositions'.
3 A line of reasoning	A line of reasoning is a set of reasons, presented in logical order. It is like a path leading the audience through the reasons, in steps, towards the desired conclusion. It should be ordered so that it leads clearly and logically from one reason to the next. In a poor line of reasoning, it is difficult to see how each reason contributes to the conclusion.
4 Conclusion	Arguments lead towards a conclusion. The conclusion would normally be the position that the author wants you to accept. However, it is possible that the conclusion stated does not support the position that the author is advocating.
5 Persuasion	The purpose of an argument is to persuade the audience to a point of view.
6 Signal words and phrases	These help the audience follow the direction of the argument.

Locating the conclusion

There are short-cuts we can take to help us locate the main conclusion within a passage more quickly. These are only indicators of where to look as the author may not have chosen to use any of these methods to signal the conclusion.

Clues to finding the conclusion	
1 Start of passage	Conclusions are often set out or indicated early in the message, such as in the first or second sentence, or in the initial paragraph.
2 The end of a passage	Conclusions are frequently located towards the end of a message, such as in the final or penultimate sentence or paragraph.
3 Interpretive summary	Look for a summary that interprets the line of reasoning and/or makes deductions, often towards the end of a text where all the evidence is brought together. However, note page 49: summaries are not always conclusions to arguments.
4 Signal words	Look for words used to indicate that a conclusion is about to follow. See page 158.
5 Challenges and recommendations	These often form part of the conclusion. They often contain the author's position or point towards it.
6 Words indicating a deduction	Look for words that express a probable or possible outcome or interpretation.

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Summary

This chapter has looked at ways of identifying an argument. We have seen that an argument begins with a point of view or a perspective. In an argument, the author is attempting to convince the audience of that point of view.

Just because the author intends to do this, it does not mean that their position is clear. In this chapter, and in Chapter 5, we look at how authors can obscure their position through presenting it badly. Furthermore, Chapters 6 and 7 examine how authors may conceal their real position in order to persuade the audience to accept their argument.

If we can identify argument, we are in a better position to recognise when somebody is trying to persuade us of their own point of view. This can make us more alert to their line of reasoning and put us in a better position to analyse their argument. Knowing the basic components of an argument can help us to identify an argument. In checking whether material is an argument or not, we need to look for reasons that aim to persuade, and the conclusion that the author wants us to accept.

The key components of an argument are summed up in the table on page 43. These, and the 'Clues to finding the conclusion', can help to focus our reading on the main messages of a passage. Using these can make reading faster and more efficient.

An argument in critical thinking consists of reasons organised into a 'line of reasoning' or 'overall argument', with the intention of persuading a potential audience. It also consists of a conclusion, which makes a deduction based on the reasons. The line of reasoning that leads to the conclusion may be surrounded by other material that is not directly relevant to the conclusion, and which may distract the reader from the argument. Good critical reading strategies, including skill in identifying the main features of an argument, may be needed to distinguish the argument from other material.

Information about the sources

The work of barristers: Boyle, F. (1997) *The Guardian Careers Guide: Law* (London: Fourth Estate).

Children's cognitive development: Donaldson, M. (1978) *Children's Minds* (London: Fontana).

The Sahara: Sattin, A. (2004) *The Gates of Africa: Death, Discovery and the Search for Timbuktu* (London: HarperCollins).

Sea grasses: Green, E.P. and Short, F.T. (2004) *World Atlas of Sea Grasses* (Berkeley, CA: University of California Press).

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Chapter 4

Is it an argument?

Argument and non-argument

Learning outcomes

This chapter gives you opportunities to:

- understand the distinction between argument and disagreement
- recognise forms of non-argument such as summaries, explanations and descriptions
- distinguish analytical writing from descriptive writing
- select relevant material from extraneous material

Introduction

We saw in Chapter 3 that an argument consists of particular features. However, other messages may also contain some of those features without being an argument. This chapter looks at messages that are sometimes confused with argument, such as disagreement, description, summary and explanation.

Being aware of what is not an argument helps critical analysis by enabling you to categorise different types of material. This, in turn, helps you to find your way around a text or other material more effectively. The most important material is often contained within the argument, so it helps if you can find it quickly.

Critical thinking involves distinguishing what is really relevant from other forms of information. When considering argument, it is easy to be distracted by surrounding material so that you miss the point. If you can distinguish the argument from other material, you can:

- focus your attention more accurately and make better use of your time;
- ensure that your own response is directed to the most appropriate material;
- save the effort of criticising a point of view unnecessarily simply because you missed the point;
- select relevant information more effectively to cite in your own writing and reports.

It is also beneficial to your own student assignments to be able to check whether what you have written constitutes an argument. Frequently, students achieve lower grades because they think they have written a strong argument when what they have produced is, in practice, a summary of what they have read, or a description of a process or a situation.

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Argument and disagreement

Argument is not the same as disagreement. You can disagree with someone else's position without pointing out why you disagree or persuading them to think differently. In critical thinking, there is a distinction between a position, an agreement, a disagreement, and an argument.

Key terms

- **Position** A point of view.
- **Agreement** Concurring with someone else's point of view.
- **Disagreement** Holding a different point of view from someone else.
- **Argument** Using reasons to support a point of view, so that known or unknown audiences may be persuaded to agree. An argument may include disagreement, but is more than simply disagreement if it is based on reasons.





Example

- *Position*: Genetic engineering really worries me. I don't think it should be allowed. [No reasons are given so this is simply a position.]

- *Agreement 1*: I don't know much about genetic engineering but I agree with you.

Or

- *Agreement 2*: I know a lot about this subject and I agree with you. [No reasons are given so these are simply agreements.]

- *Disagreement*: That doesn't convince me. I think genetic engineering is really exciting. [No reasons are given so this is simply a disagreement.]

- *Argument 1*: Genetic engineering should be curtailed because there hasn't been sufficient research into what happens when new varieties are created without natural predators to hold them in check.

Or

- *Argument 2*: The possibilities for improving health and longevity through genetic engineering offer hope to sufferers of many conditions that currently don't have an effective cure. We should be pushing ahead to help these people as quickly as we can.

The arguments above use reasons for the position held, to persuade others to the point of view. Note that these are simple arguments: they don't have extended lines of reasoning and they don't present any evidence to support their case. Without these, the power of the argument would have to depend on other factors such as tone of voice, body language, or insider knowledge about the listener, such as that they had a vested interest in the outcome.

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Activity: Argument and disagreement

Activity

Identify for each whether the author is presenting:

A an argument, and if so, say why;

B a disagreement.

Passage 4.1

Bilingualism and multilingualism confer many benefits. Speakers of more than one language have a better understanding of how languages are structured because they can compare across two different systems. People who speak only one language lack this essential point of reference. In many cases, a second language can help people to have a better understanding and appreciation of their first language.

A B

Passage 4.2

Complementary therapies are an increasingly popular supplement to other forms of treatment. Those who use these therapies argue that treatments such as reflexology, homeopathy and shiatsu complement the care provided by the medical profession. Indeed, some people claim that these therapies are more effective than traditional medicines. Anecdotal cases of miraculous cures abound and there are those who believe such methods can compete on equal terms with medical approaches. This just isn't convincing.

A B

Passage 4.3

Several young people die each year training for the construction trades. Legislation is in place to cover health and safety at work, but some employers argue that this is too expensive to implement and onerous to monitor. They say that young people are not responsible enough at work and that there is nothing further they can do to prevent their deaths. That cannot be a good argument.

A B

Passage 4.4

People are less politically aware now than they have been at any time in the past. For hundreds of years, people took great personal risks to fight for causes that would benefit other people more than themselves. This rarely happens today. As late as the 1980s, there were frequent rallies with people in one country demonstrating to show solidarity with people elsewhere. Now, rallies are more likely to be for personal gain such as better salaries or student grants rather than for political issues of wider application. Even low-risk activities such as voting in elections attract low turn-outs.

A B

Passage 4.5

Sea-levels have risen and fallen for generations, as have temperatures. Research suggests that global warming, if it is indeed occurring, is primarily the result of natural changes in the earth's temperature and the effects of solar winds. It is now claimed that industrialisation and the burning of hydro-carbons have little effect upon climatic change. My contention is that arguments against global warming are dangerous.

A B

Passage 4.6

I cannot agree with people who say that smacking children does them no harm. Of course it harms them, both physically and emotionally. Hitting another person is assault and it would not be tolerated against an adult. Many adults have no sense of the cruelty of smacking precisely because they were smacked themselves as children and erroneously regard this as normal. They then go on to assault other vulnerable people, perpetuating a vicious cycle.

A B

Answers: see p. 282.

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Descriptions

Descriptions give an account of *how* something is done, or *what* something is like. They do not give reasoned accounts of how or why something occurred nor do they evaluate outcomes. In reports and academic writing, description should be factual, accurate and free of value judgements. Description is sometimes confused with critical analysis as both can investigate an issue in detail. Descriptive detail is not intended to persuade to a point of view but aims, rather, to give the audience a more thorough impression or understanding of the item or issue being described.

Example

The solution was placed in a test-tube and heated to 35° centigrade. Small amounts of yellow vapour were emitted. These were odourless. Forty millilitres of water were added to the solution, which was then heated until it began to boil. This time, grey steam was emitted. Water droplets gathered on the side of the test-tube.

This describes the steps taken in an experiment. Careful description of methodological procedures is an important part of writing up any kind of experimental research. No reasons are given for what happened. That critical analysis of the results would be in a separate part of the report.

Example

The painting depicts several figures gathered around a cottage and in the fields. These figures are dressed in peasant dress. All of them are located in the shadows either of the house or of the trees. It is not possible to make out any individual features on their faces or in their clothing. By contrast, the figures of the noblemen who commissioned the painting are dressed in fine and individualised apparel. These figures are all located in the foreground of the painting, in full sunshine, and their facial features are clearly distinguishable.

This passage describes some salient features of a landscape painting. The details that the author has chosen to select suggest a point of view. However, this is not made explicit. If a conclusion was added, these details might provide useful propositions to support an argument about the way rich and poor people are depicted differently in art at a particular time and place. However, the passage does not contain a conclusion and so is a description rather than an argument.

Example

Usually, when people see an object that is familiar to them, such as an elephant, a tree, a bowl, a computer, they grasp immediately what it is. They recognise the overall pattern that the object makes and don't need to work out from other sensory information such as sounds, smell and colour, what the whole object might be. However, people with a condition known as visual agnosia cannot see a whole pattern in this way: they cannot recognise objects visually. If they traced the outline of the object with their hand, they might recognise an elephant, but they can't see an elephant. They can see, and they know they are seeing something, but they can't see an elephant.

In this instance, the author is describing what the condition of visual agnosia is like. The passage is a report of the facts, as far as they were known at the time of writing. The author is not trying to persuade the audience to a point of view. You can check this by looking through the passage for an argument and reasons to support it. The word 'however', which is often associated with a change in the direction of an argument, is used here to indicate a change in the direction of the description of how vision works.

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Non-arguments: Explanations and summaries

Non-arguments can look like arguments, especially if they:

- result in a final conclusion;
- use the same signal words as an argument in order to help the flow of the writing.

Explanations

Explanations can appear to have the structure of an argument. They may include statements and reasons, leading to a final conclusion, and be introduced by signal words similar to those used for arguments. However, explanations do not attempt to persuade the audience to a point of view. They are used to:

- account for why or how something occurs;
- draw out the meaning of a theory, argument or other message.

Example

It was found that many drivers become drowsy when travelling and that long hours at the wheel were a major cause of accidents. As a result, more stopping places were set up along motorways to enable drivers to take a break.

The above example explains why more stopping places were set up along motorways.

Example

The children ate the mushrooms because they looked similar to those found in supermarkets and on the dinner table. They hadn't been taught to discriminate between safe and dangerous fungi and hadn't been told not to eat mushrooms found in hedgerows.

The above example explains why children ate dangerous mushrooms. If there were an additional sentence, such as 'therefore we need to educate children about fungi', this would become an argument, and the explanation would become a reason.

Summaries

Summaries are reduced versions of longer messages or texts. Typically, a summary repeats the key points as a reminder of what has been said already, drawing attention to the most important aspects. A conclusion may include a summary of what has been said already. New material is not usually introduced in a summary.

In the example below, the text is a list of instructions for making a cake. It does not constitute an argument. The final sentence is merely a summary of what has already been stated. The word 'therefore', which often indicates the conclusion of an argument, here simply introduces the final summary.

Example

For this cake, you need equal weights of self-raising flour, margarine and sugar. Add one egg for approximately each 50 grams of flour. Place all the ingredients in a bowl and beat furiously for three minutes. Blend the ingredients well. Pour into a greased tin and cook in the oven at 190°C for 20 mins until it is risen, golden brown and coming away from the sides of the tin. Different ovens may require different timings. Leave to cool before adding decoration such as jam and cream. Therefore, to make the cake, simply buy the ingredients, mix well, cook at 190°C, leave to cool and decorate to taste.

The passage below is a summary of Passage 3.18 (p. 41).

Example

Csikszentmihalyi argues that there is unhappiness around because we do not focus enough on how we want the world to be. Because of this, we act selfishly and focus on short-term gains, ignoring the longer-term consequences for other people and the environment. His answer is to live more in harmony with the wider world around us.

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Activity: What type of message?

Activity



Read the passages, and identify whether each is an example of an argument, a summary, an explanation or a description. How do you know?

Passage 4.7

The solar system is an inhospitable place not just for humans but also for machines. Despite this, over 8000 satellites and spacecraft were launched into space from more than 30 countries between 1957 and 2004. Over 350 people have hurtled through space, not all returning to earth. Launch sites based near the equator, such as that at Kourou in Guyana, enable rockets to make best use of the earth's rotation.

Passage 4.8

New-born babies may lack the capacity to monitor their own breathing and body-temperature during the first three months of life. Babies who sleep alongside their mothers could benefit from learning to regulate their breathing and sleeping, following the rhythm of the parent. These babies wake more frequently than those who sleep alone. Moreover, mothers who sleep next to their babies are better able to monitor their child for movement during the night. Consequently, it may be safer for new-born babies to sleep with their parents.

Passage 4.9

The article outlined the difference between individual yawns and infectious yawning. It referred particularly to research by Professor Platek which suggests that only humans and great apes yawn sympathetically. The article went on to say that people who yawn more easily in response to other people's yawns are also more likely to be good at inferring other people's states of mind. Finally, the article indicates some social benefits of yawning, suggesting that contagious yawning might have helped groups to synchronise their behaviour.

Passage 4.10

The village was located near the outer reaches of the city. The city was starting to encroach upon it, swallowing it up, road by road. It would not be long before the village disappeared altogether, to become part of the huge conurbation forming on the eastern seaboard. To the west, hills enclosed the village, trapping it between the city and the mountains beyond. A single road led out from the city, through the village and into the mountains.

Passage 4.11

Both of the toy mice were the same size and shape so the dog was confused. Although one mouse was red and one was blue, Misty was unable to tell which mouse was his toy simply by looking. Like other dogs, he needed to sniff them both, using his sense of smell to tell them apart, because he couldn't discriminate between different colours.

Passage 4.12

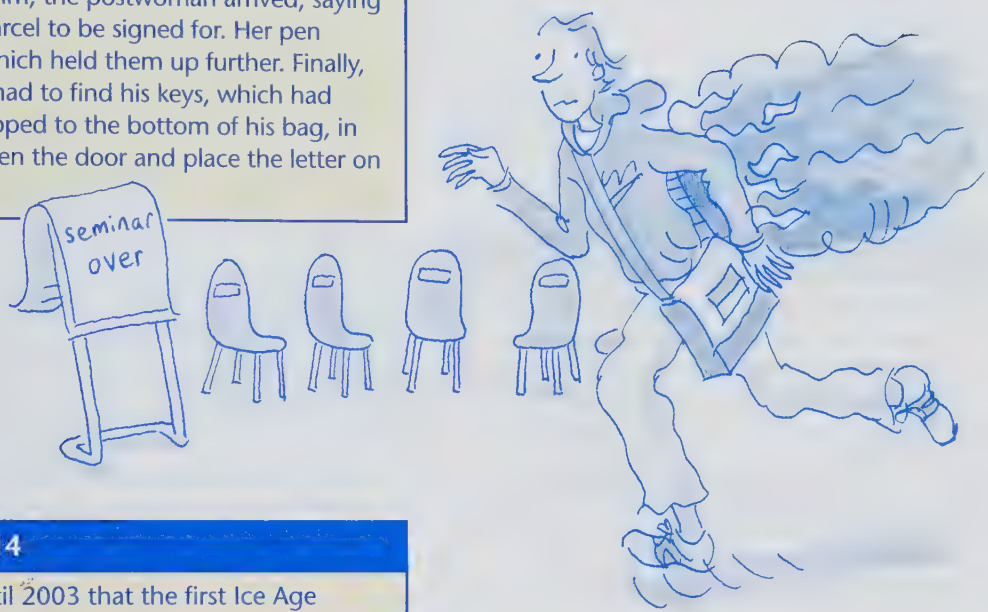
Shakespeare's *Romeo and Juliet* is set in Verona in Italy. At the beginning of the play, Romeo is pining for another young woman, but quickly falls for Juliet at a ball. Although their two families are hostile to each other, Romeo and Juliet enlist the services of their friends and a friar to bring about their marriage. Unfortunately, in a tragic turn of events, they each kill themselves, believing the other to be already dead.

Answers: see p. 282.

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Passage 4.13

There were many reasons why the student was an hour late for the seminar. First of all, a pan caught fire, causing a minor disaster in his kitchen. It took 20 minutes to restore order. Then, he couldn't find his housekeys. That wasted another 10 minutes of his time. Then, just as he closed the door behind him, the postwoman arrived, saying there was a parcel to be signed for. Her pen didn't work which held them up further. Finally, of course, he had to find his keys, which had once more slipped to the bottom of his bag, in order to re-open the door and place the letter on the table.



Passage 4.14

It was not until 2003 that the first Ice Age engravings of horses, red deer and bison were discovered at Cresswell Crags in Nottinghamshire, England. However, the oversight occurred partly because it was assumed that such work was not to be found in Britain. Indeed, in the initial survey of the cave, the experts did not notice the art that surrounded them.

Passage 4.15

The bas relief images of horses, bisons and red deer found in Cresswell Crags, England, bear remarkable similarities to those found in Germany. It is unlikely that two separate cultures would have produced drawing of such similarity if there were no links between them. This suggests that there were greater cultural links between continental Europe and Britain during the Ice Age than was formerly believed.

Passage 4.16

Recently, Ice Age specialists were excited to find evidence of some cultural links between Ice Age peoples across Europe. On a return visit to Cresswell Crags in England, they found images of horses, bison, and red deer similar to those already found in Germany. There is much controversy about other figures found on cave walls, which some experts believe to be images of dancing women, whereas others remain unconvinced.

Answers on p. 282.

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Extraneous material

Usually, arguments are not provided separately from other material. They may be surrounded by:

- introductions;
- descriptions;
- explanations;
- background information;
- summaries;
- other extraneous materials.

Example

Satellite imaging has been used to match water temperature swirls drawn on a map of ocean currents made as long ago as 1539. The map was produced by a Swedish cartographer, Olaus Magnus. It had been thought that the rounded swirls, located between pictures of serpents and sea monsters, were there for purely artistic reasons. However, the size, shape and location of the swirls match changes in water temperature too closely for this to be a coincidence. The map is likely to be an accurate representation of the ocean eddy current found to the south and east of Iceland. It is believed that the map-maker collected his information from German mariners of the Hanseatic League.



Analysis of the example

The overall argument in the example is that an old sea map is likely to be an accurate chart of part of the ocean.

Description The passage opens with a description of the method used to test the map: *Satellite imaging has been used to match water temperature swirls drawn on a map of ocean currents …*

Background information *a map of ocean currents … made as long ago as 1539. The map was produced by a Swedish cartographer, Olaus Magnus. It had been thought that the rounded swirls, located between pictures of serpents and sea monsters, were there for purely artistic reasons.*

Reason given to support the conclusion Note that the reason follows logically from the description of the swirls and is well placed to refute the idea that the swirls were primarily there for artistic reasons: *the size, shape and location of the swirls match changes in water temperature too closely for this to be a coincidence.*

Conclusion The conclusion follows on logically from the reason: *The map is likely to be an accurate representation of the ocean eddy current found to the south and east of Iceland.*

Explanatory detail The passage finishes with information that helps to explain how the map-maker gained information to make the map: *It is believed that the map-maker collected his information from German mariners of the Hanseatic League.*

Developing the skill

When you can identify different kinds of material, you will find that you can categorise parts of the text quickly as you read. You may be able to scan a text and pick out the argument. If not, it can be useful to highlight, underline or mark the conclusion and the reasons. Extract these and note them down in your own words.

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Activity: Selecting out the argument (1)

Activity

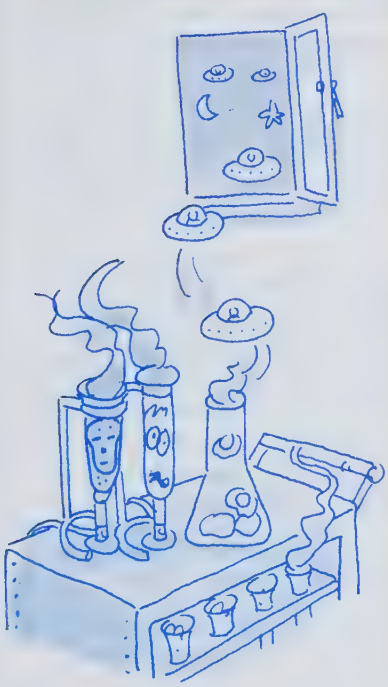
Read Passage 4.17 and identify:

(1) the conclusion;
(2) reasons given to support this;
(3) the author's consideration of opposing arguments;

and other types of message such as:

(4) the introduction;
(5) description;
(6) explanation;
(7) summary;
(8) background information and other extraneous material.

An analysis of the passage is given on the following page.



Passage 4.17

Is there anyone out there?

In some countries, the idea that there is life on other planets would make people laugh or sneer. In others, the inhabitants not only believe in life elsewhere in the universe but make efforts to communicate with it. There are certainly doubters and believers on this issue. One traditional argument for the existence of extraterrestrial life, known as the plenitude theory, is that there are so many star systems in the universe that it is unlikely that only earth would bear intelligent life. Indeed, it could be considered the folly of human arrogance to think that we are the only intelligent life in all of space. Not so, argue those who subscribe to contingency theory. Their argument, and it is a compelling one, is that life is a happy accident, a serendipitous phenomenon. They claim that the processes which led to the evolution of life are so complicated that it is extraordinary they occurred even once. They consider it extremely unlikely that the same set of processes could ever occur again. Thus, we have very divergent theories on whether there is life out there or not. It is unlikely that there is extraterrestrial life. For over 100 years, radio waves have been used to track space for signs of life and so far have uncovered nothing. If there was intelligent life out there, it is probable that we would have identified some sign of it by now. The most convincing current argument for extraterrestrial life comes from convergence theory. Convergence theory refers to situations when two different species are faced with a problem and independently arrive at the same solution. For example, both bats and birds evolved wings in order to fly. Similarly, octopus and squid have camera-like eyes. The species evolved separately, arriving at these adaptations independently. This suggests that although there may be infinite possibilities in the universe, nature tends to repeat itself. Morris (2004) has argued that where nature has produced something once, it is likely to produce it again. However, Morris himself recognises that even the basic conditions for life may be rare in the universe. Nature may be willing but the conditions might not be right. It is probable that the exacting conditions required for life are unlikely to be found more than once. It is unlikely that other planets will be exactly the right distance from their sun, with the right gravity, the right combination of chemicals and physics, with water and atmosphere. Although convergence theory indicates that nature tends to reproduce the same outcomes, and plenitude theory argues that the multiplicity of star systems increases the likelihood of extraterrestrial life, the arguments are not convincing. The conditions for life itself are so fragile and complex that it is remarkable that life occurred even once, much less that it could be repeated elsewhere.

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Activity: Selecting out the argument (2)



Analysis of Passage 4.17 *Is there anyone out there?*

The numbers in brackets refer to the tasks set in the activity box on page 53.

The Argument

(1) **Conclusion** It is unlikely that there is extraterrestrial life (line 17). The final sentence summarises the argument that supports this conclusion.

(2) **Reason 1** For over 100 years, radio waves have been used to track space for signs of life and so far have uncovered nothing (lines 18–20).

(2) **Reason 2** This uses the refuted argument referred to in (3), that it is probable that the exacting conditions required for life (chemicals and physics, water and atmosphere) are unlikely to be found again (lines 35–40).

(3) **Author's consideration of opposing theories** Alternative theories such as convergence and plenitude theories are refuted in lines 34–41, and the refutation is harnessed as a reason to support the conclusion.

Other types of message

(4) **Introduction** Lines 1–5. This sets the scene, indicating that there are widely divergent views on the issue. See also (8) below.

(5) **Description** Lines 11–16 describe contingency theory. They list the key points of the theory. Although the author does describe this argument as 'compelling', no reasons are given to show why it is compelling, so this is description, not argument or explanation. In this case, the description is also likely to be a summary of longer accounts of the theory.

(6) **Explanation** Lines 23–33 explain convergence theory. Unlike lines 11–16, these lines do more than simply list or describe what the theory says. Instead, they give examples to help clarify what is meant by the theory and draw out general principles from those examples: 'this suggests that …' (line 29). They also bring out what is significant about the theory: 'This suggests that although there may be infinite possibilities in the universe, nature tends to repeat itself.'

(7) **Summary of the material so far** Lines 16–17 summarise: 'Thus, we have very divergent theories on whether there is life out there or not.'

(8) **Background information** Lines 5–8 present background information to help set the scene: 'One traditional argument … bear intelligent life.' The argument isn't introduced until line 18. Further background information is presented in lines 10 to 16: 'Not so, argue those who subscribe to contingency theory … processes could ever occur again.'

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Summary

This chapter has looked at ways of distinguishing argument from other types of message that might be confused with arguments, either because of the interpretation of the word 'argument' in everyday language, or because a message bears the appearance of an argument.

Critical thinking is sometimes confused with disagreement. However, in critical thinking, an argument is a way of presenting a set of reasons to support a conclusion and to persuade others to a point of view. This may involve an element of disagreement, but does not necessarily do so. Conversely, in critical thinking, a disagreement that does not involve reasoning is not an argument.

Descriptions give an account of how something is done, or what something is like. They can be detailed, and so are sometimes confused with critical reasoning, which can include detailed analysis. Descriptions do not give reasoned accounts of how or why something occurred nor evaluate outcomes. In reports and academic writing, description should be factual, accurate and free of value judgements. Brief and succinct descriptions can play an important role in introducing a subject, before beginning an evaluation of it.

Explanations and summaries can appear to have the structure of an argument as they may include reasons, conclusions and signal words similar to those used for arguments. However, explanations do not attempt to persuade the audience to a point of view. They are used to account for 'why' or 'how', or to draw out the meaning, rather than to argue 'for' or 'against'. Summaries may be a shorter version of an argument, but their function is to reduce the length of the message.

Being able to identify both what is an argument and what is not, can speed your reading as you can search out the key points in a text more quickly. It can also help comprehension, as you are more likely to identify the salient points for your purpose. These skills will be looked at in more detail in Chapters 9 (reading) and 10 (writing).

Information about the sources

The nature of happiness: Csikszentmihalyi, M. (1992) *Flow: The Psychology of Happiness* (London: Random House).

Social class in eighteenth-century painting: Barrell, J. (1980) *The Dark Side of the Landscape: The Rural Poor in English Painting, 1730–1840* (Cambridge: Cambridge University Press).

Arnheim, R. (1954, 1974) *Art and Visual Perception: The Psychology of the Creative Eye* (Berkeley: University of California Press).

Sudden Infant Death Syndrome: Trevathan, W., McKenna, J. and Smith, E.O. (1999) *Evolutionary Medicine* (Oxford: Oxford University Press).

Contagious yawning: Platek, S. et al. (2003) 'Contagious Yawning: the Role of Self-awareness and Mental State Attribution', *Cognitive Brain Research*, **17** (2): 223–7.

Farrar, S. (2004a) 'It is Very Evolved of Us to Ape a Yawn', *Times Higher Educational Supplement*, 12 March 2004, p. 13.

Cresswell Crags cave art: Farrar, S. (2004b) 'It's Brit Art, but Not as We Know It', *Times Higher Educational Supplement*, 16 July 2004.

Research on Olaus Magnus's sea charts: Farrar, S. (2004c) 'Old Sea Chart is So Current', *Times Higher Educational Supplement*, 16 July 2004.

Theories about extraterrestrial life: Morris, S. (2004) *Life's Solution: Inevitable Humans in a Lonely Universe* (Cambridge: Cambridge University Press).

Pagel, M. (2004) 'No Banana-eating Snakes or Flying Donkeys are to be Found Here', *Times Higher Educational Supplement*, 16 July 2004.

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Chapter 5
How well do they say it?
Clarity, consistency and structure

Learning outcomes

This chapter gives you opportunities to:

- check arguments for clarity and internal consistency
- identify logical consistency in an argument
- check for logical order
- understand what is meant by joint and independent reasons
- identify intermediate conclusions and understand their use

Introduction

In Chapter 3, we saw that there are normally six features to check for when searching for an argument, as summarised in the table on page 43:

- author's position;
- propositions and reasons;
- a line of reasoning;
- conclusion;
- persuasion;
- use of indicator and signal words.

However, on their own, these features merely help us to identify whether an author is using an argument. They don't tell us about the quality of the argument, such as whether it is well structured and consistent. This chapter looks at how authors construct clear, consistent and logical arguments. You will have opportunities to look in more depth at how an argument is structured as a line of reasoning through the use of joint and independent reasons, interim conclusions and logical order.

By understanding how an argument is structured, you can:

- use the structure of the argument to focus reading;
- improve comprehension by understanding how one part of an argument links to another;
- apply this understanding to model your own arguments, such as for essays, reports, presentations, and dissertations.

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How clear is the author's position?

Stating the point

Clarity is important in constructing a good argument. Sometimes an author can present a great deal of interesting information but their point of view, or position, becomes lost in the detail. If the author's position is clear, then it is more likely that their audience will grasp what they are trying to say, and make the effort to follow an argument through to the end.

In a good argument, the author's position will be apparent through a number of means, such as:

- the introductory sentences;
- the final sentences;
- the conclusion;
- the overall line of reasoning;
- an overall summary of the argument;
- careful selection of facts so the argument is not lost.

Activity

Read the following passages. For each, consider:

- Is the author's position clear?
- What makes the author's position clear or unclear?

Passage 5.1

The brain of an elephant is five times larger than that of humans. Some people believe elephants are very intelligent but, even if that were true, are they really five times brighter than humans? But maybe we are looking at this the wrong way. After all, is it fair to compare the brain size of a large animal with that of a small creature? Perhaps it is relative size that matters? Human brains weigh as much as 2.5 per cent of body weight whereas elephants' brains are less than half of a per cent of their total body weight. Proportionally, the brain of a human is 10 times greater than that of an elephant. Maybe it is the ratio of brain to body size that matters? If that were the case, then the shrew, with its heavier brain, would be brighter than humans and elephants – and yet shrews do little more than eat.

Passage 5.2

Individuals have free will and so can control their own destiny. On the other hand, groups also have an identity. Research by Campbell (1984), for example, has shown that girls who mix with boys are more likely to have seen a fight and become involved in a fight than girls who mix mostly with girls. This suggests that aggressive behaviour is affected by the social environment and isn't just about character. In everyday life, our sense of self is such that we believe we are making independent decisions. We are aware we have choice and we make decisions for ourselves. Groups can also force decisions upon members, sometimes without them realising.

Passage 5.3

This report researched whether a new sports centre should be constructed in region X. Market research suggests that there is little popular demand for another sports centre in the area. However, very few people in the region use sports facilities to improve their health. The government is trying to encourage more personal responsibility for health and fitness. A sports centre would be useful in promoting this objective. People in the area are not aware of health issues and are not interested in sport. There may be government subsidies available.

Answers: see p. 283.

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Internal consistency

Clarity and internal consistency

One important aspect of presenting a clear authorial position is creating a consistent argument, so that all parts of the line of reasoning contribute to the conclusion. Nothing then contradicts or undermines the main message. Inconsistencies make an argument hard to follow, leaving the audience uncertain about what the author is trying to persuade them to believe.

Example 1

Apples are good for your teeth. Acid corrodes. Apples consist mainly of acid so they can't be good for teeth.

Here, the message lacks internal consistency. The reader is left wondering whether apples are good for your teeth or not.

Including opposing arguments

A strong line of reasoning will usually give consideration to alternative points of view, including those that appear to contradict the main argument. A good argument manages such apparent contradiction by:

- making it clear throughout the line of reasoning what position it wants the audience to take;
- making it clear when it is introducing an alternative point of view (see signal words on page 158);
- countering arguments to show why the alternative point of view is less convincing;
- resolving any apparent contradictions by showing how the main argument holds true.

Example 2

Apples are better for your teeth than refined sugar snacks. Some people argue that apples are an acid and that acid damages tooth enamel. However, any food, if left on the teeth, is bad for them. Refined sugars are particularly damaging to teeth. Compared with the sugary snacks most people eat, apples provide a more beneficial alternative and have long been recommended by dentists.

Here, the argument is internally consistent: *apples are better for your teeth than refined sugar products.* All the reasons support this. The opposing view (that acids corrode teeth) is included but its importance is minimised.

It is worth noting that the main argument is strong partly because it is worded in a more tentative way so that it is easier to defend. It is easier to argue that something is 'more beneficial than ...' rather than making an absolute statement such as 'Apples are good ...', which may not hold true in every circumstance.

Precision

Example 2 demonstrates that arguments may need to be very precisely worded. Imprecise wording is a common cause of inconsistency, as in Example 3.

Example 3

Apples are good for your teeth and have long been recommended by dentists. It may seem strange that this is the case, given that apples consist of acid and acid corrodes enamel. However, the acid is relatively harmless, and certainly apples are more beneficial than alternative snacks made of refined sugar, such as sweets and cakes.

Here, the argument is relatively well structured and is more consistent than Example 1. However, it is still not a consistent argument. The author's opening statement is: 'Apples are good for your teeth.' However, by the end of the passage, the author is arguing that the acid is 'relatively harmless' and that 'apples are more beneficial than alternative snacks'. An argument about the relative benefits is not the same as the absolute statement that 'apples are good', so the message is not internally consistent.

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Activity: Internal consistency

Activity

Read through the following passages.

- Identify whether each is: A internally consistent, or B inconsistent, and why.
- For the inconsistent ones, consider how you could adapt them to make them consistent.

Passage 5.4

All drugs which enhance performance should be banned from sport as they confer an unfair advantage on those who take them. Anyone caught taking them should be automatically banned from national and international competition. Sportspeople who take such drugs are not acting in the spirit of fair competition. On the other hand, if someone needs drugs on medical grounds, they should be allowed to compete as they did not intend to cheat.

Passage 5.5

Trainers should discourage sportspeople from taking performance enhancement drugs as these can have serious effects upon their health. Some of these drugs have resulted in distorted body shapes, skin conditions, and increased aggression. The long-term effects of some of these drugs are unknown. On the other hand, some individuals with conditions such as asthma need medication which contains those drugs. For them, taking the drugs may be more beneficial than not taking them. Therefore, it would be wrong to ban performance enhancement drugs altogether.

Passage 5.6

Reality TV is not delivering what the public wants. Too many programmes are cheaply made, turning a camera on the experiences of ordinary people who are duped into wanting their short period of fame. As a result, investment in quality programmes is declining. There is much less variety on television. The promise of choice heralded by digital TV has not materialised. Far from exercising choice, last night almost the whole nation switched on to watch the final episode of the latest reality show. What has happened to television drama, good comedy programmes and well-researched documentaries?

Passage 5.7

The countryside is a lost cause. The green fields and woodlands known as 'green belts' that surround our cities are essential to maintain the beauty of the countryside. Over 8 per cent of the countryside is now built up. Green belts are ever more essential to provide lungs to our growing cities, helping them to 'breathe'. Unfortunately, the countryside is rapidly disappearing as the extensive building of new homes stretches out of the cities. Before long it will be gone and once that happens, it will be difficult, if not impossible, to ever restore the complex ecosystems of lost woodlands and hedgerows.

Passage 5.8

Christopher Columbus was courageous in attempting to sail west to find the East Indies as, before then, everyone believed the world was flat and that he would sail over the edge. Fourth-century Christian writers such as Lactanius and Indicopleustes described the world as rectangular, but their views were not widely known. Leading medieval scholars such as Augustine, Aquinas and Albertus knew the world was round, but their minds were on higher religious issues. In Columbus's time, the scholars of Salamanca had made more accurate calculations than Columbus and, although they knew the shape of the earth, they realised Columbus had under-estimated the distances involved. They opposed his voyage but he persisted. Without his courage, the Americas might never have been discovered.

Answers: see p. 283–4.

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Logical consistency

In clear and consistent arguments, the reasons support the conclusion that the author draws from them. When evaluating an argument, we need to check whether the reasons given by the author do indeed support the conclusion. In other words, we need to check that the argument adds up. When we do this, we are checking for logical consistency.

Sometimes, authors lose track of their own arguments and draw a conclusion that does not follow from the reasons given. Sometimes, there may not be good reasons for the argument and we may feel the author is clutching at straws in the hope we won't notice the lack of logic. For Example 1, consider why the reason does not support the conclusion.

Example 1

There was a murder near the station last night. There are always young lads hanging around there. One of them probably did it. The local council should ban young people from hanging around the station.

In Example 1, the conclusion is that young people should be banned from hanging around the station. The reason given to support the conclusion is that one set of young people is often found near a station where a murder took place. This reason does not support the conclusion because there is nothing to show that:

- those young people did commit the murder;
- even if they did so, other young people would do the same;
- a general ban on young people would prevent future murders.

This is partly a question of lack of evidence. However, it is also faulty reasoning, as the conclusion does not follow from the reasons presented. An alternative conclusion might have been that if the young people were in the vicinity when the murder took place, they might have seen or heard something that would help to solve the case. For Example 2, see if you can identify the conclusion and the reasons given to support it before reading on.

Example 2

Behaviour is better in schools in rural areas than in inner city schools. Children brought up in the country have more responsibility for contributing to the family livelihood and care for vulnerable animals. This fosters a more mature attitude and a respect for life in general. Children in inner city schools often have more material possessions but value them less. They show less respect for parents and teachers. Children from the cities should be sent to school in rural schools. This would lead to more children who are respectful and well behaved.

In this case, the conclusion is provided in the last two lines: if children were sent from city to country schools, their attitude and behaviour would improve. The main reason given is that children in rural areas have better behaviour and attitudes.

However, the alleged better behaviour of children in the countryside is attributed to the responsibilities they have at home, not to the schools themselves. As city children would not gain such responsibilities simply by going to rural schools, it does not follow logically that moving school would lead to a change in their behaviour. The reasons given in the example provide better grounds for an alternative conclusion: the behaviour of city children might improve if they were given more responsibilities.

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Activity: Logical consistency

Activity

Read through the following passages. Decide whether each is logically consistent or not. Give your reasons.

Passage 5.9

The deepest parts of the oceans are known as the abyssal zone. The bathyl zone, which is that part of the abyssal zone found on the continental shelf, is too deep even for light to penetrate. Despite this absolute darkness, animal life still thrives there. Humans form part of the animal kingdom. As animals survive in the bathyl zone, this proves that we do not need light in order to survive

A consistent **B inconsistent**

Passage 5.10

Accidents happen on building sites when workers don't take sufficient care of health and safety. Many employees are lax in following health and safety guidance. This means that there will be a rise in accidents on building sites over the next year.

A consistent **B inconsistent**

Passage 5.11

Although subjects such as sports, media and popular culture involve theoretical understanding of the application of scientific principles, these subjects often have lower status at universities and with the public than subjects such as history and the classics, which are less intellectually demanding. This is partly because the former subjects attract more students from working-class backgrounds. Students who take these subjects go on to earn less than those who take more traditional subjects. This perpetuates working-class people in lower-income jobs. Therefore, working-class students should be encouraged to take traditional subjects, such as history.

A consistent **B inconsistent**

Passage 5.12

Layers of sediment are laid down over time, and build up to fill the valleys and seas until they form a sequence of rocks. The oldest rocks are always at the bottom, unless the beds of rock have been overturned, such as by folding or faulting. When there is too much molten lava under the earth or in a volcano, molten rock is forced through the layers of sediment. These are known as igneous intrusions and they harden into volcanic dikes that cut through many layers of sedimentary rock. Therefore, where an igneous intrusion cuts through a sequence of sedimentary rock, it is always more recent than the surrounding layers.

A consistent **B inconsistent**

Passage 5.13

It is impossible to find any place where there is absolute silence. Now, everywhere you go there are mobile phones ringing, people shouting, car horns blaring, music pouring from ghetto-blasters or ringing out in its irritating tinny tones from personal stereos. There is no place where you can go that does not have a sound of some kind breaking the silence. Noise pollution is definitely on the increase.

A consistent **B inconsistent**

Passage 5.14

Computers can now compete with humans in complex games such as chess and beat them. This was believed impossible until the end of the last century. Since then, computer memories have become ever larger and faster. Now, very large memories can be stored in tiny spaces. Computers do not feel emotions, a faculty which is needed in order to empathise with other people. Nonetheless, computers will one day be able to outperform humans at everything.

A consistent **B inconsistent**

Answers: see p. 284.

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Independent reasons and joint reasons

If an author gives two or more reasons to support a conclusion, these may be either:

- joint reasons; or
- independent reasons.

Joint reasons

In this case, the reasons are connected in some way and mutually reinforce each other.

Example

It is important that employers in Britain actively encourage older people to remain within the work-force. First of all, as the population ages, there won't be enough young people entering the work-force to meet the needs of the economy. Secondly, the economy benefits from the skills and experience that older people have accrued over their lifetimes. Moreover, older people often have rare skills and useful attitudes that cannot be taught or acquired quickly.

Here, the conclusion is in the first sentence. The reasons given all relate to the skills needs of the economy, and support each other:

- there won't be enough younger people to do the work;
- older people have relevant skills and experience;
- their skills and attitudes are often rare and difficult to acquire.

Independent reasons

The author may use several reasons to support the conclusion, each of which may be valid in its own right but have nothing to do with the other reasons given.

Example

It is important that employers in Britain actively encourage older people to remain within the work-force. Older people often have rare skills and useful attitudes that are wasted when they leave the work-force early. Moreover, staying on longer in full-time or part-time work is believed to be good for the health. Besides, it is unrealistic to expect savings and pensions to be sufficient to meet the needs of people retired for 30 years or more.

Here, all the reasons support the argument but are independent of each other:

- the first is economic (rare skills);
- the second relates to health concerns;
- the third relates to personal finance.

It is useful to identify whether each separate reason is sufficient in its own right to support the argument. Lots of weak reasons do not add up to a good argument, as is demonstrated in the example below.

Example

It is important that employers in Britain actively encourage older people to remain within the work-force. Firstly, older people have a right to a better standard of living. Secondly, many of them will emigrate if they do not remain active here. Thirdly, older people like to meet younger people and rarely get the opportunity outside of the workplace.

The three reasons may all be true in their own right. Having several reasons makes it sound like there must be a good case. However, an employer might consider that these are social issues that do not make a good business case for retaining older employees.

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Activity: Independent reasons and joint reasons

Activity

For each of the passages, identify whether joint or independent reasons are used to support the conclusion. The conclusions are written in italics.

Passage 5.15

Young people over the age of 16 should be allowed to vote. They pay taxes so should have a voice on how their money is spent. They can fight and die for their country so should be entitled to have a voice in the country's political process. If they have political obligations, they should also have political rights.

Passage 5.16

Expeditions leave behind a range of litter, broken equipment and other unwanted items that are gradually ruining the landscape. Few useful discoveries result from the vast numbers of expeditions now taking place. Furthermore, local economies are distorted by the requirements of expedition teams. Expeditions are sometimes unsafe and survival cannot be guaranteed. *Therefore, the number of expeditions to the Arctic should be greatly reduced.*

Passage 5.17

Telling lies is sometimes justifiable. Lies can be hurtful, but the truth can hurt even more. People do not always need to hear the truth – a fantasy can sometimes provide a practical coping mechanism for dealing with difficult circumstances. Moreover, it isn't possible always to tell the truth because it isn't clear what constitutes the 'truth'. For example, exaggeration is a form of lie but it also holds something of the truth. Lies are an important part of social bonding: we lie to maintain friendships and to keep social situations harmonious.

Passage 5.18

The author travelled with the band on tour. She visited their homes, stayed in the same hotels, and attended family parties and funerals. Having had her own band for several years, she knows the life of a rock band from the inside. However, as she was never a member of this band and was not in competition with it, she is able to give an objective account of its highs and lows, its music and the lives of the artists. *As a result, the book gives us a faithful representation of the life of the rock band.*

Passage 5.19

Knowledge management is increasingly important for business. Without it, resources are wasted. For example, companies often make poor use of the training and experience of their staff, failing to cascade it to their other employees. Furthermore, businesses that do not manage knowledge well may appear less up-to-date, and therefore less attractive, to potential customers. With the growth of electronically accessible information, businesses need strategies to help staff cope emotionally with information overload.

Passage 5.20

It took a long time for the world to appreciate the art of Magritte because he gave the public so few clues about how to interpret his work. His art calls heavily upon the unconscious, but he steadfastly refused investigation into aspects of his own life that might have helped others to understand the workings of his own unconscious. He refused to talk even about the basic events of his early life. As he didn't agree with interpretations of art based on personal problems and experiences, he offered little to encourage public interpretations of that nature.

Answers: see pp. 284–5.

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Intermediate conclusions

In longer and more elaborated lines of reasoning, there may be several sets of reasons to support the overall conclusion. In a well-constructed argument, these will be ordered so that:

- similar reasons are grouped together into sets;
- each set of reasons supports an intermediate conclusion;
- all the intermediate conclusions support the main line of reasoning.

The author may draw an intermediate conclusion on the basis of each set of reasons. This helps the reader to hold in mind the different stages of the argument. Intermediate conclusions help to structure an argument, acting as stepping stones between one stage of an argument and the next.

Example

Smokers should be given more freedom to smoke and more personal responsibility for the choices they make. Many know that cigarettes carry serious health risks, but these are risks that consenting adults are willing to take. Most smokers plan to give up before the risk becomes extreme. Adults should be allowed to make up their own mind about whether they smoke or not, without warnings on cigarette packaging. Smokers pay at least as much tax and insurance as anyone else. They also pay additional taxes through levies on cigarettes and are often required to pay higher insurance. Despite this, some medical practitioners refuse them health care. Smokers should have the same rights to health care as any other tax-payer. They should also have the same access to public spaces. In some countries, it is becoming almost impossible to find a place to smoke. Smokers are forced outside no matter what the weather. They are becoming social pariahs where once smoking was the most social of activities.

In the example above, the conclusion is at the beginning of the passage: *Smokers should be given more personal responsibility for the choices they make.*

In the version of the example reproduced below, the intermediate conclusions are underlined. Note that they can be used either to introduce a new set of reasons or to summarise reasons already introduced.

There are three sets of reasons in this passage, each linked to an intermediate conclusion which is underlined.

Reworked example

Many know that cigarettes carry serious health risks, but these are risks that consenting adults are willing to take. Most smokers plan to give up before the risk becomes extreme. Adults should be allowed to make up their own mind about whether they smoke or not, without warnings on cigarette packaging.

Smokers pay at least as much tax and insurance as anyone else. They also pay additional taxes through levies on cigarettes and are often required to pay higher insurance. Despite this, some medical practitioners refuse them health care. Smokers should have the same rights to health care as any other tax-payer.

They should also have the same access to public spaces. In some countries, it is becoming almost impossible to find a place to smoke. Smokers are forced outside no matter what the weather. They are becoming social pariahs where once smoking was the most social of activities.

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Intermediate conclusions used as reasons

Different types of intermediate conclusions

An intermediate conclusion can have two purposes:

- summative;
- to serve as a reason.

Summative

Summing up the argument at intermediate points clarifies the argument by providing it in more manageable bites. It can also reinforce the message, reminding the audience of the overall argument. The example on page 68 uses this approach. In a good argument, the author will:

- organise reasons into logical groups;
- use a sentence or a paragraph to summarise each set of reasons; this summary serves as an intermediate, or interim, conclusion.

To serve as a reason

An intermediate conclusion can also serve as a reason. The author may need to establish a solid case for an intermediate conclusion before it can serve as a reason. In other words, one set of reasons is used to establish an intermediate conclusion, and then that interim conclusion becomes a reason to support the overall conclusion (as in the table below).

Example

Universities want objective methods of marking students' work but objectivity is time-consuming. Lecturers spend a great deal of time checking their interpretations of students' answers. As there is only one correct answer for multiple-choice questions, there are no opportunities for subjective judgements, making the system fairer. These tests can be marked at speed, and objectively, by a computer. Multiple choice offers a quicker and fairer way of marking. With increased numbers of students, universities want to make better use of lecturers' time. Therefore, universities should make more use of multiple-choice tests.

Here, the overall conclusion is that universities should make more use of multiple-choice tests.

The interim conclusion is that *Multiple choice offers a quicker and fairer way of marking.*

The author of the example needed to establish that multiple choice is a quick and objective way of marking in order to argue that universities should use it. The reasons given to support the interim conclusion are that as there is only one correct answer for a multiple-choice question:

- it can be marked objectively;
- it can be marked quickly.

The structure of an argument using intermediate conclusions

Individually minor reasons …	→	act as supporting detail for intermediate conclusions …	→	which form more significant reasons …	→	to support the main argument or conclusion
reason a reason b reason c	}	All three reasons support intermediate conclusion 1	→	Intermediate conclusion 1 then becomes Reason 1	}	These two reasons then support the overall conclusion
reason d reason e	}	Both reasons support intermediate conclusion 2	→	Intermediate conclusion 2 then becomes Reason 2		

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Activity: Intermediate conclusions

Activity

Identify the main argument and the intermediate conclusions for Passage 5.21.

Passage 5.21

Although most smokers say they enjoy smoking, many smokers wish they didn't smoke. 'It feels as if I am setting light to my money', wrote one correspondent. Cigarettes can account for up to a half of an individual's total spending. As people are borrowing more money in general, and paying interest on it, the overall cost of cigarettes is sometimes hidden. However, as many smokers are all too aware, smoking does not make good financial sense. The effects on long-term health are equally devastating. Just as smokers are often building up debts in the bank, they are also accruing unseen deficits in terms of their health. It is easy to forget the health implications of smoking. Warnings about illness and death can seem a long way away. Unfortunately, once cancer of the bowel, the lung, the throat, or the stomach sets in, it is often too late to take any action. Moreover, these diseases can strike unexpectedly whilst people are still young. Smokers spread strong, unpleasant odours all around them, affecting other people without their consent. Smoking impairs the sense of smell so smokers do not realise how much they are inflicting awful odours on others. Some believe that smoking outdoors washes all those nasty odours away, but this is clearly not the case. Furthermore, studies of the houses of people who always smoke outdoors have found that the chemicals found in cigarettes are over seven times as prevalent as in the houses of non-smokers. Noxious chemicals linger, affecting the health of other people, sometimes fatally. Whether outdoors or in, smoking doesn't simply kill the smoker, it kills other people and this should not be permitted. The government should take strong action to raise awareness of the risks of smoking and to ban it in public places

Activity

Identify two intermediate conclusions used as reasons in each of the passages. In each case, the overall conclusion is in the final sentence.

Passage 5.22

It is a legal offence to assault other people. Hitting and slapping people are forms of assault and cause psychological, if not physical, damage. They should always be considered as examples of legal assault. Although this rule is applied to adults, it is often not recognised in the case of children. Slapping is defended as a useful and necessary form of discipline. It is also argued that children are not independent beings. This is not a valid argument. Children may be dependent on adults but they are still people. *Therefore, slapping a child should also count as legal assault.*

Passage 5.23

Many people speak out in discussion too quickly because they are anxious about leaving a silence. When questioned, people often acknowledge that they spoke early in order to ensure there was no gap in the discussion. They are not used to silences in conversation and don't know how to manage them skilfully. They can find silences in discussion to be unnerving and embarrassing. However, silence can be productive. First of all, it allows time for reflection so that speakers can construct a more considered and accurate response, making a more useful contribution to the debate. Secondly, it gives more people the opportunity to speak first. *For more productive discussions, we need to be skilled in managing silences.*

Answers: see pp. 285–6.

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Summative and logical conclusions

It is important to note the difference between a summative conclusion, and a logical conclusion.

Summative conclusions

Summative conclusions are simply conclusions that draw together previous information into a shorter overall summary. For example, if a text presents two main points of view, a summative conclusion would give a short synopsis of these. Summative conclusions tend to draw a piece of writing or debate to a close, without making a judgement, as in Example 1.

Example 1

What causes stomach ulcers?
It used to be assumed that stomach ulcers occurred as a result of stress. People who worked too hard or worried too much were assumed to produce excess stomach acid which would, in turn, cause ulcers. Many still hold this view. On the other hand, research has indicated that 70 per cent of stomach ulcers could be caused by the bacterium *H. Pylori*, which changes the stomach lining so that it is more vulnerable to the effects of stomach acid. This bacterial infection can be treated with antibiotics, rather than forcing the patient to reduce his or her stress levels. Hence, whilst some believe that stomach ulcers are caused by stress, others now believe that they are caused by infection.

In Example 1, the conclusion is in the last sentence and simply summarises what has gone before. In this instance, the author states the two opposing points of view, and does not use the evidence to draw a logical conclusion about which is the most likely explanation for stomach ulcers. As this example does not have a logical conclusion, it is not an example of an argument. This is an example of a summary with a summative conclusion.

Logical conclusions

A logical conclusion is a deduction based on reasons. It is more than simply a summary of the arguments or the evidence. It will include one or more judgements, drawn from an analysis of the reasons given.

Example 2

How can we predict when volcanoes will erupt?
Predicting a volcanic eruption is not an exact science. Monitoring summit activity often cannot help us predict flank activity such as eruptions down the sides of the volcano. Scientists monitoring Mount Etna in Sicily thought they had established a link whereby such flank activity was preceded by summit activity for a period of a few months. However, in 1995 summit activity began but there was not a flank eruption for a further six years. They decided Etna's eruptive cycle was more complicated than they had first thought in terms of the relationship between summit and flank activity. This may be true of other volcanoes too. Consequently, a period of summit activity cannot necessarily be used as a predictor for flank activity.

In Example 2, the conclusion is signalled by the word 'consequently'. The author deducts a conclusion from the reasons, so this is an example of an argument. The conclusion is that when the summit of a volcano shows a lot of activity, this does not necessarily mean that lava will start pouring down the side of the volcano. This is clearly based on a judgement that more recent research on Etna undermines earlier research which had suggested a closer link between its flank and summit activity.

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Activity: Summative and logical conclusions

Activity

Identify whether the conclusions in the passages are summative or logical conclusions. In each case, say whether the passage forms an argument.

Passage 5.24

Are criminals born or made?

In the 1960s, Jacobs et al. suggested a strong genetic component in criminal behaviour. On the other hand, the psychologist Bowlby argued that criminal behaviour is caused by upbringing rather than genetics, and noted that a significant number of criminals grew up in families where they experienced abuse or a lack of emotional warmth. More recently, Wilson and Hernstein suggested that a person is more likely to commit a crime if they have genes that predispose them towards criminality as well as facing additional stressors such as childhood abuse or substance misuse in adulthood. Although genes may predispose people towards crime, this is not a cause. As many criminals have experienced abuse and childhood neglect, it is fairer to argue that crime is the result of environment rather than genes, and that criminals are 'made' rather than 'born'.

Passage 5.25

Are 'reality' shows good for television?

In recent years the number of 'reality' shows on television has grown substantially. They are cheap to make and producers argue that viewers want to see 'real people' on their screens. However, critics complain that reality shows are made at the expense of original drama or current affairs programmes and that the overall quality of television is being reduced. Consequently, some people argue that reality shows are good for TV because they are cheap and popular whilst others argue that they result in poor quality television.

Passage 5.26

What is the true cost of cancelling debt? The Jubilee organisation has called for the cancellation of Third World debt. Concerns have been raised that this will mean serious losses that either commercial banks or Western governments will be forced to meet. Rowbotham suggests that debt could actually be cancelled with little cost to anyone. He argues that the dominant form of money in modern economies is bank credit. Although banks have accountancy rules about balancing assets and liabilities, credit does not exist in a physical form. It is not money sitting around in a vault waiting to be used or loaned – it is numerical or 'virtual' money. Consequently, if banks were not obliged to maintain parity between assets and liabilities they could cancel Third World debt without having to move the equivalent amount of money from the reserves to cover this. Therefore, the cancellation relates to 'virtual' money and the banks would experience no real financial loss if Third World debt were to be cancelled.

Passage 5.27

Does organic food taste better?

Supporters of organic produce argue that as well as being healthier than commercially produced food, it tastes better. Fillion and Arazi (2002) carried out blind tastings of organic and non-organic juices and milk with trained panelists. They concluded that although organic juice tasted better, there were no taste distinctions between organic and conventional milk. However, supporters of organic produce maintain that it is 'common sense' that organic food tastes better as it has been produced under healthier conditions. Hence, although scientific support for organic produce tasting better is limited, consumers who choose organic are convinced it does.

Answers: see p. 286.

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The line of reasoning, or the overall argument, should lead forwards with a clear direction, rather than hopping from one point to another in a random way, or leading the audience round in circles. In Example 3, the author moves from one point to another without direction or logical order.

Example 3

Pets add to the quality of life. Any benefits outweigh the costs. However, they can destroy household furniture. Stroking pets is thought to reduce stress. Property values can be affected by the odour animals leave behind them in carpets and curtains. Many people find talking to a pet helps them sort out personal problems. Problems with pets can be sorted out, so they are not insuperable.

The author of Example 3 could have constructed a more logical argument by:

- grouping similar points together;
- presenting reasons that support their argument first, so as to establish a good case for it;
- considering opposing reasons after they have established their own case, demonstrating why these are not significant or are less convincing.

Note the difference in Example 4, which takes a similar position to that above.

Example 4

Pets add to the quality of life. This is evident in several ways. For example, stroking pets can reduce stress. Many people find talking to a pet helps them sort out personal problems. There are some disadvantages to having an animal about the house such as damaged furniture and unpleasant odours. However, these problems can easily be overcome. The benefits of having a pet outweigh the disadvantages.

Dealing with poor logical order

If you are trying to follow a jumbled argument such as the one in Example 1, it can help to order the arguments for yourself:

- as lists of arguments 'for' and 'against'; or
- as 'arguments that support the conclusion' and 'arguments that do not support the conclusion'.

Consider how you could do this for Example 5, before reading the box below it.

Example 5

Nuclear power stations are not a viable source of energy for the future. Nuclear reactors are more expensive to build than fossil fuelled power stations. Fossil fuels such as coal, gas and oil are a dwindling resource so nuclear fuel offers a useful alternative for the future. Nuclear reactors are also very expensive to decommission so may not be efficient over the longer term. Coal costs may rise as fossil fuels become harder to find, making nuclear fuel more attractive. No truly safe way of storing nuclear fuel has yet been found. Research into alternative fuels has been under way for some time, with some success. Solar power and use of methane from waste are just two alternatives to fossil fuels.

Arguments for nuclear power stations	Arguments against nuclear power stations
• Fossil fuels will become more expensive as reserves dwindle. • Fossil fuels are likely to run out.	• More expensive to build. • More expensive to decommission. • No truly safe way of storing nuclear waste. • Other alternatives to fossil fuels exist.

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Activity: Logical order

Activity

The following passage is not ordered logically. This makes it difficult to follow its line of reasoning. You do not need to be a specialist in the subject to identify how the argument could be better constructed.

Write a short list of the ways the passage is poorly organised – then order the sentences into a more logical sequence yourself. The sentences are numbered to help you write out a preferable order.

Answers: compare your response with those on pp. 287–8.

Passage 5.28

Circadian rhythms

1: In experiments, human volunteers spent several weeks underground in constant light. 2: At first, their natural clock and sleep patterns were disrupted. 3: After a few weeks, they reverted back to the natural circadian rhythm with a 24-hour clock more or less in line with the outside world. 4: Our natural clocks are helped to adjust by exposure to sunlight and do respond to patterns of light and dark. 5: Our bodies remain more responsive to biological rhythms than to the demands of clock time or the distractions of the outside world.

6: Since the mapping of human genes as part of the genome project, we have a greater understanding of circadian rhythms and their role in genetic conditions. 7: Some families have genetic conditions which make them less sensitive to circadian rhythms. 8: This may help explain patterns of sleep disturbances found in those families. 9: Our work patterns, leisure patterns, architecture, lighting, food, drugs and medication compete with our natural clocks. 10: These biological rhythms are known as circadian rhythms and we know they are particularly strong in birds. 11: In humans they are particularly controlled by the suprachiasmatic nucleus (SCN) in the anterior hypothalamus at the base of our brains. 12: If this part of the brain is damaged, a person loses all sense of a natural 24-hour clock, where sleep coincides with night-time. 13: In other people, circadian rhythms are much stronger than was expected. 14: Astronauts, who lose this connection to the sun's rhythms for a long time, find it hard to adjust. 15: Many require medication to help them sleep.

16: Night-workers, even after 20 years on shift patterns, do not adjust circadian rhythms to suit the demands of night working. 17: Certain illnesses such as peptic ulcers and heart disease, as well as increased risk of car crashes, are much more common to night-shift workers. 18: As the long-term effects of disrupting circadian rhythms are yet to be discovered, we should take care to ensure the health of shift-workers and those with genetic conditions that make them less sensitive to the biological 24-hour clock. 19: It may be that conditions associated with mental ill-health, such as schizophrenia and bi-polar disorders, are also linked to malfunctions in circadian rhythms.

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Summary

This chapter looked at some ways of evaluating how well an argument is presented. A well-presented argument is not necessarily the correct argument, but it can be more convincing. The benefits of understanding how to present an argument well are that you are better able to:

- construct your own arguments in a convincing way;
- identify when you are being convinced by an argument because of the way it is being presented, rather than the quality of the evidence and the inherent merits of the case.

The chapter opened by looking at the author's position. This isn't always evident in an argument. However, if you can identify the author's underlying position, it is easier to anticipate the logical conclusion and reasons which support it. This aids comprehension and can help to evaluate the quality of the argument. The author's position is usually reflected in the conclusion. It is much easier to construct your own arguments if you are clear what your position is, and draw up a conclusion that reflects it. If you cannot do this, then your thinking may be muddled and further work is needed to establish what you really think and why.

Many of the other themes covered in this chapter follow on from having a clear authorial position. A clear position helps to sort ideas so that those that support the argument are easily distinguished from those that contradict it. This assists with internal consistency as a strong argument will present apparently contradictory information in such a way that it does not undermine the main argument. Indeed, a well-managed consideration of apparent contradictions can strengthen the main argument.

Once it is clear which information supports the argument, it is easier to order the argument in a logical way, so that similar points are grouped together. This helps the audience to see how the different components of the argument link together. A good argument presents materials in a logical order – that is, one which makes the best sense of the material, so that each point seems to follow on quite naturally from the one that precedes it. There can be more than one way of presenting an argument in a logical order. The important point to bear in mind is that the argument should be presented so that it leads the audience forward in an ordered way through the key points in a sequence that is clear, structured, and makes sense. This is examined further in Chapter 10.

Information about the sources

Brain size: Greenfield, S. (1997) *The Human Brain: A Guided Tour* (London: Phoenix).

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Circadian rhythms: Foster, R. (2004) *Rhythms of Life* (London: Profile Books).

Telling lies: Stein, C. (1997) *Lying: Achieving Emotional Literacy* (London: Bloomsbury).

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Chapter 6

Reading between the lines

Recognising underlying assumptions and implicit arguments

Learning outcomes

This chapter offers you opportunities to:

- recognise assumptions underlying arguments and to identify hidden assumptions
- evaluate when an argument is likely to be based on false premises
- understand what is meant by an 'implicit argument' and recognise such arguments when they occur
- understand what is meant by 'denoted' and 'connoted' meanings and be able to identify these within an argument

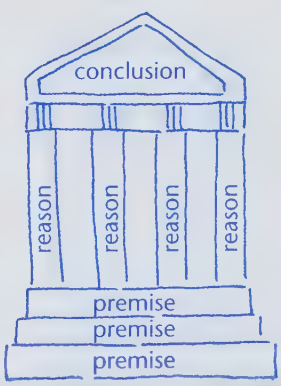
Introduction

In earlier chapters, we looked at explicit features of an argument. However, not all aspects of an argument are expressed explicitly. Arguments are often based on unstated assumptions and latent methods of persuasion. This chapter looks at some of the reasons for this, and provides practice in identifying hidden assumptions and implicit arguments.

The premises upon which an argument is based are not always immediately obvious either. These can often contain implicit assumptions or be based on incorrect information. If the premises are not sound, the argument can fall down, no matter how well it is argued. This means that a consideration of the premises of the argument is just as important as a consideration of the reasoning.

This chapter also looks briefly at latent messages used to reinforce an argument. The connotations of a message can add to its ability to persuade.

If we can recognise connoted messages, we are in a better position to see how the argument is structured, and to decide whether we agree with its underlying point of view.



Conclusion: should be well supported

Reasons: the pillars of the argument

Premises: Underlying beliefs, assumptions, foundations, theories

Good conclusions are well-supported by strong reasons based on solid premises.

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Assumptions

What is an assumption?

In critical thinking, 'assumptions' refers to anything that is taken for granted in the presentation of an argument. These may be facts, ideas or beliefs that are not stated explicitly but which underlie the argument. Without them, the same conclusion would not be possible.

Proper use of assumptions

Most arguments contain assumptions. In effect, the author invites the audience to accept something as true rather than proving it. Often, this is to save time and to simplify the argument. We don't need to have everything proved to us. When assumptions are made properly, the author has decided that it is reasonable the audience will know what is meant and is likely to agree.

Example

Holidays are a time for relaxation and enjoyment. This year, thousands of people will have their holidays ruined by oil slicks along our beaches.

Here, there are a number of assertions which we may not even recognise as assumptions because we agree with the sentiments of the passage. The conclusion is that thousands of people's holidays will be ruined. The underlying assumptions include:

Assumption 1: that holidays are for relaxation and enjoyment. This may seem obvious but the original meaning of holidays was 'holy days', which were intended for religious observation. Some people still use holidays in that way. Others may use them for seeing family or, in the case of students, finding temporary work.

Assumption 2: that thousands of holiday-makers will want to go to the beach.

Assumption 3: that those holiday-makers who go to the beach will not like oil on the beach.

Assumption 4: that oil on the beach in itself can ruin a holiday.

Assumption 5: that the audience will understand words such as *holiday*, *beach*, *relaxation*, *enjoyment*, *ruined*, *our*, and *oil slick* and that these do not need to be defined.

All of these are reasonable assumptions. The facts may not be true for every individual: some people may enjoy their holiday even with oil on the local beach. However, the assertions have sufficient general applicability to be fair assumptions. We would not expect the author to provide proof that most people who go to the beach for their holidays want to relax on an oil-free beach. We might be irritated if the author spent time proving such assertions or defining words that we are likely to know.

Taking the context into account

In critical thinking, it is important to identify what are reasonable assumptions and what are not. This can depend on the context, such as the intended audiences: will they share the same assumptions and background knowledge? If the example about oil on the beach was written in a book aimed at people learning English, there might be words such as *oil slick* which the author would need to explain.

Similarly, if the phrase 'our beaches' referred to a small part of local coastland but the article appeared in a national publication, then it would be wrong to assume the audience would be aware that only some local beaches were affected.

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Activity: Identify the underlying assumptions

Activity

For each passage, identify the underlying assumption. Remember that an assumption is not necessarily incorrect or unreasonable.

Passage 6.1

Students of the late twentieth century regularly campaigned against nuclear weapons. Students rarely demonstrate against nuclear weapons any more. Students must be less political than they used to be.

Passage 6.2

House prices rose quickly in the 1980s in many countries. There was a big slump in the 1990s and lots of house-buyers lost money. House prices are now rising very quickly again. House-buyers can expect to lose a lot of money.

Passage 6.3

Children are costing parents more. They demand more of their parents' time, expecting to be taken to activities after school, whereas in the past, parents' own interests took priority. Parents are under more pressure to provide clothes and shoes with expensive designer labels, toys, trips and even more costly brands of breakfast cereal in order for their children to be accepted by their peers. Advertising aimed at children should be banned in order to reduce this excessive peer pressure.

Passage 6.4

According to overture.com, more people search for information about the modern scientist, Emeagwali, on the internet than any other scientist. The number of pages downloaded are the equivalent to a best-selling book. Everybody must have heard about his discoveries by now.

Passage 6.5

Large companies move jobs to other countries where labour is less expensive. When wages rise in one country, the companies look for cheaper options overseas, taking the work to a new set of employees and making the former work-force redundant. Services such as call-handling can be offered from thousands of miles away. Soon, there will be no jobs left in the former high-wage economies.

Passage 6.6

Consumers are keen to eat more healthily. Information on packaging helps people to identify what food contains so they can make more informed judgements about what they eat. However, many people now refuse to eat food if the label refers to any E numbers. This demonstrates that simply putting such information on the label is not necessarily helpful: people need to know what it means.



Answers: see p. 289.

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Identifying hidden assumptions

Why identify implicit assumptions?

It is useful to identify the assumptions that underlie an argument as the overall argument can then be better understood and evaluated.

Careless use of implicit assumptions

Implicit or hidden assumptions are often used to support a conclusion. However, these may be made in such a careless way that they do not support the conclusion.

Example 1

Holidays are a time for relaxation and enjoyment. People need this time to recuperate from the stresses of work and family life. This year, thousands of people will have their holidays ruined by oil slicks along our beaches. Therefore, people who have already booked their holidays should receive compensation for the stress that these holidays will bring.

The assumption here is that people are entitled to compensation for stress caused by a spoilt holiday. If this assumption was not being made, then there would be no sense in arguing that people in a particular situation should receive such compensation. The passage also carries the assumption that people are entitled not to feel stress at holiday time:

- Holidays are needed to overcome stress.
- If there is stress during a holiday, there should be compensation.

There is also an assumption that if a holiday goes wrong after it was booked, someone somewhere must pay for this. However, this is only likely to be the case in certain circumstances. The passage is not well reasoned as it makes assumptions that are not explained clearly or well based in fact.

Non-sequiturs

'Non-sequitur' means 'doesn't follow on'. Sometimes, we can guess that there must be a hidden assumption because the conclusion seems to jump out of nowhere, rather than following on from the sequence of reasons.

Example 2

The number of people in prisons continues to rise each year and is much higher than it was over a hundred years ago. Many prisons are now overcrowded. Rehabilitation of criminals would be a much better option.

The conclusion here is that *Rehabilitation of criminals would be a much better option.* This may be the case but it doesn't follow on logically from the reasons that preceded it. The conclusion is a 'non-sequitur'. Overcrowded prisons and a larger prison population may be facts but these do not give information about the relative virtues of rehabilitation versus time in prison. That would require a different set of reasons, such as those given in Example 3.

Example 3

Research shows that, far from curing people of crimes, prison teaches criminals about how to succeed at a wider range of crimes – and how not to get caught next time. On the other hand, methods such as further education, increased social responsibilities and coming face to face with their victims have worked in individual cases to change people away from a life of crime. Prison does not have to be the only option.

Here, the conclusion may or may not be correct, but it does follow logically from the sequence of reasons. The author here gives reasons why prison does not work and why rehabilitation can.

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Implicit assumptions used as reasons

Authors may use hidden assumptions as reasons to justify their argument. In effect, they 'jump to conclusions'. We can check for this by:

- looking for gaps in the argument;
- then working out what the missing link is in the chain of reasons;
- then checking to see whether the conclusion would still be supported without those hidden assumptions.

Example

Examinations are a typical way of assessing what students have learnt and we are all familiar with the stress they can bring. How many of us have dreaded hearing those words 'put your pens down', signalling the end of the exam? If students had more time in examinations, they would finish their last questions with less hurry. This would bring them better marks. Students with disabilities can claim additional time so they have an unfair advantage during exams.

The conclusion here is: *Students with disabilities have an unfair advantage during exams.*

Three reasons are given to support this:

Reason 1: If students had more time in examinations, they would finish their last questions with less hurry.

Reason 2: (an interim conclusion used as a reason): If they finished in a less hurried way, they would get better marks.

Reason 3: Students with disabilities can claim additional time.

The implicit assumption, used as a hidden reason to support the conclusion, is that students with disabilities use additional time to complete their final question with less hurry. Without this assumption, there is a gap in the argument.

Furthermore, the effects of coping with a disability, such as sitting through an examination in extreme pain, or dictating an answer to a scribe, or translating back and forth between a signed language for the deaf and the language of the examination, were not considered in the example. It may be just as true that the additional time does not compensate sufficiently for some disabilities, much less confer an advantage. We would need more evidence to know whether any student would benefit unfairly from additional time.

Sometimes there may be several implicit assumptions. This is especially typical of spoken arguments, where we tend to jump more easily from a statement to a conclusion, leaving many assumptions unstated.

Example

Old people are scared of being robbed. They shouldn't keep their money under the bed, then.

The hidden or implicit assumptions in the example are:

- that old people in general fear being robbed, rather than only certain individuals;
- that elderly people keep money under their beds;
- that they are robbed because of this;
- that there is a link between their fear of being robbed and their keeping money under their beds.

There would need to be more evidence to support all of these assumptions. For example, we don't know how common it is for elderly people to worry about being robbed, or what percentage of them conduct their finances through organisations such as banks and building societies. However, it is more likely that senior citizens are scared of being robbed for a range of different reasons, such as the difficulty of recouping stolen money when living on a pension, or the media attention given to the occasional brutal attacks on older people.

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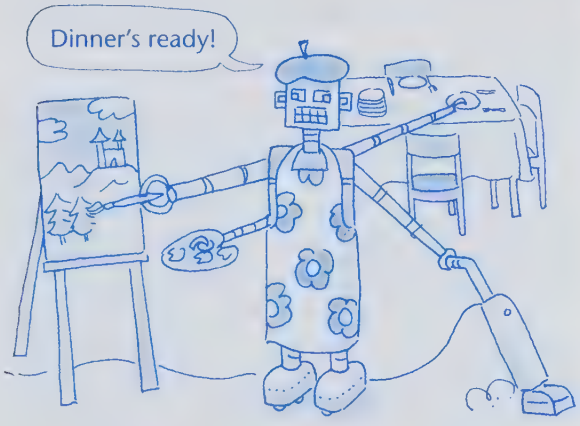
Activity: Implicit assumptions used as reasons

Activity 6

Read the following passages. In each case, identify:
(1) the conclusion;
(2) the implicit assumptions used as reasons to support the conclusion.

Passage 6.7

It has long been the hope of many people that robots would revolutionise mundane chores and hard labour such as construction work and housework. The first humanoid robot was designed by Leonardo da Vinci as long ago as 1495. We have gone for hundreds of years with little progress in gaining humanoid robots to assist around the house and construction site. Labour-saving robots are just a dream. As there has been so little advance on humanoid robots assisting with housework and construction, it will probably never be achieved.



Passage 6.8

The Electoral Commission found that intimidation was used to influence how some voters used their postal vote in the local elections. We should call an end to postal voting. This will ensure a return to fair elections.

Passage 6.9

People used plants as a method of curing illness for centuries before the advent of modern medicines. The same plants are often used by the pharmaceutical industry as the basis for the medicines we use even today. Medicines are now expensive to produce and purchase. It would be better if we returned to traditional methods, using leaves and roots of plants rather than mass-produced pharmaceuticals.

Passage 6.10

We should continue to improve sanitation and diet in order to further increase our life expectancy. People in the past had much shorter life expectancies than today. The life expectancy of pre-industrialised societies tended to be an average of 30 years. Today, people in developed countries can expect to live to over 70 years. Men, in particular, live much longer now.

Passage 6.11

Most new catering businesses collapse within the first year. Entrepreneurs tend to underestimate how long it takes to establish a client base. They run out of operating funds before they have a chance to establish themselves in the market. Many new restaurant owners give clients over-generous portions, often in a misguided attempt to lure them back to the restaurant. Therefore, in order to keep their businesses afloat, new restaurant owners should delay installing new kitchens until the restaurant is established.

Passage 6.12

Many people in the world are under-nourished or do not get enough to eat. More should be done to reduce the world's population so that food supplies can go round.

Answers: see pp. 289–91.

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False premises

Predicating an argument on premises

An argument is based on reasons which are used to support the conclusion. However, when an argument is being formulated, it is also based on beliefs, theories or assumptions, known as *premises*. We say that an argument is *predicated* on its premises. *Predicated* means 'based on'. The examples below show how these terms are used.

Example 1

Usually, only 70,000 people attend the summer festival. A recent report has argued that, this year, the organisers need to order sufficient facilities for 500,000 people. People will want to attend the location that day to see the rare solar eclipse.

Here, the argument that the organisers need to order facilities for half a million people is predicated on the premise that many people will be so interested in the solar eclipse that they will come to the festival to see it. In this example, there are underlying assumptions about the popularity of a solar eclipse.

Example 2

The airport authorities have argued that they need additional security because the proportion of football fans using the airport has risen in the last year.

Here, the argument that there is a need for increased security is predicated upon the premise that football fans automatically create more of a security risk at airports.

False premises

As the basis of an argument, the premises act like the foundations of a building. If the premises are not well founded, the argument can come tumbling down. When the underlying assumptions are incorrect, we say the argument is based, or predicated, on false premises. Usually, we need some knowledge of the circumstances, such as data or the outcome of an event, in order to recognise false premises.

Example 3

A report prior to the festival argued that the organisers needed to provide facilities for 500,000. This was based on the false premise that the public would wish to see the solar eclipse at the same location as the festival. On the day, however, the public stayed home and watched the eclipse on television. Only the usual 70,000 attended.

After the event, it was easy to see that the whole argument was predicated on incorrect assumptions – or false premises.

Example 4

The proportion of football fans using the airport has risen in the last year. The airport used to be used primarily by oil rig workers before work moved further up the coast. In order to maintain the same volume of travellers, the airport is now offering cheap family deals for football fans travelling with children.

Example 2 assumed a particular type of football fan. In Example 4, when we find out more about the fans, we can see there is no obvious reason why families travelling to a football match would create a higher security risk.

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Activity: False premises

Activity 6

For each of the following passages, make a judgement about whether the argument is likely to be based on sound or false premises. Give reasons for your answers.

Passage 6.13

Conflicts in the Middle East are likely to affect how much oil is produced in the next few months. When there is a shortage of oil, petrol prices usually rise. Therefore, the price of petrol is likely to rise this year.

Passage 6.14

Getting wet in the rain gives you a cold. The builders worked for several hours in pouring rain. Therefore, they will get colds.

Passage 6.15

Five per cent of people got married last year, and 5 per cent the year before. This means that 10 per cent of people get married every two years. Therefore, in 20 years' time, everybody will be married.

Passage 6.16

Most new restaurants struggle to survive. In order to break even after the first year of opening, we need to earn £2500 pounds a week. To make this, we need to fill every table every night. Other local restaurants fill about half their tables during the week. We have a good menu so we are likely to get a full restaurant every night. This means we will break even.

Passage 6.17

Online television will increase the number of channels from which viewers can choose. The more choice there is, the better the quality of the programmes that are produced. Therefore, online television will lead to better television programmes.

Passage 6.18

Bollywood, the Mumbai-based film industry in India, produces around 900 films every year, far more than any rival. These are being distributed to more countries than ever before. Indian films used to appeal mostly to home audiences but now attract large non-Indian audiences. India has diversified into art-films that win international acclaim. Therefore, the Indian film industry is gaining worldwide appeal.

Passage 6.19

National identities are strongly entrenched. When you are on a beach overseas, you can tell which country people come from just by watching their behaviour. French people, for example, play *boules* in the sand, whilst Englishmen are noticeable for walking round without any clothing on their upper bodies. So, there must be something in their genetic make-up that makes the people of a country behave in a similar way.

Passage 6.20

Cities are too polluted by cars' exhaust fumes and chemicals pumped into the air. In the countryside, the air is free of pollution. People ought to stop living in cities as it is healthier to live in the countryside.



Answers: see p. 291.

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Implicit arguments

Explicit and implicit arguments

When an argument follows recognisable structures, the argument is explicit. Most of the arguments introduced in the book so far have been explicit.

When it doesn't obviously follow the familiar structure of an argument, the argument is implicit. Implied arguments may lack:

- an obvious line of reasoning;
- a stated conclusion;
- the appearance of attempting to persuade.

Why use implicit arguments?

An argument can be more powerful when it does not appear to be an argument or when there does not appear to be an attempt to persuade an audience. When an argument is explicit, the audience is likely to analyse it in detail, evaluating the strength of the reasoning and the quality of the evidence. This may not suit the purposes of the author.

If a set of statements leads directly to an unstated conclusion, the audience is more likely to draw the desired conclusion for themselves. An argument can be more convincing if the audience thinks they are drawing their own conclusions. It follows that implicit arguments are most likely to be used for purposes such as:

- catching someone unawares or persuading people through an appeal at an unconscious level, for example, in advertising;
- persuading someone to do something they don't really want to do;
- putting an idea into another person's head without appearing to do so;
- threatening others or creating the idea of threatening circumstances;
- maligning other people without actually mentioning their faults;
- suggesting a consequence without stating it, in an attempt to mislead or to make the audience feel they thought of it themselves.

Example

Huge cash prizes of over a million pounds! Your number has been selected out of over 3.4 million entries to win one of our cash prizes! Ring now on this number to find out more.

In this example, the implicit argument is that the recipient of the message has won a large cash prize, probably of over a million pounds. The message doesn't actually say whether all the cash prizes are over a million pounds. It also doesn't state whether the recipient has been selected as a winner of any kind: we only know the number has been selected 'to win'. This may turn out to be a number entered into a draw. Many people are encouraged to respond to such messages, only to find they have paid more in phone bills than the prize is worth.

Ideological assumptions

An implicit argument may be one that is simply not recognised as implicit because it represents what is taken for granted in the author's society or culture – in its body of beliefs or 'ideology'. For example, it was assumed until very recently that men should not express emotion or were incapable of coping with children. This didn't always need to be stated when it underlay an argument, because everyone 'knew' it was true. Implicit arguments can be a society's equivalent of a 'blind spot'.

Subjects such as cultural and media studies today analyse texts to bring out such 'taken for granted', or ideological, aspects so that we are becoming more aware of our hidden assumptions.

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Activities: Implicit arguments

Activity

Identify the implicit arguments in the following passages.

Passage 6.21

Employees would do very well to bear in mind that all forms of trade union and association, other than for sports and recreation, are not viewed favourably. Employees are not to discuss their rates of pay with other workers.

Passage 6.22

When our candidate says he fought for his country, he really did fight for his country. When our candidate says that he hasn't stolen from the nation, he really hasn't. And when our candidate makes electoral promises about taxes, he will keep them.

Passage 6.23

There were three hundred copper pipes loaded on lorries in the parking bay at the factory on Saturday afternoon when the manager and other staff left. The pipes had disappeared by Sunday morning. Julian and Ian worked late on Saturday. Both can drive the lorries. Neither has given an alibi for Saturday night.

Passage 6.24

People in our country believe in honesty and decency. We don't believe in stealing or cheating the state. Now, officials are allowing two thousand people to emigrate to here from other countries.

Passage 6.25

Most people in this country want the death penalty. This country is a democracy. In a democracy, what most people want should count. This country does not have the death penalty.

Activity **Ideological assumptions**

What are the implicit social or ideological assumptions in the following passages?

Passage 6.26

I don't see why Ernest should be speaking when there are adults present. He is barely 20 and at an age when he should be attending to his seniors. A child should not force himself forward in this way.

Passage 6.27

Anna is eight years old now and it is time she was sent away to work. The farm at Nexby requires a pair of hands to help gather hay and feed the pigs and chickens. They will take her on and pay her room and board. She will only work from 6 a.m. until 6 p.m. every day. They are good people and will see that she does not fall into sin through idleness. Anna will be allowed home most years for Christmas day.

Passage 6.28

Now that Mr Potts has died, we will have to decide on the future of his estate. As he left only three daughters and has no living sons, the estate will have to pass to his dead brother's son, Mr Andrew Potts.

Passage 6.29

It is quite unreasonable to expect women to be employed to read the news. Some of the news is quite upsetting. It isn't all cakes, bazaars and cats stuck up trees. Newscasters often have to report on war, death and political unrest, which require a serious and steady approach.

Answers: see p. 292.

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Denoted and connoted meanings (1)

Any message may carry both denoted and connoted meanings.

Denoted meanings

The denoted meaning is the manifest meaning – the one that is most apparent on the surface.

Denoted meanings
The denoted message is the literal or explicit meaning.

Example 1

Denoted meaning

'Today! £100 reductions on all computers!'

The denoted message is: If you buy any computer at the place where the message appears, the price will be reduced by £100.

Example 2

Denoted meaning

You, too, could have a life in the sun.

The denoted message in Example 2 is: You could live where there is sunshine.

Connoted meanings

However, an argument may also contain latent messages in persuading us to a point of view. These tend to act on our unconscious as we are not necessarily aware that they are being used. Messages that act on the unconscious can be particularly powerful, so it is important to be aware of when an argument sounds convincing because of its connotations rather than its line of reasoning.

The connotations of a message can add to its effectiveness in persuading. If we can recognise connoted messages, we are in a better position to see how the argument is structured, and to decide whether we agree with its underlying point of view.

Connoted meanings
The connoted message carries additional unstated, or latent, meanings and implications. These may be obvious to the reader in some cases, but are often concealed and may need to be teased out.

Example 1a

Connoted meanings

The connotations of Example 1 are:

- These computers are bargains.
- If you don't buy the computer today, you are unlikely to get the £100 reduction so it is best to buy quickly.

Example 2a

Connoted meanings

The connotations of Example 2 are:

- A life in the sun is a desirable state that not everyone can achieve.
- If you do what we suggest, this opportunity will become available to you.



Arguing by association

One common way of creating connotations is by associating the item under discussion with another. This way, the author doesn't have to argue explicitly that an item is a certain way, but implies it through the second item.

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Denoted and connoted meanings (2)

Example 3

That's a great car you got for your birthday. I got these trainers for mine. These trainers are like gold.

The denoted meaning in Example 3 is that the person received trainers as a birthday gift. The connotations of the messages are more complicated. By associating the trainers with gold, the trainers appear to be rare and therefore more valuable. This confers some importance to the gift and/or to the receiver of the gift. This may be because the trainers really are rare. Alternatively, the author may be trying to create the illusion that the gift of trainers is just as good as the more obviously expensive gift of the car.

Products which have no connections with gold often contain the word 'gold' in the name. Alternatively, marketing materials locate a golden image such as a wedding ring prominently where it will catch the eye. The association with gold immediately suggests excellence, wealth, or scarcity. Terms such as 'golden age' suggest a better time. A golden wedding ring suggests a lasting relationship. This may encourage the audience to associate the product with the romance of weddings. The idea of a lasting relationship is useful when encouraging the idea of a long-term relationship between the audience as purchasers and the product being sold.

Latent messages

Latent messages may rely on connotations. In everyday life, we may be familiar with latent messages through the notion of 'reflected glory'. Most of us are familiar with people who don't argue explicitly: 'I am important', but imply it by mentioning all the important people they have met, or significant jobs held by friends and family. Latent messages are used a great deal in advertising and political campaigning. The product being sold, or the candidate for election, or a political argument is linked with items and ideas that carry positive meanings. Rival political opponents and their campaign messages are associated with negative messages.

Latent messages often depend upon shared social, cultural and ideological values. As we saw above, if the audience is able to make the links for themselves, the intended message can be more powerful. One well-chosen key word or concept can evoke multiple associations, producing an effective latent message.

Latent messages may be conveyed through a number of means such as:

- Playing patriotic music in the background of a political broadcast, to suggest that a particular party is the most patriotic.
- Using an image of a bird flying in an open sky, to suggest freedom and unlimited choice as a consequence of acting in the way that the argument suggests.
- Baking bread when showing viewers around a house that is for sale, to suggest a feeling of home and well-being.

Stereotyping

When an idea or a set of people are continually linked to a small number of associations, such as adjectives, job roles or forms of behaviour, this is known as stereotyping. The more that the group is linked to that set of associations, the harder it is to conceptualise members of that group as individuals.

Example 4

On the left, we have the men's bathrooms, no doubt for the doctors, and over there are the ladies' bathrooms for the nurses.

For decades in Britain, there was a stereotype that doctors were men and nurses were women. Such stereotypes are now challenged. Stereotyping often accompanies the 'in-group' and 'out-group' behaviour described on page 98.

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Activities: Associations and stereotypes

Activity Word associations

For the table below, identify which set of concepts is associated with each key word.

Key word		Associated concepts	
1	mountain	A	innocence, caring, love, tenderness, softness
2	child	B	danger, bravery, speed, unstoppable
3	fruit extract	C	romance, marriage, happiness, being special or chosen
4	wall of fire	D	man being independent; a place women aren't meant to go
5	monkey	E	healthiness, vitamins, well-being, flowing hair
6	ring	F	natural freshness, refreshingly cool, outdoors, hardiness
7	shed or den	G	humour, playfulness, tricks

Activity Associations that sell

Look at several television advertisements. Identify which words or concepts are used the most to sell different types of product. What are the associations of the words used?

Key word or concept used in advert	Associations given to the word or concept

Activity Stereotypes

Identify which stereotypes are being perpetuated in the following statements.

1 We'll decorate the room pink as they have two girls.
2 There are uniforms here for the pilots, and ladies, your stewards' costumes are over there.
3 We had better make sure there is roast beef on the menu so that the British tourists have something they are able to eat.
4 We should have expected that he couldn't control his temper, seeing he has red hair.
5 We'll play some Reggae for the visitors from the Caribbean and some flamenco music for those from Spain.
6 We should have expected trouble as there were so many football fans in the crowd.
7 There's no point providing washing machines in student halls of residence. It would be better to give them a big laundry bag so they can carry their laundry home to their parents to clean.
8 They won't be interested in fashion or computers: they are both retired now.

Answers: see p. 293.

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Activity: Denoted and connoted meanings

Activity

For each of the passages, identify:

- the denoted meaning;
- the connoted meaning;
- the use of association to create a latent message.

Then read the commentary opposite.

Passage 6.30

Although my client has been a bit naughty in the past, her behaviour has now changed. Her children have been through difficult times in the last few months. Her son has been seriously ill and her daughter was very distressed by her grandfather's death. During the period of trial contact with her children, my client has been like a rock to them. They are now reliant on her support.

Passage 6.31

All the other parties change their policies as the wind blows. Only our party has a constant and clear direction. We have our leader to thank for this, as she is the only captain who can steer a clear course through the storms currently facing our country.

Passage 6.32

It shouldn't be difficult to persuade people to take the new scheme on board. We just need to persuade the community leaders to approve our suggestions and the rest of the community will follow like sheep.

Commentary

Passage 6.30

The denoted meaning is that the client's behaviour had been poor but has now improved. She has shown she can provide good support for her children.

The connotations. The word 'naughty' is one associated with children's behaviour and therefore carries the connotation that the woman's bad behaviour wasn't very serious in adult terms. The mother is associated with the concept of a 'rock' to create the impression of a supportive and dependable mother. 'Rock' carries connotations of firmness, stability, reliability, and providing good support.

Passage 6.31

The denoted meaning is that whereas other parties change their policies, the author's party is constant in its direction irrespective of events.

The connotations. Other political parties are associated with the wind, which is changeable and unreliable. The connotation is that the parties are also unreliable. This creates a greater sense of contrast with the author's party, which is presented as steady even in a storm, rather than in mere wind. The party leader is associated with a captain of a ship. This carries connotations of 'command over the elements', and of steering a steady path towards the shore. This is not an unusual comparison, so, for some people, this association will carry further connotations of previous leaders who were successfully compared to captains of ships in the past.

Passage 6.32

The denoted message is that it will not be difficult to persuade the community to accept the new scheme if the community leaders approve it.

The connotations. The passage associates the people in the community with sheep, an animal that is considered to have little mind of its own. The connotation is that communities have little mind of their own and do whatever community leaders tell them.

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Summary

This chapter looked at some aspects of an argument that are not always made explicit, such as assumptions, implicit arguments, underlying premises and the connotations of material used in establishing the argument.

All arguments are dependent on assumptions of some kind. Assumptions may be latent within an argument for good reasons, such as an expectation that the audience will recognise them and know what they mean. This is useful in keeping an argument brief and succinct, avoiding explanations of what everyone is likely to know already.

At times, the author may assume that the audience will share assumptions or have particular knowledge when this is not the case. At other times, authors may choose not to make the underlying assumptions and persuasive techniques obvious, in order to better persuade the audience to their point of view. Assumptions can also act as reasons to support the conclusion. It is important to be able to identify latent persuasion and reasoning in order to be able to evaluate the strength and validity of the argument.

An argument may appear to be well reasoned but if it is based on false premises, it is not a good argument. As the premises are not usually made explicit, it is usually necessary to read between the lines to identify these. To identify whether premises are sound may require some knowledge of the subject, as well as exercising judgement about the likelihood of the premises being well founded. Often this requires us to call upon our common sense and experience, but we may need to research further to check whether the argument is valid.

Finally, the chapter looked at denoted and connoted meanings. The denoted meaning is the overt or explicit message, which we are more likely to recognise. However, an argument may also contain latent, or connoted, messages to persuade us to a point of view. These tend to act on our unconscious, and we are not necessarily aware that they are being used. Messages that act on the unconscious can be particularly powerful, so it is important to be able to detect latent messages. We can then evaluate whether an argument sounds convincing because of its connotations and hidden messages rather than its line of reasoning.

Information about the sources

For information about the scientist Emeagwali, see www.emeagwali.com.

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Chapter 7

Does it add up?

Identifying flaws in the argument

Learning outcomes

This chapter gives you opportunities to:

- consider a range of flaws that may be contained within an argument
- practise identifying flaws in arguments
- recognise the difference between cause and effect, correlation and coincidence
- understand what is meant by necessary and sufficient conditions, and be able to distinguish between the two
- identify a range of ways in which language can be used to distort an argument

Introduction

Chapter 3 demonstrated that an argument has several components: an author's position, a line of reasoning that uses reasons to support a conclusion, and the intention to persuade. In Chapters 4 to 6, we saw that an argument can collapse even if it appears to have those components. We have already seen how an argument may be weakened by poor structure, logical inconsistency and hidden assumptions. This chapter will look at some other ways of evaluating the strength of an argument. It will enable you to consider many common types of flaws that can occur, such as: confusing cause and effect; failing to meet necessary conditions; attacking the character of a person rather than evaluating their reasoning; misrepresentation; and using emotive language.

Arguments may be flawed because:

- The authors didn't recognise that their own arguments were flawed. This chapter can help you to recognise flaws in your own arguments so you can improve your reasoning.
- The authors intended to mislead their audiences and deliberately distorted the reasoning, or misused language to create particular responses. This chapter can help you to be more alert to flaws in other people's arguments.



Cats have whiskers.
My grandfather has whiskers.
Therefore, my grandfather must be a cat.

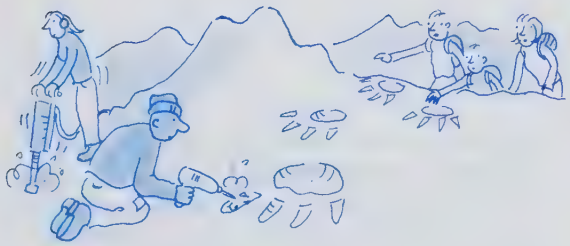
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Assuming a causal link

It is flawed reasoning to assume that because two things are found together, or occur at the same time, there must be a link between them. One example of this is assuming a link to be one of cause and effect: that one thing must be the 'cause' of another, or, in effect, jumping to a particular kind of conclusion.

Example 1

Wherever dinosaur imprints are found in rocks, there are geologists around. Therefore, geologists must make the imprints.



The assumption here is that as geologists and dinosaur prints occur in the same place, the geologists create the prints. The underlying assumption is that the dinosaur prints must be fake. If this were not the case, the author couldn't draw the conclusion that geologists must make the prints. The more logical assumption is that the prints attract the geologists as they are a natural subject for geologists to research when they are dating rocks. Other evidence is likely to prove they pre-dated the arrival of the geologists by a great many years.

Example 2

The entire family was ill last night. They all ate fish at the restaurant yesterday. Therefore, the fish must have been contaminated.

Here, the cause of the illness is linked to eating fish. The underlying assumption is that nothing else could have made the family ill. Without this assumption, the author couldn't draw the conclusion that the fish was bad. More evidence than this would be needed to prove that bad fish was the cause of the illness, such as:

- whether anybody else who ate fish from the same batch became ill;
- what the nature of the illness is;
- what else might have caused the illness;
- an examination of the fish remains.

Activity

For each passage, identify the assumed causal link or links.

Passage 7.1

Life expectancy is much higher in Western countries than in the past. Obesity is also much higher. Therefore, obesity must increase our life expectancy.

Passage 7.2

A prisoner who protested his innocence by sitting on the prison roof has been released. This is the second time that a prisoner who has protested in this way has been released. Roof-top protests must be a good way of securing release from the prison.

Passage 7.3

The man's body was found in the kitchen. A bloody knife was found nearby. The lock on the door had been broken. Somebody must have broken in and killed the man.

Answers: see p. 294.

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Correlations and false correlations

When trends are related, this is referred to as *correlation* – that is, 'related to each other'. Sometimes, there is a causal link between the correlated trends, and at other times there is not.

Example

As the temperature rises, people drink more water.

Here, the two trends of rising temperature and increased water consumption can be correlated. Drinking water is an effect caused by the increase in temperature.

Example

As the temperature fell, people were more likely to use the indoor swimming pool.

Here, the two trends of falling temperature and increased likelihood of swimming indoors can be correlated. Use of the indoor swimming pool was an effect caused by the fall in temperature. Here, the trends move in opposite directions (one falls as the other increases) so there is an inverse correlation, but the link is still one of cause and effect.

Correlations with 'third causes'

In other cases, there is not a causal link between trends that are correlated. For example, sales of ice cream may rise between May and August each year and so may sales in sandals. The trends move in the same direction and there is a relationship of some kind between the two. This means we can say that increased sales of both ice cream and sandals are correlated. It is reasonable to expect that when sales of sandals rise, there will also be a rise in ice cream sales.

However, increased sales of ice cream don't cause the higher sales of sandals, nor vice versa. If a novel brand of ice cream is launched to coincide with winter festivities, ice cream sales could rise without there being any effect on sandal sales. A third factor, warmer weather in summer, is responsible for the sales of each.



False correlations

A correlation assumes some kind of mutual relationship. Just because trends move in the same direction, this does not mean there is a correlation between them, as there may be no relationship. If a correlation is assumed where none exists, this is a false correlation.

Example

The number of car crimes has increased. There used to be only a few colours of car from which purchasers could choose. Now there is much more variety. The wider the choice of car colours, the higher the rate of car crime.

It is possible that there is some link between the two trends but it isn't likely. The connection between the two trends is likely to be coincidental rather than correlated.

Checking the relationship

When there appears to be a correlation between trends, it is important to check the ways in which they are linked:

- Are the patterns and trends coincidental rather than there being a direct link between them?
- Are they directly linked as cause and effect?
- Are they linked by a third cause?

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Activity: Identify the nature of the link

Activity

For each of the passages, identify whether:

A The reasons support the conclusion through causal links.

B The conclusion only works if there is an assumption or assumptions that are not covered by the reasons. Identify the assumption(s) made.

C There is no link between the reasons and the conclusion.

Passage 7.4

Reason 1: Sugar destroys teeth.
Reason 2: Children eat a lot of sugar.
Reason 3: Children's teeth decay quickly.
Conclusion: Children's teeth decay quickly because of the sugar they eat.

Passage 7.5

Reason 1: More students use the internet for research and for submitting their work than in the past.
Reason 2: The overall number of students has risen but the number of teaching staff has not.
Reason 3: The proportion of students plagiarising the work of other people is likely to have remained the same.
Conclusion: Students are now more at risk of being discovered plagiarising.

Passage 7.6

Marie Curie, Einstein, and Darwin had long hair. They were all great scientists. Therefore, to be a great scientist you need long hair.

Passage 7.7

Reason 1: The price of football tickets has risen.
Reason 2: Football players receive higher wages than ever before.
Conclusion: Spectators are paying more to watch matches in order to pay footballers' high wages.

Passage 7.8

Reason 1: Hedgehogs enjoy eating ice cream.
Reason 2: Hedgehogs eat ice cream if it is stored in containers that they can break open.
Reason 3: Fast food outlets report a lot of ice cream wastage recently.
Conclusion: Hedgehogs must be breaking in to eat the ice cream at fast food outlets.



Passage 7.9

Reason 1: Dubai's population doubled every 10 years between 1940 and 2010 and is continuing to rise.
Reason 2: The port created in 1979 at Jebel Ali provided a prosperous free trade zone that brought in people from all over the world.
Reason 3: Many projects for improving the economic infrastructure, from sporting events and theme parks to world-class technology parks and international finance centres, have encouraged people to settle in Dubai.
Reason 4: Large-scale property development is under way, offering better opportunities for foreign nationals to own property in Dubai.
Conclusion: Dubai's population is increasing because of the opportunities it provides to foreign nationals.

Answers: see p. 294.

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